

Chapter 6: Shunt FACTS Controllers: TCR, TSR, TSC, SVC and STATCOM

Learning Objectives

After working through this chapter, the reader should be able to:

- explain, from the Thevenin equivalent of a bus and from the two-machine power-angle relationship, why controllable shunt reactive current is the most direct means of supporting voltage and raising transfer capability;
- derive the instantaneous current, fundamental susceptance, total RMS current, individual harmonics and THD of a thyristor-controlled reactor, and translate correctly between the zero-crossing and peak-referenced firing-angle conventions found in the literature;
- analyse six-pulse, twelve-pulse and segmented TCR arrangements, including the sequence behaviour of their harmonics and the cancellation produced by phase-shifting transformers;
- design thyristor-switched reactor and capacitor stages, including the series detuning reactor, the magnification factor, transient-free and minimum-transient firing instants, valve voltage stress after turn-off, and binary-weighted banks;
- construct the V-I and susceptance characteristics of FC-TCR, TSC-TCR and transformer-coupled SVCs, locate their operating points on a network characteristic, and design the voltage regulator against the effective short-circuit ratio;
- derive the voltage-source model of the STATCOM, its V-I and P-Q capability, its converter waveforms for multi-pulse, three-level and PWM topologies, its dq current control and its DC capacitor requirement;
- compare the five devices quantitatively under voltage depression, harmonic, loss, response and cost criteria, and relate them to current research questions.

The five controllers treated in this chapter are classified in Table 6.1 by power stage, control action and the nature of their output.

Table 6.1. Classification of shunt FACTS controllers.

Device	Power stage	Control action	Output range	Nature of output
TCR	Reactor + antiparallel thyristors	Phase (firing-angle) control	Continuous, inductive only	Variable susceptance, harmonics generated
TSR	Reactor + antiparallel thyristors	On/off at voltage peak	Stepped, inductive	Fixed susceptance per stage
TSC	Capacitor + small reactor + thyristors	On/off at matched voltage	Stepped, capacitive	Fixed susceptance per stage
SVC	Combinations of TCR, TSR, TSC, FC, filters	Coordinated stage switching + TCR trim	Continuous, capacitive and inductive	Variable susceptance; $Q \propto V^2$ at limits
STATCOM	VSC or MMC with DC capacitor	Converter voltage magnitude and phase	Continuous, symmetric about zero	Controllable current source; $Q \propto V$ at limit

6.1 Reactive Power, Voltage and the Case for Shunt Compensation

6.1.1 The voltage-reactive current coupling at a bus

Seen from any bus, a transmission network can be reduced to a Thevenin source E_{th} behind an impedance $R_{th} + jX_{th}$. For transmission networks $X_{th} \gg R_{th}$, and if a shunt device draws a purely reactive current I (taken positive when inductive, that is, lagging the bus voltage by 90°), the bus voltage magnitude obeys

$$V = E_{th} - X_{th} I \quad (6.1)$$

where V is the RMS bus voltage, E_{th} the RMS open-circuit (Thevenin) voltage, X_{th} the Thevenin reactance and I the RMS reactive current drawn by the shunt device, all in per unit on a common base. Equation (6.1) is the *system load line*. It slopes downward with gradient $-X_{th}$: capacitive current ($I < 0$) raises the bus voltage and inductive current lowers it. Two consequences follow immediately. A given voltage correction ΔV requires a reactive current $\Delta V / X_{th}$, so weak buses (large X_{th} , low short-circuit ratio) need little current while stiff buses need a great deal. The same equation will later locate the steady operating point of every shunt compensator in this chapter, simply by intersecting it with the compensator's own V-I characteristic.

The more general approximate expression for voltage drop across a short line, $\Delta V \approx (RP + XQ)/V$, shows the same thing from the power side: where X/R is large, the reactive term dominates, and local reactive injection is the most effective lever on voltage magnitude.

6.1.2 Ideal midpoint compensation of a two-machine system

Consider the lossless two-machine system of Figure 6.1, with equal end voltages $V_s = V_r = V$, a total series reactance X , and a shunt compensator at the electrical midpoint.

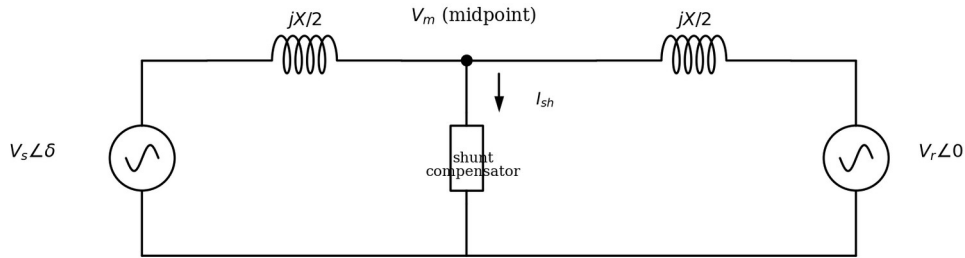


Figure 6.1. Two-machine system with a shunt compensator at the electrical midpoint.

Without compensation the transmitted power is

$$P_u = \frac{V^2}{X} \sin \delta \quad (6.2)$$

where P_u is the uncompensated active power, δ the angle between the end voltages and X the total line reactance. The uncompensated midpoint voltage follows from the phasor average of the two end voltages,

$$V_{m0} = V \cos \frac{\delta}{2} \quad (6.3)$$

where V_{m0} is the midpoint voltage magnitude with no shunt current. This sag of the midpoint as δ rises is the physical reason the uncompensated line saturates at $\delta = 90^\circ$.

Now let the compensator hold $V_m = V$. Because the compensator exchanges no active power, the same P flows through each half-line, and by symmetry each half carries angle $\delta/2$ across reactance $X/2$. Applying Equation (6.2) to one half,

$$P_c = \frac{V \cdot V}{X/2} \sin \frac{\delta}{2} = \frac{2V^2}{X} \sin \frac{\delta}{2} \quad (6.4)$$

where P_c is the transmitted power with ideal midpoint regulation. The maximum, $2V^2/X$, occurs at $\delta = 180^\circ$: in this idealized symmetric case the steady-state limit doubles. The phasor diagram in Figure 6.2 shows the geometry.

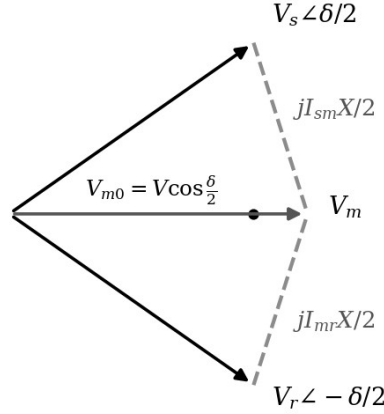


Figure 6.2. Phasor diagram of ideal midpoint compensation: the shunt device lifts $V_{m0} = V \cos(\delta/2)$ to $V_m = V$.

The current entering the midpoint from the sending side is $I_{sm} = (V e^{j\delta/2} - V)/(jX/2)$, and the current leaving towards the receiving end is $I_{mr} = (V - V e^{-j\delta/2})/(jX/2)$. Their difference must be supplied by the compensator. Carrying out the subtraction,

$$I_{sh} = I_{mr} - I_{sm} = -j \frac{4V}{X} \left(1 - \cos \frac{\delta}{2} \right) \quad (6.5)$$

which is purely reactive and, with the midpoint voltage as reference, lags it by 90° in the direction drawn from the device into the network, so the device behaves as a capacitor. Its magnitude and the corresponding reactive power are

$$I_{sh} = \frac{4V}{X} \left(1 - \cos \frac{\delta}{2} \right) = \frac{8V}{X} \sin^2 \frac{\delta}{4}, \quad Q_{sh} = \frac{4V^2}{X} \left(1 - \cos \frac{\delta}{2} \right) \quad (6.6)$$

where I_{sh} is the RMS shunt current and Q_{sh} the capacitive reactive power the compensator must inject. At $\delta = 180^\circ$ these reach $4V/X$ and $4V^2/X$, finite but large: four times the maximum power of the uncompensated line. Real compensators are therefore rated well below this ideal point, and the shape of their limiting characteristic (constant susceptance for an SVC, constant current for a STATCOM) decides how the power-angle curve bends once the limit is reached. Figure 6.3 anticipates that comparison, which is derived in Sections 6.5.8 and 6.6.9.

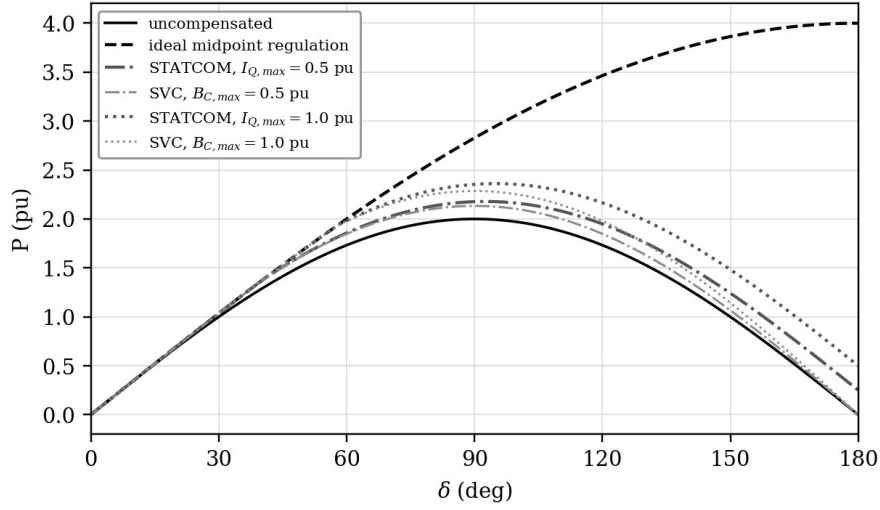


Figure 6.3. Power-angle curves for $X = 0.5$ pu: uncompensated line, ideal midpoint regulation, and midpoint SVC and STATCOM with finite ratings.

6.1.3 Voltage stability, transient stability and damping

At the receiving end of a radial corridor, the nose of the P-V curve is reached when the network can no longer supply the reactive losses that a rising load imposes. A shunt compensator moves the nose outward by supplying that reactive power locally, but only while it stays within its limits. Once an SVC reaches its capacitive limit it becomes a fixed capacitor whose output falls as V^2 , and it can then hasten collapse rather than arrest it. The current-limited behaviour of a STATCOM ($Q \propto V$) is more benign. This single distinction recurs throughout the chapter and underpins the comparison in Section 6.6.11.

For transient stability, the equal-area criterion explains the benefit. During the first swing after a fault, a compensator driven to its capacitive limit raises the post-fault power-angle curve, enlarging the decelerating area. For small-signal oscillations, modulation of the compensator's reactive output in phase with rotor speed deviation (through bus voltage modulation) adds damping torque. Both functions are treated as supplementary controls in Sections 6.5.7 and 6.6.10 and in depth in Chapter 11.

6.1.4 Evolution of shunt compensation: a short literature survey

Mechanically switched capacitors and reactors, and rotating synchronous condensers, were the only options until the late 1960s. Saturated-reactor compensators, built for arc-furnace flicker reduction, were the first static devices, but their slow, fixed slope and high losses limited them. The line-commutated thyristor changed the picture in the early 1970s. Gyugyi's systematic treatment of thyristor var generators [1] and the collection edited by Miller [2] established the TCR-FC and TSC-TCR configurations and the analytical tools still used today, including the susceptance-versus-firing-angle law, the harmonic formula, and the transient-free switching conditions for capacitors. Mathur's edited volume [3] consolidated SVC application and control practice.

The second wave began with forced-commutated converters. A 20 MVA force-commutated var generator was field tested in Japan from 1980 [4], an 80 MVA GTO-based installation entered commercial service at Inuyama in 1991 [5], and the ± 100 MVA Sullivan STATCOM of the Tennessee Valley Authority, built under EPRI sponsorship and commissioned in 1995, became the reference case for transmission STATCOMs in North America [6]. Hingorani and Gyugyi [7] gave these devices the vocabulary used in this book (STATCOM, SSG, SVG) and introduced the V-I and loss comparisons reproduced in Section 6.6. Mathur and Varma [8], Song and Johns [9], Padiyar [10], Acha *et al.* [11], Zhang *et al.* [12] and Sen and Sen [13] extend the modelling, control and power-flow representation. Since about 2010, modular multilevel converters (Chapter 5) have displaced multi-pulse transformers in new STATCOMs, and research has moved towards energy-storage-equipped and grid-forming STATCOMs (Section 6.7).

6.1.5 Thyristor valves and firing-angle conventions

All thyristor-based shunt devices use the *bidirectional valve*: two antiparallel thyristor strings, each string consisting of many series-connected devices for voltage blocking, with parallel strings where current demands it. Each device carries an RC snubber and a grading resistor for static and dynamic voltage sharing; modern valves use light-triggered thyristors with integrated break-over-diode protection and deionized-water cooling. Valve design is revisited in Example 6.21.

A source of genuine confusion in textbooks, lecture notes and examination papers is the reference for the firing angle. Two conventions coexist, illustrated in Figure 6.4. In this chapter α is always measured from the zero crossing of the voltage across the branch, so that full conduction of a reactor corresponds to $\alpha = 90^\circ$ and blocking to $\alpha = 180^\circ$. Several classic texts and many course notes instead measure a delay β from the voltage peak, with $0 \leq \beta \leq 90^\circ$. The two are related by

$$\beta = \alpha - 90^\circ \quad (6.7)$$

where α is the zero-crossing-referenced firing angle and β the peak-referenced delay angle. Every formula must be used with the reference for which it was derived. Table 6.2 lists the equivalent forms of the main TCR expressions, and Example 6.19 shows the numerical consequences of mixing them.

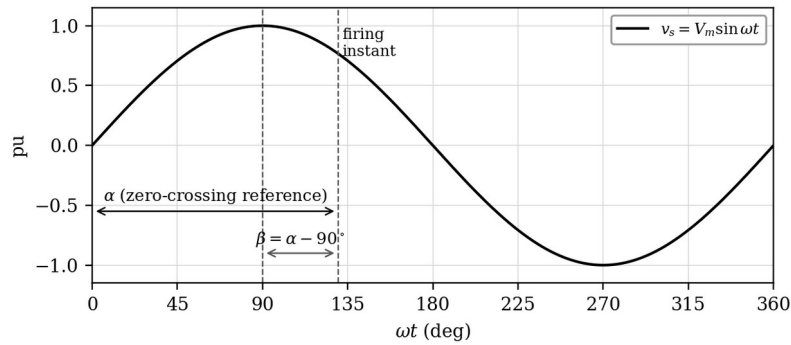


Figure 6.4. Zero-crossing (α) and peak-referenced (β) firing-angle conventions.

Table 6.2. Equivalent TCR expressions in the two firing-angle conventions.

Quantity	Zero-crossing form ($90^\circ \leq \alpha \leq 180^\circ$)	Peak-referenced form ($0 \leq \beta \leq 90^\circ$)
Conduction angle	$\sigma = 2(\pi - \alpha)$	$\sigma = \pi - 2\beta$
Branch current	$\frac{V_m}{\omega L}(\cos \alpha - \cos \omega t)$	$\frac{V_m}{\omega L}(\sin \omega t' - \sin \beta)$, $v = V_m \cos \omega t'$
B_{TCR}/B_{max}	$\frac{2(\pi - \alpha) + \sin 2\alpha}{\pi}$	$1 - \frac{2\beta}{\pi} - \frac{\sin 2\beta}{\pi}$
Full conduction / blocked	$90^\circ / 180^\circ$	$0^\circ / 90^\circ$
Third harmonic maximum	$\alpha = 120^\circ$	$\beta = 30^\circ$

6.2 The Thyristor-Controlled Reactor (TCR)

6.2.1 Operating principle

A single-phase TCR (Figure 6.5) is a linear, usually air-cored, reactor of inductance L in series with a bidirectional thyristor valve. Thyristor T_1 is gated once per positive half-cycle at angle α and T_2 half a cycle

later. Each thyristor stops conducting only when its current falls naturally to zero, so the firing angle sets the start of each current pulse while the circuit itself sets its end. The reactor is normally split into two halves on either side of the valve: if one half fails short, the other still limits the fault current, and the valve is never exposed to the full bus voltage across a single shorted winding.

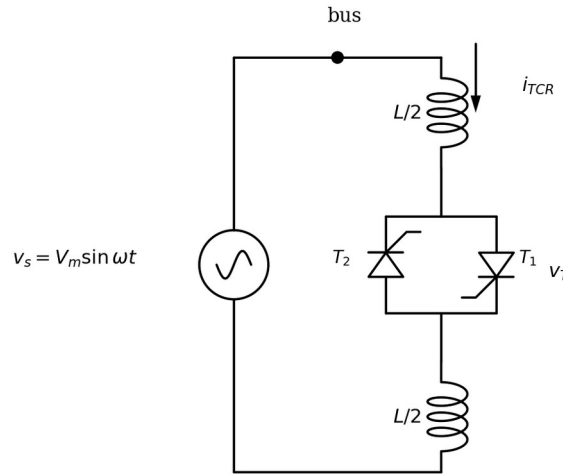


Figure 6.5. Single-phase TCR with split reactor and antiparallel thyristor valve.

Figure 6.6 shows the branch current and valve voltage at four firing angles. At $\alpha = 90^\circ$ the pulses from T_1 and T_2 join into a continuous sinusoid lagging the voltage by 90° , exactly the current of the bare reactor. As α increases, conduction starts later and, by symmetry of the inductive circuit, ends equally early, so each pulse is centred on the voltage zero crossing at $\omega t = \pi$ (and 2π for the negative pulse) and shrinks towards zero as $\alpha \rightarrow 180^\circ$. Firing earlier than 90° is not used: conduction would then extend past the next firing instant of the opposite thyristor, the currents would become asymmetric, and a DC component would appear.

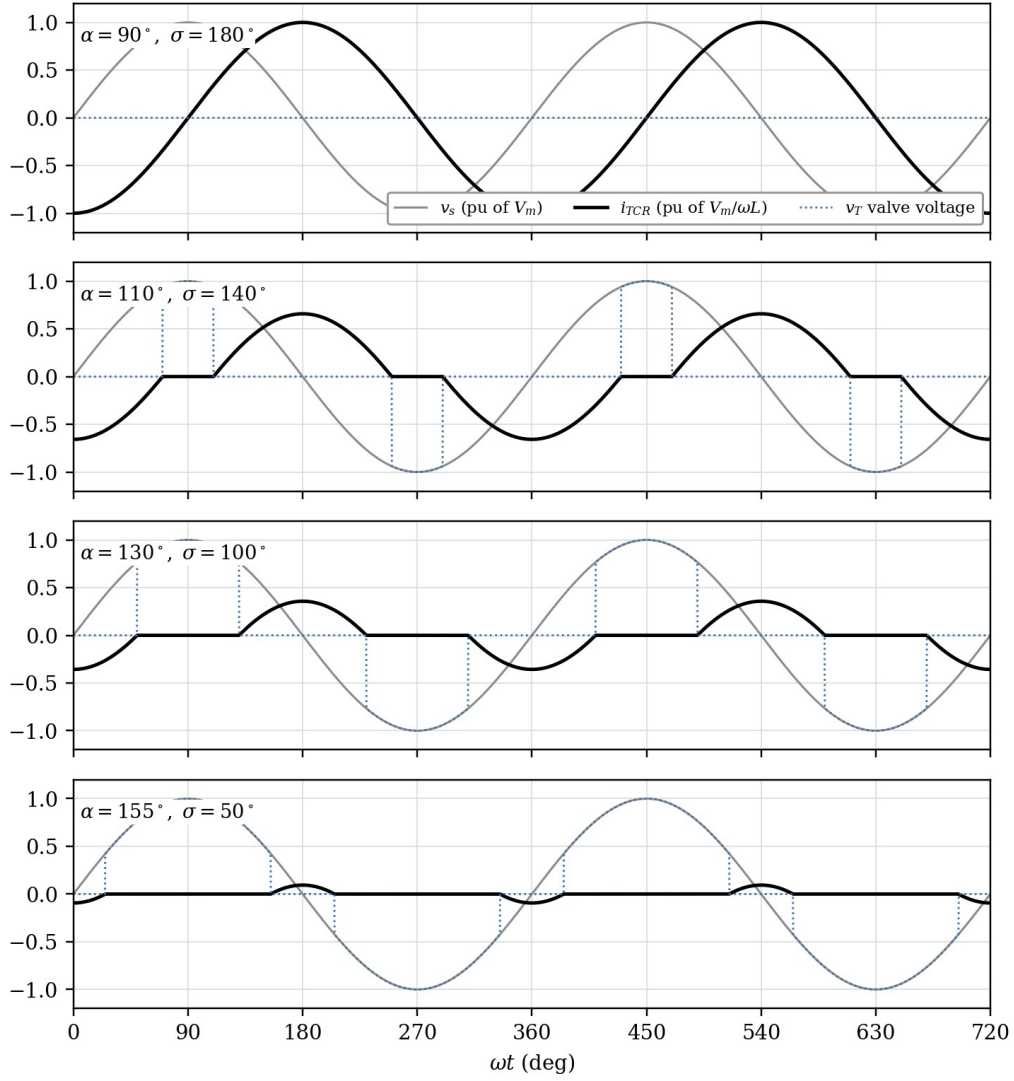


Figure 6.6. TCR source voltage, branch current and valve voltage at $\alpha = 90^\circ$, 110° , 130° and 155° .

The valve voltage v_T equals the source voltage whenever both thyristors are off and is zero while either conducts. At the instant of firing the valve therefore blocks $V_m \sin \alpha$, which is the full peak value when α approaches 90° ; the valve must be rated for the full peak branch voltage plus switching overvoltage margins.

6.2.2 Instantaneous current

Let the branch voltage be $v_s = V_m \sin \omega t$, where V_m is its peak value and $\omega = 2\pi f$ the angular frequency. While T_1 conducts, the circuit is a pure inductance:

$$L \frac{di}{dt} = V_m \sin \omega t \quad (6.8)$$

where L is the branch inductance and i the instantaneous branch current. Integrating, $i = -(V_m/\omega L) \cos \omega t + K$. In periodic steady state the pulse starts from zero at $\omega t = \alpha$, which fixes K :

$$i(t) = \frac{V_m}{\omega L} (\cos \alpha - \cos \omega t), \alpha \leq \omega t \leq 2\pi - \alpha \quad (6.9)$$

Setting Equation (6.9) to zero gives the second root $\omega t = 2\pi - \alpha$, the natural extinction angle. The conduction angle of each thyristor is therefore

$$\sigma = 2(\pi - \alpha) \quad (6.10)$$

where σ is the conduction angle in radians. The peak instantaneous current, at $\omega t = \pi$, is $(V_m/\omega L)(1 + \cos \alpha)$, which falls from $V_m/\omega L$ at full conduction to zero at blocking. The negative pulse from T_2 is the mirror image half a cycle later, so the waveform has half-wave odd symmetry, $i(\omega t + \pi) = -i(\omega t)$, and contains only odd harmonics.

6.2.3 Fundamental current and equivalent susceptance

Because the positive pulse is symmetric about $\omega t = \pi$ and the waveform has half-wave odd symmetry, the fundamental contains only a cosine term. Its coefficient is twice the integral over one pulse:

$$a_1 = \frac{2}{\pi} \int_{\alpha}^{2\pi-\alpha} \frac{V_m}{\omega L} (\cos \alpha - \cos \theta) \cos \theta d\theta \quad (6.11)$$

where $\theta = \omega t$ and a_1 is the peak cosine coefficient of the fundamental. The two integrals are $\int_{\alpha}^{2\pi-\alpha} \cos \theta d\theta = -2 \sin \alpha$ and $\int_{\alpha}^{2\pi-\alpha} \cos^2 \theta d\theta = (\pi - \alpha) - \frac{1}{2} \sin 2\alpha$. Substituting and collecting terms,

$$a_1 = -\frac{V_m}{\omega L} \cdot \frac{2(\pi - \alpha) + \sin 2\alpha}{\pi} \quad (6.12)$$

The negative cosine coefficient means the fundamental lags $v_s = V_m \sin \omega t$ by 90° : the TCR is purely inductive at fundamental frequency. Writing the peak fundamental as $I_{1m} = V_m B_{TCR}(\alpha)$ defines the equivalent susceptance

$$B_{TCR}(\alpha) = B_{max} \frac{2(\pi - \alpha) + \sin 2\alpha}{\pi} = B_{max} \frac{\sigma - \sin \sigma}{\pi}, B_{max} = \frac{1}{\omega L} \quad (6.13)$$

where B_{TCR} is the fundamental-frequency inductive susceptance magnitude and B_{max} its value at full conduction. The second form uses Equation (6.10) and $\sin 2\alpha = -\sin \sigma$. The RMS fundamental current and the reactive power absorbed by one branch of RMS voltage V are

$$I_1 = V B_{TCR}(\alpha), Q_{TCR} = V^2 B_{TCR}(\alpha) \quad (6.14)$$

where I_1 is the RMS fundamental current and Q_{TCR} the absorbed reactive power (a positive magnitude). For a delta-connected three-phase TCR, V is the line-to-line voltage and the three-phase absorption is $3V^2 B_{TCR}$; for a star connection V is the phase voltage.

Figure 6.7 plots B_{TCR}/B_{max} . The curve is steep near $\alpha = 90^\circ$ and flat near 180° . Differentiating Equation (6.13),

$$\frac{d}{d\alpha} \left(\frac{B_{TCR}}{B_{max}} \right) = -\frac{4 \sin^2 \alpha}{\pi} \quad (6.15)$$

so the incremental gain of the actuator varies by a factor of almost four (3.97) between $\alpha = 95^\circ$ and $\alpha = 150^\circ$, and by nearly fifteen between 95° and 165° . Practical controllers never use α directly as the output of a linear regulator. They compute a susceptance reference and pass it through a *linearizer*, an inverse of Equation (6.13) implemented as a look-up table or a few Newton iterations (Example 6.19), so that the loop gain stays constant over the operating range.

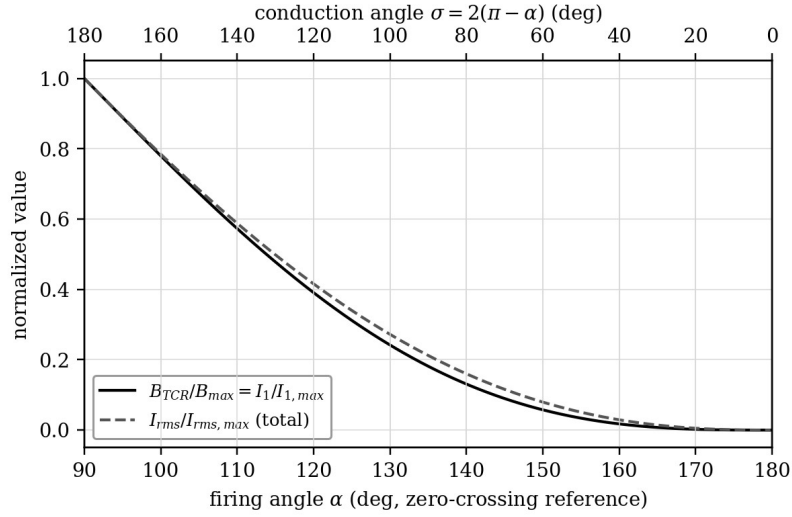


Figure 6.7. Normalized fundamental susceptance (equivalently fundamental current) and total RMS current of the TCR versus firing angle, with the conduction angle on the upper axis.

6.2.4 Total RMS current and total harmonic distortion

Thermal design of the reactor and the valve is governed by total RMS current, not by its fundamental. Squaring Equation (6.9), integrating over both pulses and dividing by 2π :

$$I_{rms}^2 = \frac{1}{\pi} \left(\frac{V_m}{\omega L} \right)^2 \int_{\alpha}^{2\pi-\alpha} (\cos \alpha - \cos \theta)^2 d\theta \quad (6.16)$$

The integral evaluates to $2(\pi - \alpha)\cos^2 \alpha + 4\sin \alpha \cos \alpha + (\pi - \alpha) - \frac{1}{2}\sin 2\alpha$, which simplifies to

$$I_{rms} = \frac{V_m}{\omega L} \sqrt{\frac{(\pi - \alpha)(1 + 2\cos^2 \alpha) + \frac{3}{2}\sin 2\alpha}{\pi}} \quad (6.17)$$

where I_{rms} is the total RMS branch current. At $\alpha = 90^\circ$ this reduces to $V_m/(\sqrt{2}\omega L)$, as it must. In the peak-referenced convention the same expression reads $(V_m/\omega L) \sqrt{\left[(\pi - 2\beta) \left(\frac{1}{2} + \sin^2 \beta \right) - \frac{3}{2}\sin 2\beta \right] / \pi}$, which is the form found in several lecture notes. The current THD follows from

$$THD_I = \sqrt{\left(\frac{I_{rms}}{I_1} \right)^2 - 1} \quad (6.18)$$

where $I_1 = a_1/\sqrt{2}$ in magnitude. Figure 6.8 shows that branch-current THD rises *monotonically* with α : about 11% at 100° , 36.3% at 120° , 95% at 150° , and it grows without bound as the pulses narrow towards zero width. There is therefore no finite “maximum THD” of a TCR branch current. What does have a maximum is each individual harmonic, measured on a fixed base, which is the quantity that matters for filter design. The delta line current, from which triplen harmonics are absent, has a much lower THD over the useful range.

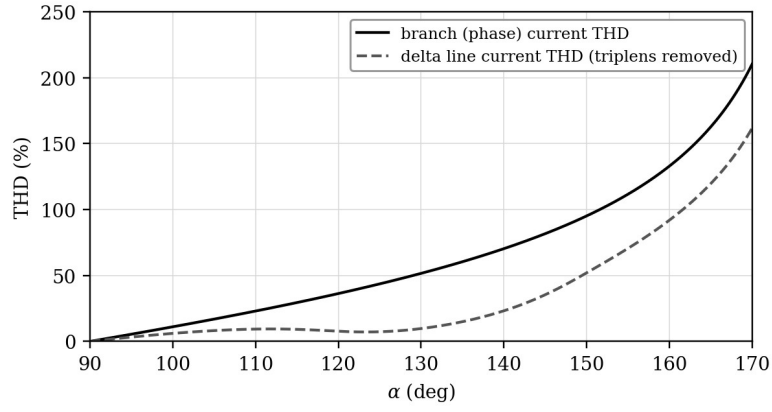


Figure 6.8. THD of TCR branch current and of delta line current versus firing angle.

6.2.5 Harmonic spectrum

For odd $n \geq 3$, the cosine coefficient is obtained exactly as for the fundamental, now with $\cos n\theta$ inside the integral. Using product-to-sum identities and the half-wave symmetry,

$$a_n = \frac{4V_m}{\pi\omega L} \left[\frac{\sin(n+1)\alpha}{2(n+1)} + \frac{\sin(n-1)\alpha}{2(n-1)} - \frac{\cos\alpha \sin n\alpha}{n} \right] \quad (6.19)$$

where a_n is the peak value of the n th harmonic. Combining the three terms over the common denominator $n(n^2 - 1)$ gives the compact form

$$I_n = \frac{4}{\pi} \frac{V_m}{\omega L} \left| \frac{\cos\alpha \sin n\alpha - n \sin\alpha \cos n\alpha}{n(n^2 - 1)} \right|, n=3, 5, 7, \dots \quad (6.20)$$

where I_n is the peak n th harmonic branch current; dividing by $\sqrt{2}$ gives its RMS value. The same result in the peak-referenced convention is $(4V_m/\pi\omega L) |\sin\beta \cos n\beta - n \cos\beta \sin n\beta| / [n(n^2 - 1)]$.

Where is each harmonic largest? Differentiating the numerator of Equation (6.20) with respect to α gives $(n^2 - 1) \sin\alpha \sin n\alpha$, so the extrema lie at $\sin n\alpha = 0$, that is $\alpha = k\pi/n$ within $90^\circ < \alpha < 180^\circ$. For the third harmonic this is $\alpha = 120^\circ$ ($\beta = 30^\circ$), where I_3 reaches 13.78% of the full-conduction fundamental; for the fifth it is 108° and 144° . Figure 6.9 plots the low-order harmonics, and Table 6.3 lists their maxima, which are the numbers used to rate filters.

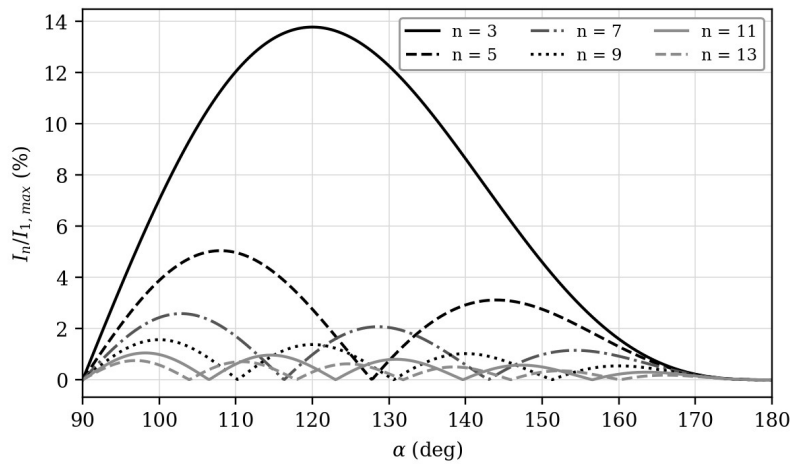


Figure 6.9. Odd harmonic branch currents of a TCR as a percentage of the full-conduction fundamental.

Table 6.3. Maximum amplitude of each TCR harmonic, as a percentage of the full-conduction fundamental, and the firing angle at which it occurs (balanced operation). The values apply to branch and line currents alike, except that triplens are absent from the line current of a delta connection.

n	3	5	7	9	11	13	15	17	19	21	23	25
$I_{n,max}$ (%)	13.78	5.05	2.59	1.57	1.05	0.75	0.57	0.44	0.35	0.29	0.24	0.20
α at max	120.0°	108.0°	102.9°	100.0°	98.2°	96.9°	96.0°	95.3°	94.7°	94.3°	93.9°	93.6°

Example 6.1: Delta-connected TCR: rating, firing angle and harmonics.

A balanced, delta-connected, six-pulse 50 kvar TCR operates from a 415 V, 50 Hz supply. Find (a) the branch reactance and inductance; (b) the branch and line currents at full conduction; (c) the firing angle for 25 kvar; (d) the total RMS branch current and its THD at that angle; (e) the 3rd, 5th and 7th harmonic branch currents; (f) the line-current harmonics; (g) which harmonics reach the supply.

Solution. (a) Each branch sees the line voltage $V = 415$ V and absorbs one third of the rating, so $Q_{3\phi} = 3V^2/X_L$; $X_L = 3(415)^2/50\,000 = 10.3335\ \Omega$ and $L = X_L/\omega = 32.893$ mH.

(b) $I_b = V/X_L = 415/10.3335 = 40.161$ A; $I_\ell = \sqrt{3}I_b = 69.560$ A.

(c) Half rating requires $B_{TCR}/B_{max} = 0.5$, so from Equation (6.13), $[2(\pi - \alpha) + \sin 2\alpha]/\pi = 0.5$. Solving numerically, $\alpha = 113.83^\circ$ ($\beta = 23.83^\circ$ in the peak-referenced convention).

(d) The base current is $V_m/\omega L = \sqrt{2} \times 415/10.3335 = 56.796$ A. Equation (6.17) at 113.83° gives $I_{rms} = 0.36706 \times 56.796 = 20.847$ A. The fundamental is $I_1 = 0.5 \times 56.796/\sqrt{2} = 20.080$ A, so Equation (6.18) gives $THD = \sqrt{(20.847/20.080)^2 - 1} = 27.9\%$.

(e) From Equation (6.20) with $V_m/\omega L = 56.796$ A (peak), dividing by $\sqrt{2}$: $I_3 = 5.271$ A, $I_5 = 1.786$ A, $I_7 = 0.283$ A RMS (13.12%, 4.45% and 0.70% of the full-conduction fundamental of 40.161 A).

(f) The 5th and 7th appear in the line multiplied by $\sqrt{3}$: $I_{5\ell} = 3.094$ A and $I_{7\ell} = 0.489$ A. The line fundamental is $\sqrt{3} \times 20.080 = 34.780$ A.

(g) The third harmonic (5.27 A in each branch) is zero-sequence and circulates in the delta; only $n = 6k \pm 1$ reach the supply under balanced conditions.

Note. This example deliberately combines rating, firing-angle control and harmonic interpretation. The purpose is to connect the branch-level TCR equations with what the supply actually sees through the delta connection. Always distinguish branch harmonics from line-current harmonics.

6.2.6 The three-phase six-pulse TCR

Three single-phase TCRs connected in delta (Figure 6.10) form the standard six-pulse TCR. Each branch is fired with the same α referred to its own line-to-line voltage, so the three firing patterns are displaced by 120° .

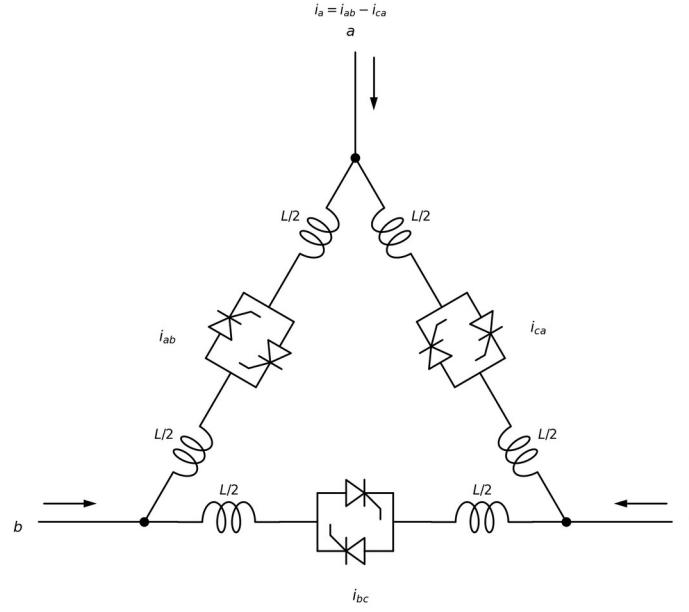
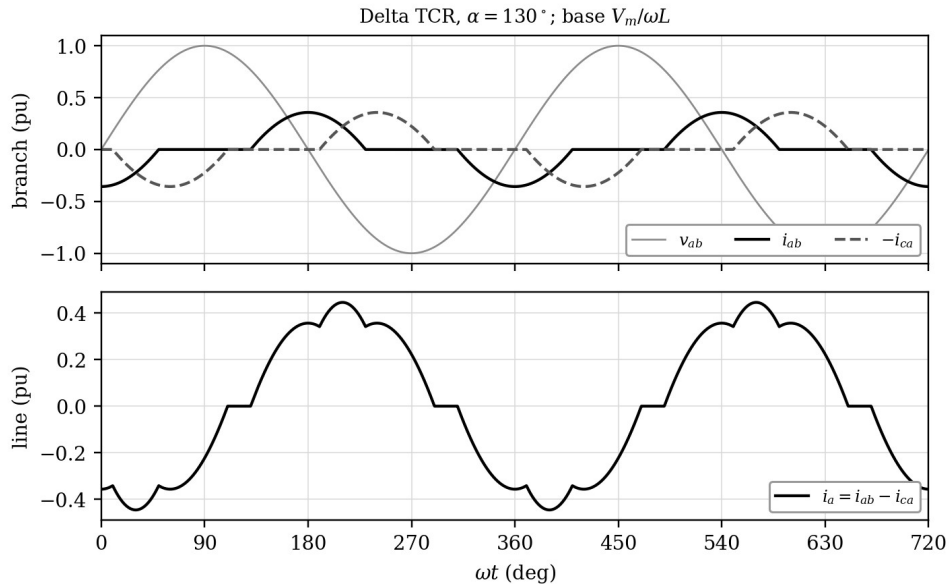


Figure 6.10. Delta-connected six-pulse TCR with split reactors.

Line currents follow from Kirchhoff's current law at each vertex, for example

$$i_a = i_{ab} - i_{ca} \quad (6.21)$$

where i_a is the line current and i_{ab} , i_{ca} the branch currents with arrows $a \rightarrow b$ and $c \rightarrow a$. Applying Equation (6.21) harmonic by harmonic is revealing. The n th harmonic of i_{ca} is displaced from that of i_{ab} by $n \times 120^\circ$. For triplen orders ($n=3, 9, 15, \dots$) this displacement is a multiple of 360° , so the triplens in the three branches are in phase: they form a zero-sequence set that circulates around the delta and cancels in every line current. For $n=6k-1$ (5, 11, 17, ...) the set has negative-sequence rotation; for $n=6k+1$ (7, 13, 19, ...) positive-sequence. Both appear in the line current multiplied by $\sqrt{3}$, exactly as the fundamental does. The line current therefore carries the characteristic harmonics $n=6k \pm 1$ with the same percentages as in Table 6.3, and Figure 6.11 shows its typical waveform.


 Figure 6.11. Branch and line currents of a balanced delta TCR at $\alpha = 130^\circ$ computed by KCL.

Ideal cancellation assumes identical reactors, balanced sinusoidal voltages and equal firing. In practice, small asymmetries let *non-characteristic* harmonics, triplens included, reach the network, and unequal firing of T_1

and T_2 produces a DC component that can drive the coupling transformer into saturation. Typical reactor quality factors lie between 40 and 100, so a small in-phase current representing losses is also present.

6.2.7 The twelve-pulse TCR

Two six-pulse TCRs of equal rating, supplied from two transformer secondaries whose voltages are displaced by 30° (for example a Y-Y and a Y- Δ winding, Figure 6.12), cancel the 5th, 7th, 17th, 19th, ... harmonics on the primary side. The reason is that a Y- Δ transformer shifts positive-sequence quantities by $+30^\circ$ and negative-sequence quantities by -30° . The group supplied through the Y- Δ winding sees voltages displaced by -30° , so its n th harmonic current is displaced by $-30n^\circ$ relative to the Y-Y group; transferring it back to the primary adds $+30^\circ$ for positive-sequence orders ($n=6k+1$) and -30° for negative-sequence orders ($n=6k-1$). The net displacement is $-30^\circ(n-1)$ or $-30^\circ(n+1)$ respectively, which equals 180° (cancellation) for $n=5, 7, 17, 19, \dots$ and 0° or 360° (addition) for $n=11, 13, 23, 25, \dots$. The primary current therefore contains only $n=12k \pm 1$.

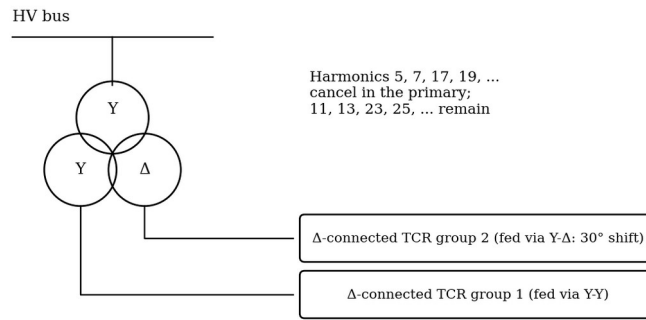


Figure 6.12. Twelve-pulse TCR arrangement with Y-Y and Y- Δ secondaries.

Figure 6.13 and Figure 6.14 confirm this by direct computation. The waveforms were obtained by applying Equation (6.9) to both groups and referring the second group through the Y- Δ relation $i_A = (i_{a2} - i_{b2})/\sqrt{3}$ for unity line-voltage ratio. The twelve-pulse arrangement leaves only high-pass filtering duty, at the price of a more complex transformer and two valve groups. It was used where filter cost or network resonance made six-pulse operation unattractive, much as in HVDC converter stations [19].

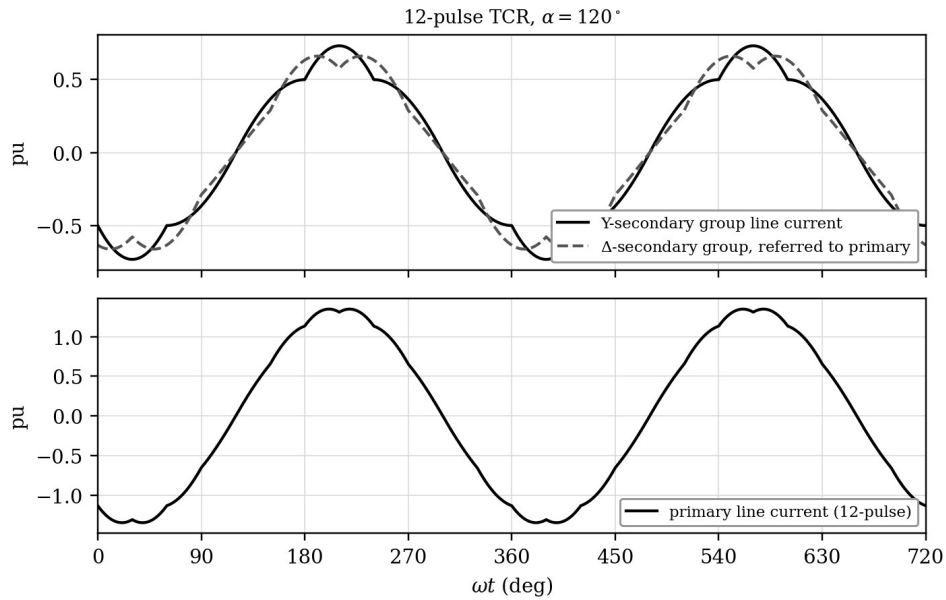


Figure 6.13. Line currents of the two six-pulse groups referred to the primary and their twelve-pulse sum at $\alpha = 120^\circ$.

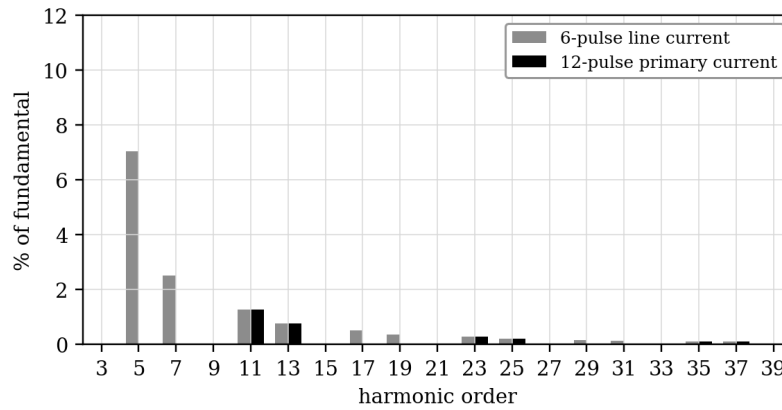


Figure 6.14. Harmonic content of six-pulse line current and twelve-pulse primary current at $\alpha = 120^\circ$.

Example 6.2: Twelve-pulse TCR.

A twelve-pulse TCR is fed from a 415 V primary through a three-winding transformer with Y-Y and Y- Δ secondaries, each 415 V line-to-line. Each secondary supplies a delta TCR of 15 mH per branch, and $\alpha = 120^\circ$. Find (a) the total rating; (b) the branch fundamental and RMS currents; (c) the secondary line fundamental; (d) the primary fundamental; (e) the 5th and 7th harmonics in each secondary line; (f) the 5th and 7th in the primary; (g) the 11th and 13th in the primary.

Solution. (a) $X_L = 4.7124 \Omega$; each group $3 V^2 / X_L = 109.64$ kvar; total 219.28 kvar.

(b) Base 124.54 A. $I_{1b} = 0.391 \times 124.54 / \sqrt{2} = 34.43$ A; $I_{rms,b} = 36.63$ A.

(c) $\sqrt{3} \times 34.43 = 59.64$ A in each secondary.

(d) With 1:1 line-voltage ratios, the two fundamentals add in phase on the primary: $2 \times 59.64 = 119.28$ A. The absorbed power is 85.74 kvar.

(e) $I_{5\ell} = 4.205$ A and $I_{7\ell} = 1.502$ A in each secondary.

(f) The Y- Δ group's 5th and 7th are shifted by 180° relative to the Y-Y group's when referred to the primary (Section 6.2.7), so both cancel: zero.

(g) The 11th and 13th add: $2 \times 0.7645 = 1.529$ A and $2 \times 0.4621 = 0.924$ A.

Note. Twelve-pulse cancellation is a connection-dependent harmonic result, not a statement that the individual six-pulse groups are harmonic-free. Transformer phase shift and balanced operation are essential to the cancellation mechanism.

6.2.8 The segmented (sequentially controlled) TCR

A TCR of total rating Q can be split into m parallel branches of rating Q/m . Only one branch is phase controlled at a time; the others are either fully conducting (acting as TSRs at $\alpha = 90^\circ$) or blocked. Because the controlled branch carries at most $1/m$ of the rated current, every harmonic amplitude, expressed on the total-rating base, is reduced by the factor m . Figure 6.15 shows a four-section TCR delivering 60% of its rating with two sections fully on and the third at $\alpha = 119.5^\circ$. The price is more valves and reactors, which is why segmentation is reserved for installations with tight harmonic limits. Example 6.20 quantifies the benefit.

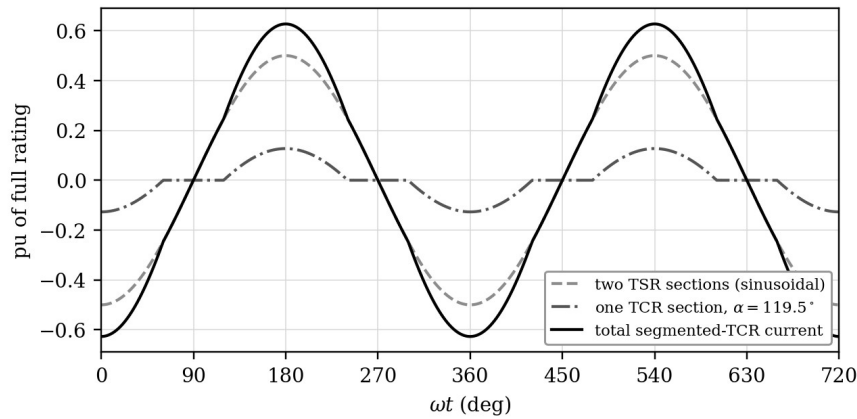


Figure 6.15. Segmented TCR with $m = 4$ supplying 60% of rating: two TSR sections plus one phase-controlled section.

6.2.9 Practical design considerations

Valve construction. Each valve consists of parallel strings of series-connected thyristors. The number in series follows from the peak repetitive voltage, including the switching overvoltage factor and a redundancy margin of one or two devices; the number of parallel strings follows from the RMS and surge currents. Each thyristor in a TCR conducts a half-cycle pulse, so at full conduction its RMS current is $I_{pk}/2$ and its average current I_{pk}/π , where I_{pk} is the peak branch current.

Reactor and losses. Air-cored reactors are linear and cannot saturate, which keeps the susceptance law exact, but they offer little overload capability; a TCR must be protected against sustained overvoltage by limiting the conduction angle. Losses are dominated by $I_{rms}^2 R$ in the reactor and the on-state loss of the valve.

Dead time. A new firing angle can only take effect at the next firing instant of either thyristor, so the TCR responds with a worst-case delay of one half-cycle (10 ms at 50 Hz) and a mean delay of roughly a quarter cycle. A decrease in conduction can be implemented at the next firing instant; an increase cannot interrupt a pulse already under way.

Filtering. Six-pulse TCRs are normally accompanied by single-tuned filters at the 5th and 7th harmonics and often a high-pass branch for the 11th and 13th. These filters are capacitive at fundamental frequency, so they double as the fixed capacitive element of the SVC (Section 6.5).

6.2.10 V-I characteristic and operating area

At fundamental frequency a TCR is a susceptance adjustable from zero to B_{max} , so its V-I characteristic is a fan of straight lines through the origin. The operating area (Figure 6.16) is bounded by the $\alpha = 90^\circ$ line, by the maximum voltage the valve and reactor can withstand, and by the maximum current. A TSR, by contrast, offers only the discrete lines corresponding to the number of stages in conduction.

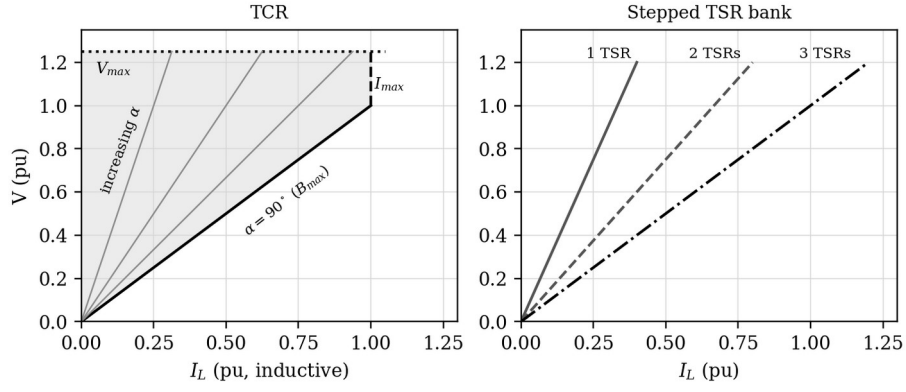


Figure 6.16. Operating area of a TCR (left) and the stepped characteristic of a TSR bank (right).

6.3 The Thyristor-Switched Reactor (TSR)

6.3.1 Principle

If the firing angle of a TCR is fixed at 90° and the valve is either gated continuously or not at all, the branch becomes a thyristor-switched reactor (Figure 6.17a). It offers a fixed inductive admittance whose reactive current is proportional to the applied voltage. Several TSRs in parallel give a susceptance adjustable in steps. Compared with a TCR of the same total rating, the TSR generates no phase-control harmonics, has lower losses and needs no linearizer, but it cannot provide continuous control. It is chosen where the inductive duty is coarse, for example to absorb line charging on light load or to suppress temporary overvoltage after load rejection, often in combination with a smaller TCR for the fine adjustment.

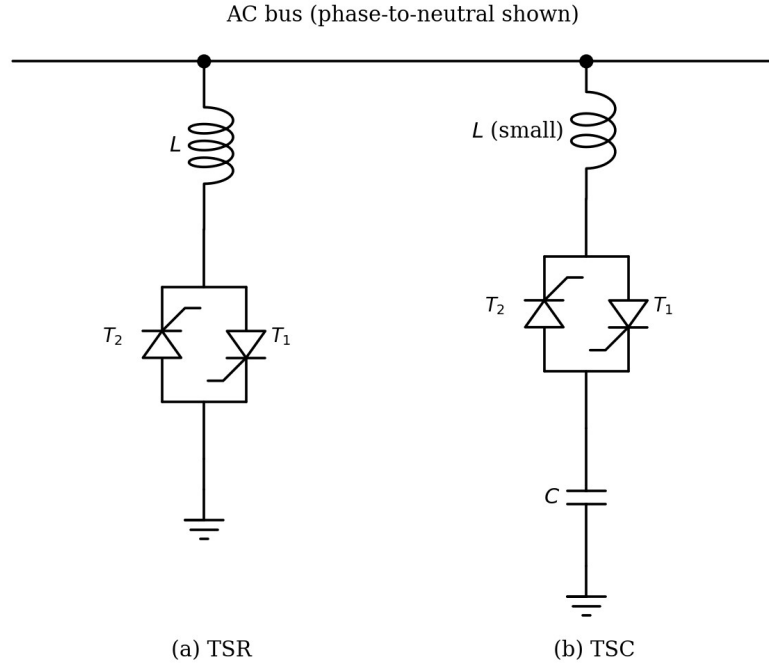


Figure 6.17. Single-phase (a) TSR and (b) TSC branches.

6.3.2 Transient-free connection

Suppose the valve of an initially de-energized reactor is closed at an arbitrary instant $\theta_c = \omega t_c$. Integrating Equation (6.8) from $i(\theta_c) = 0$ while the valve remains gated gives

$$i(\theta) = \frac{V_m}{\omega L} (\cos \theta_c - \cos \theta), \theta \geq \theta_c \quad (6.22)$$

where θ_c is the closing angle measured from the positive-going voltage zero crossing. The second term is the steady sinusoidal current; the first is a DC offset of magnitude $(V_m/\omega L)\cos\theta_c$ that, in a lossless circuit, persists indefinitely and in practice decays with the reactor time constant L/R . The offset vanishes only for $\cos\theta_c=0$, that is, closure at a voltage peak. Closure at a voltage zero gives a fully offset current whose peak is twice the steady value, as Figure 6.18 shows. The TSR is therefore switched in at $\theta_c=90^\circ$ (or 270° for the opposite valve) and switched out by removing gate pulses, after which the conducting thyristor commutates naturally at the next current zero.

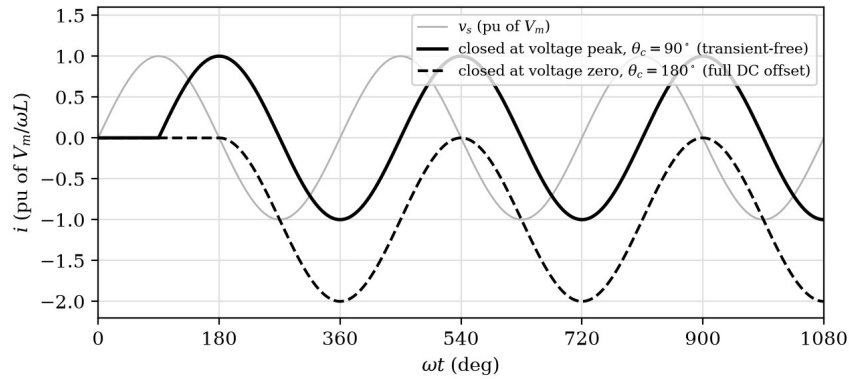


Figure 6.18. TSR current when closed at a voltage peak (transient-free) and at a voltage zero (fully offset).

Example 6.3: Thyristor-switched reactor.

A single-phase 25 kvar TSR is switched at 230 V, 50 Hz. Find (a) reactance and inductance; (b) the transient-free closing instant; (c) the peak and RMS branch currents; (d) the RMS and average current per thyristor; (e) the peak current if closed at a voltage zero; (f) the peak blocking voltage.

Solution. (a) $X = 230^2/25\,000 = 2.116\,\Omega$; $L = 6.735\text{ mH}$.

(b) At a voltage peak, $\theta_c = 90^\circ$ (Equation (6.22) with $\cos\theta_c = 0$).

(c) $I_{pk} = 325.27/2.116 = 153.72\text{ A}$; $I = 108.70\text{ A}$.

(d) $I_{T,rms} = 76.86\text{ A}$; $I_{T,av} = 48.93\text{ A}$.

(e) At $\theta_c = 0^\circ$ (or 180°) the offset equals the AC peak, so $i_{pk} = 2 \times 153.72 = 307.4\text{ A}$, decaying with L/R .

(f) 325.3 V plus margin.

Note. A TSR is a switched reactor, not a continuously controlled reactor. The numerical result should therefore be interpreted as one discrete susceptance state. Closing near the appropriate voltage crest avoids the large DC offset that would result from an arbitrary switching instant.

6.3.3 Stepped banks

With N branches of binary-weighted susceptance $B, 2B, 4B, \dots, 2^{N-1}B$, the bank provides 2^N states including zero, in uniform steps of B . Three branches of 50, 100 and 200 kvar, for instance, give eight levels from 0 to 350 kvar (Example 6.23). Each thyristor of a TSR carries half-cycle pulses of a full sinusoid, so its RMS current is $I_{pk}/2$, its average current I_{pk}/π , and it must block the full peak branch voltage when off.

6.4 The Thyristor-Switched Capacitor (TSC)

6.4.1 Why capacitor switching is the difficult case

A TSC switches a capacitor bank in or out with a bidirectional valve (Figure 6.17b). Unlike a reactor, the capacitor cannot be phase controlled: partial conduction would produce large current pulses and unacceptable

harmonics. The capacitor is therefore switched fully on or fully off, and the entire difficulty lies in choosing the switching instant.

Consider first a capacitor connected directly to an ideal voltage source through the valve. Two cases arise. If the capacitor voltage differs from the source voltage at the closing instant, Kirchhoff's voltage law demands an instantaneous voltage step across the capacitor, which requires an infinite current impulse; no valve survives this. If the voltages happen to be equal, the current still jumps instantly from zero to $C dv_s/dt$, its steady value at that instant, so di/dt is infinite even though the current magnitude stays bounded. Either way the thyristor di/dt limit is violated. Every practical TSC therefore includes a small series reactor L , sized just large enough to keep di/dt and inrush within valve capability. That reactor turns the branch into a series resonant circuit, and the switching transient must now be analysed properly.

6.4.2 Complete current of the TSC branch

With the valve conducting and resistance neglected, the branch obeys

$$L \frac{di}{dt} + v_C = V_m \sin \omega t, i = C \frac{dv_C}{dt} \quad (6.23)$$

where C is the capacitance, v_C the capacitor voltage and L the series reactor. Two parameters describe the branch. The *tuning ratio* is

$$n = \frac{\omega_n}{\omega} = \frac{1}{\omega \sqrt{LC}} = \sqrt{\frac{X_C}{X_L}} \quad (6.24)$$

where ω_n is the natural resonant frequency of the branch, $X_C = 1/(\omega C)$ and $X_L = \omega L$. The *magnification factor* is

$$k = \frac{X_C}{X_C - X_L} = \frac{n^2}{n^2 - 1} \quad (6.25)$$

The steady-state solution is found by treating the branch as a net capacitive reactance $X_C - X_L$:

$$i_{ss} = I_{AC} \cos \omega t, I_{AC} = \frac{V_m}{X_C - X_L} = k \omega C V_m, v_{C,ss} = k V_m \sin \omega t \quad (6.26)$$

where I_{AC} is the peak steady-state current. Now close the valve at $\theta_c = \omega t_c$ with the capacitor carrying a residual voltage V_{C0} and the reactor current zero. The natural response must make up the mismatch between the initial states and the forced states. Writing $\tau = t - t_c$ and solving Equation (6.23) with these initial conditions gives

$$i(t) = I_{AC} \left[\cos \omega t - \cos \theta_c \cos \omega_n \tau + n \left(\sin \theta_c - \frac{V_{C0}}{k V_m} \right) \sin \omega_n \tau \right] \quad (6.27)$$

The first term is the forced current. The second and third oscillate at ω_n ; they are the switching transient, and without damping they persist. Resistance makes them decay over a few cycles, but their initial amplitude still sets the valve current stress. The amplitude of the oscillatory part, relative to I_{AC} , is

$$\frac{I_{osc}}{I_{AC}} = \sqrt{\cos^2 \theta_c + n^2 \left(\sin \theta_c - \frac{V_{C0}}{k V_m} \right)^2} \quad (6.28)$$

and the worst instantaneous current is bounded by $I_{AC} + I_{osc}$.

6.4.3 Consequences of the series reactor: magnification and the choice of n

Equation (6.26) shows that the reactor raises both the current and the capacitor voltage by the factor k compared with the capacitor alone. Figure 6.19 plots k . For $n \geq 3$ the magnification is modest, 1.125 at $n=3$ and 1.052 at $n=4.5$, but it rises steeply for small n . The capacitor must be voltage rated for $k V_m$, and the bank rating must be computed with the net reactance $X_C - X_L$.

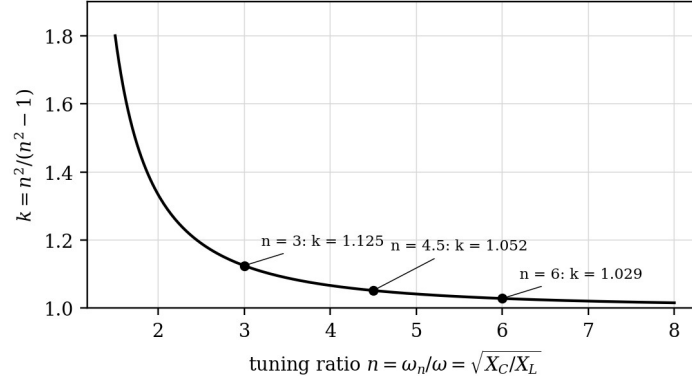


Figure 6.19. Magnification factor $k = n^2 / (n^2 - 1)$ versus tuning ratio.

The tuning ratio also sets the branch resonance at n times the fundamental. In an SVC with a TCR, n must not coincide with a characteristic harmonic, otherwise the TSC branch becomes an unintended filter that absorbs TCR harmonic current and may resonate with the network. Values between about 4 and 5, typically 4.3 to 4.8, are usual: the branch is capacitive at the 5th harmonic, yet stays clear of it. A larger L (smaller n) limits inrush better but increases magnification and cost.

Example 6.4: TSC branch design with detuning reactor.

A single-phase TSC must deliver 10 kvar at 230 V with tuning ratio $n=4.5$. Find (a) k ; (b) X_C and C so that the branch delivers 10 kvar; (c) X_L , L and the resonant frequency; (d) the branch current and capacitor voltage; (e) the capacitor's own var rating; (f) the maximum valve voltage and a thyristor rating with 50% margin.

Solution. (a) $k = 20.25 / 19.25 = 1.05195$.

(b) The net reactance is $V^2 / Q = 5.29 \Omega = X_C - X_L$. With $X_L = X_C / n^2$, $X_C = k \times 5.29 = 5.565 \Omega$ and $C = 572.0 \mu\text{F}$.

(c) $X_L = 0.2748 \Omega$; $L = 0.875 \text{ mH}$; $f_n = 225 \text{ Hz}$.

(d) $I = 43.48 \text{ A RMS}$ (61.49 A peak); $V_C = I X_C = 241.9 \text{ V RMS}$, peak 342.2 V = $k V_m$.

(e) $I^2 X_C = 10.52 \text{ kvar}$: the capacitor must be rated about 5% above the branch output.

(f) $(1+k) V_m = 667.4 \text{ V}$; with 50% margin, about 1.0 kV.

Note. The series reactor is part of the TSC design, not merely a protective accessory. It changes the branch resonant frequency, capacitor voltage, current magnification and the transient produced by switching.

6.4.4 Conditions for transient-free switching

From Equation (6.27), both oscillatory terms vanish only if, simultaneously,

$$\cos \theta_c = 0 \text{ and } V_{C0} = \pm k V_m \quad (6.29)$$

The first condition requires closure at a positive or negative peak of the supply voltage, where the forced current is zero. The second requires the capacitor to be pre-charged to the magnified steady-state voltage with the matching polarity. Figure 6.21c shows such a closure: the branch current starts smoothly with no oscillation.

These conditions are naturally satisfied when a bank that was previously in service is reconnected before it discharges: the valve turns off at a current zero, which coincides with the capacitor voltage at $\pm k V_m$, and the next same-polarity peak then provides a transient-free closing instant. Elsewhere they are not met. Distorted bus voltages shift the peaks, residual charge leaks away through the discharge resistors, and maintaining an exact pre-charge would require extra charging equipment and impose a prolonged DC stress that AC capacitors are not designed for. Practical controllers therefore use simpler rules that keep the transient small rather than zero.

6.4.5 Practical firing schemes and switching strategies

Firing scheme 1 (valve-voltage-zero firing). Fire when the instantaneous supply voltage equals the residual capacitor voltage, $V_m \sin \theta_c = V_{C0}$, so that the valve voltage is zero at turn-on and the initial di/dt is zero. Writing $\rho = V_{C0}/V_m$ and substituting $\sin \theta_c = \rho$ into Equation (6.28) gives

$$\frac{I_{osc}}{I_{AC}} = \sqrt{1 - \rho^2 \left(1 - \frac{1}{n^2}\right)}, \theta_c = \sin^{-1} \rho, |\rho| \leq 1 \quad (6.30)$$

where ρ is the residual voltage ratio. The oscillatory current never exceeds the steady-state current; it is largest (equal to I_{AC}) for a discharged capacitor fired at a voltage zero, and it falls as the residual charge increases. Differentiating Equation (6.28) with respect to $\sin \theta_c$ shows that this choice minimizes the oscillatory amplitude for $|\rho| \leq 1$. The scheme cannot be used if the capacitor is overcharged ($|\rho| > 1$), because the supply voltage never reaches V_{C0} .

Firing scheme 2 (crest firing). Fire at the crest of the supply voltage of the same polarity as V_{C0} , where the valve voltage is at its minimum $V_m - V_{C0}$. With $\theta_c = 90^\circ$,

$$\frac{I_{osc}}{I_{AC}} = n \left| \frac{V_{C0}}{k V_m} - 1 \right| = n \left| \left(1 - \frac{1}{n^2}\right) \rho - 1 \right| \quad (6.31)$$

This scheme works for an overcharged capacitor and becomes transient-free as $\rho \rightarrow k$, but its performance deteriorates rapidly as ρ falls: for a discharged capacitor the oscillatory amplitude is n times the steady current. Figure 6.20 compares the two schemes, and Figure 6.21 shows simulated closing transients for four representative cases with $n = 4.5$.

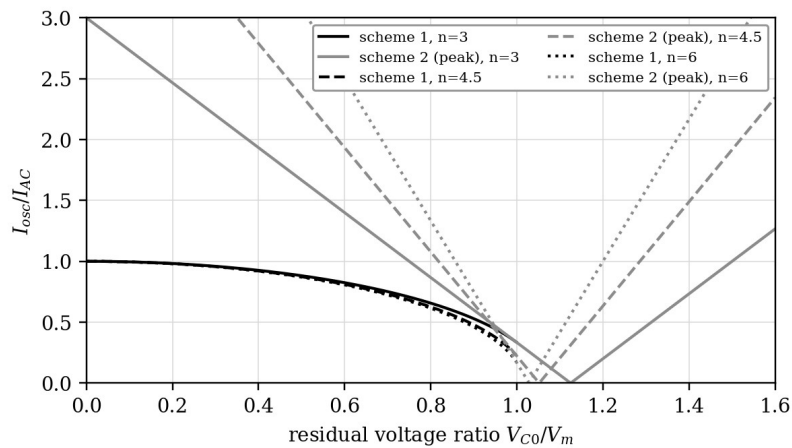


Figure 6.20. Oscillatory current amplitude relative to the steady-state peak for firing schemes 1 and 2.

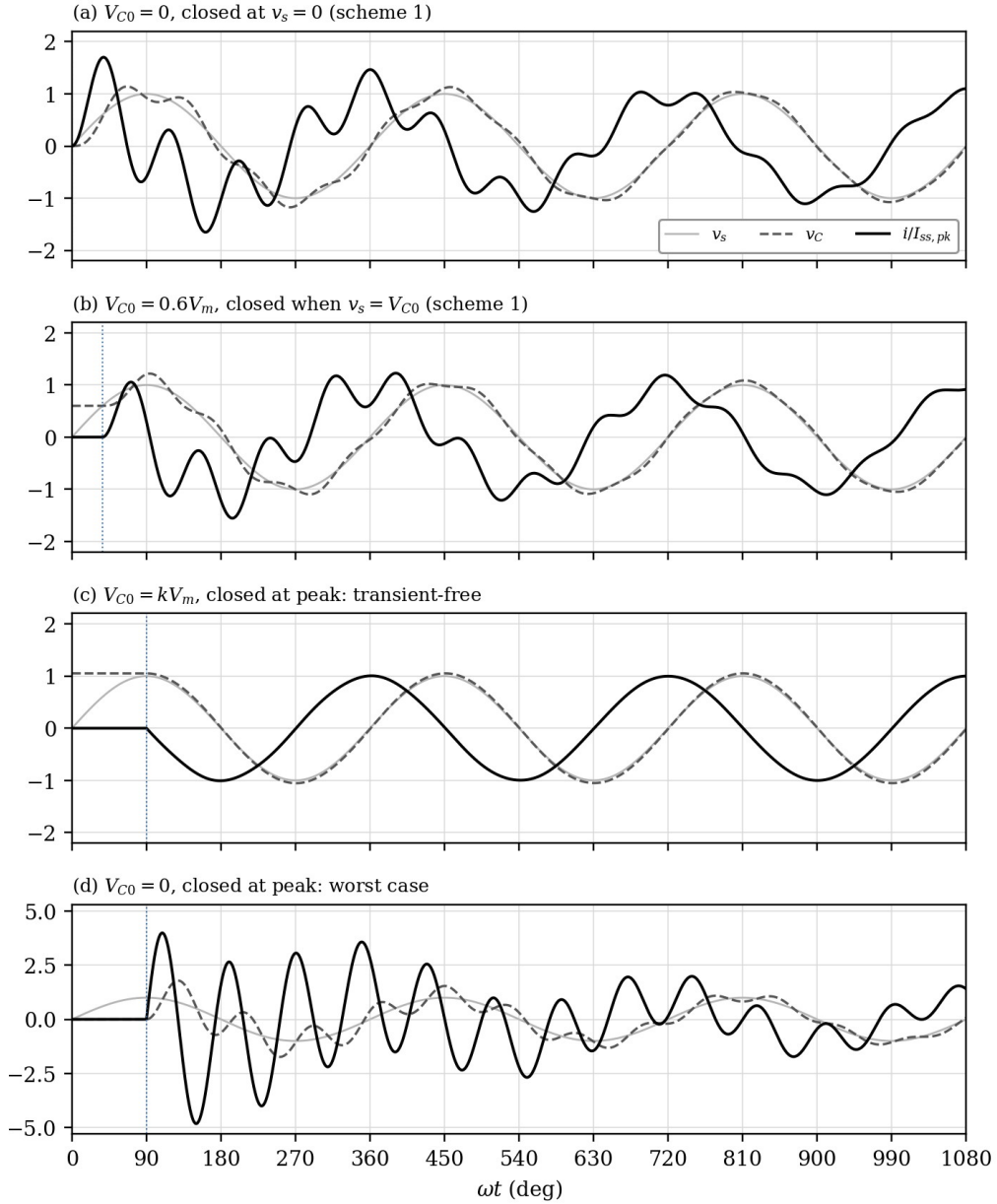


Figure 6.21. Simulated TSC closing transients ($n=4.5$, light damping): (a) discharged capacitor fired at a voltage zero, (b) $V_{C0}=0.6V_m$ fired at voltage match, (c) $V_{C0}=kV_m$ fired at the peak, transient-free, (d) discharged capacitor fired at the peak, the worst case.

These two rules combine into the switching strategies used in practice. *Strategy A*, used in transmission TSCs, applies scheme 1 when $V_{C0} < V_m$ and scheme 2 (a forced switch-on at the crest) when $V_{C0} \geq V_m$. *Strategy B*, common in industrial plants, keeps every disconnected capacitor charged to the supply crest by gating only one thyristor, and always fires at the crest. The transient is then small but not zero, because the capacitor sits at V_m rather than kV_m . Its drawbacks are the sustained DC stress, the need for periodic polarity reversal, and the risk that a transient overvoltage leaves the capacitor overcharged; these are tolerable in industrial service but not in transmission compensation.

Example 6.5: TSC with residual charge: scheme 1.

A 6 kvar, 230 V TSC with $n=5$ has $V_{C0}=150$ V. Find (a) the applicable scheme; (b) the closing angle and delay after the zero crossing; (c) I_{osc}/I_{AC} ; (d) a bound on peak current; (e) I_{osc}/I_{AC} for crest firing; (f) for a discharged capacitor at a voltage zero; (g) the residual voltage for zero transient.

Solution. (a) $\rho = 150/325.27 = 0.4612 < 1$: scheme 1.

- (b) $\theta_c = \sin^{-1} 0.4612 = 27.46^\circ$, i.e. 1.53 ms after the positive-going zero.
- (c) Equation (6.30): $\sqrt{1 - 0.2127 \times 0.96} = 0.892$.
- (d) $I_{AC} = \sqrt{2} \times 26.09 = 36.89$ A (the 6 kvar rating already includes the reactor); bound $36.89 \times 1.892 = 69.8$ A.
- (e) Equation (6.31): $5|0.96 \times 0.4612 - 1| = 2.79$, much worse.
- (f) 1.0.
- (g) $k V_m = 1.04167 \times 325.27 = 338.8$ V, closed at the positive crest.

Note. Matching the instantaneous source voltage to the trapped capacitor voltage reduces the voltage discontinuity, but the complete LC state must also be considered. A minimum-transient switching rule and an exactly transient-free state are not automatically the same condition.

6.4.6 Turn-off, trapped charge and valve voltage stress

When gate pulses are removed, the valve stops at the next current zero. Because the capacitor current leads its voltage by 90° , that current zero coincides with a crest of capacitor voltage, so the capacitor is left charged to $\pm k V_m$. The valve must then block the difference between the supply and the trapped voltage:

$$v_T(t) = V_m \sin \omega t - k V_m \quad (6.32)$$

where v_T is the valve voltage for a capacitor left at $+k V_m$. Half a cycle later the supply reaches $-V_m$ and the valve voltage peaks at $-(1+k)V_m$, slightly more than twice the supply peak. Figure 6.22 shows this, together with the slower decline when a discharge resistor is fitted. The initial small step in v_T at turn-off is the voltage across the series reactor, which collapses when the current stops.

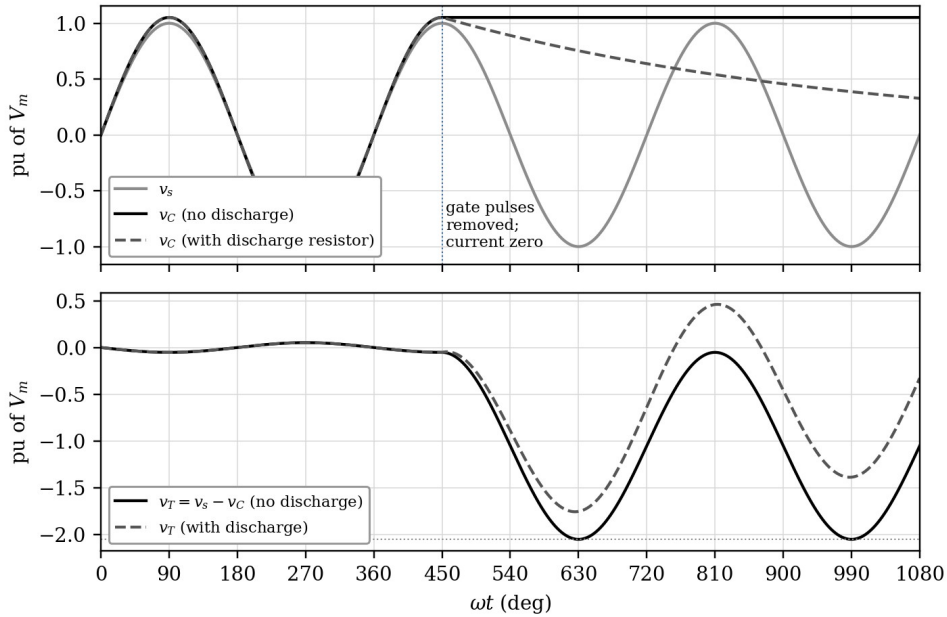


Figure 6.22. TSC turn-off: supply and capacitor voltages (top) and valve voltage (bottom) for non-discharging and discharging capacitors.

Two practical consequences follow. A TSC valve needs roughly twice as many series thyristors as a TCR valve at the same bus voltage, and it must also withstand a DC voltage component for as long as the bank is disconnected. Because the valve voltage stress is so high, TSCs are almost always placed on the low-voltage side of a step-down transformer.

6.4.7 Inrush and di/dt protection

The series reactor must also protect the valve against a mistimed firing, for instance a control fault, a misread synchronizing signal or forward break-over protection firing the valve at an arbitrary instant. At the instant of closure the reactor carries the whole voltage difference, so

$$\left. \frac{di}{dt} \right|_{t_c^+} = \frac{v_s(t_c) - V_{C0}}{L} \quad (6.33)$$

The worst case is closure when the supply is at one peak while the capacitor holds the trapped voltage of opposite polarity, giving $|v_s - V_{C0}| = (1+k)V_m$. The minimum reactor for a valve di/dt capability $(di/dt)_{max}$ is therefore $L_{min} = (1+k)V_m / (di/dt)_{max}$. In a well-designed TSC the reactor chosen for detuning ($n \approx 4.5$) normally satisfies this limit with ample margin; Example 6.25 checks it for a 33 kV bank.

6.4.8 Configurations

Three-phase TSCs are connected in delta or in ungrounded star. The delta arrangement exposes each valve to line-to-line voltage but keeps the phases independent. In an ungrounded star, the neutral floats until all three phases conduct, so the switching sequence must be coordinated: two phases are closed first at the appropriate line-to-line voltage and the third a quarter-cycle later. In low-voltage industrial compensators, one thyristor in each valve is often replaced by a diode. The diode keeps the capacitor charged to the supply crest, the thyristor is fired at the crest, and the arrangement is a cheap realization of strategy B.

Capacitor banks are divided into stages. With binary weights $B, 2B, \dots, 2^{N-1}B$ the bank gives $2^N - 1$ non-zero levels. Repeated weights such as $C, C, C/2$ give fewer distinct levels (here five non-zero levels) but allow identical, interchangeable valve and capacitor assemblies, which simplifies spares and redundancy (Example 6.26). In every case the weighting must be applied to the *net* branch susceptance including the series reactor.

6.4.9 V-I characteristic and hysteresis

Each TSC stage contributes a fixed capacitive susceptance, so a bank of N stages has the discrete characteristic of Figure 6.23: one straight line through the origin for each number of stages in service. To regulate voltage, a stage is switched in when the bus voltage falls below $V_{ref} - \Delta V/2$ and switched out when it rises above $V_{ref} + \Delta V/2$. The hysteresis band ΔV must exceed the voltage change caused by one step, $\Delta V_{step} \approx X_{th} B_{stage} V$, otherwise the controller hunts. Because the response is quantized, a TSC alone regulates voltage only within this band. Continuous control requires a TCR or a converter alongside it.

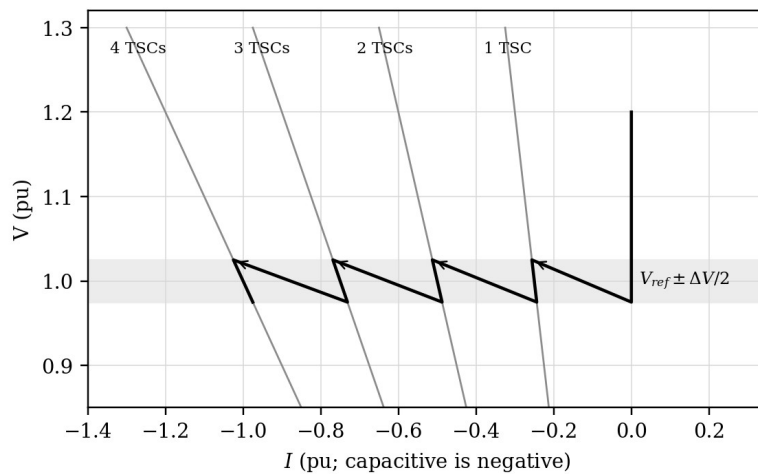


Figure 6.23. Discrete V-I characteristic of a four-stage TSC bank with hysteresis band $V_{ref} \pm \Delta V/2$.

A TSC also has a non-zero response delay. After a command to switch in, the valve must wait for the next permissible firing instant, which with trapped charge of a given polarity can be up to one full cycle. Switching out takes effect at the next current zero, within half a cycle. These delays are small compared with typical voltage-control time constants, but they matter in flicker compensation (Section 6.5.10).

6.5 The Static Var Compensator (SVC)

6.5.1 Definition and objectives

The IEEE and CIGRE definition of a static var compensator is a shunt-connected static var generator or absorber whose output is adjusted to exchange capacitive or inductive current so as to maintain or control specific parameters of the power system, typically bus voltage [17], [23]. “Static” distinguishes it from the rotating synchronous condenser: an SVC has no moving parts other than cooling equipment. Physically it is an assembly of the branches already studied (TCRs, TSRs, TSCs, fixed capacitors and harmonic filters), connected to a medium-voltage busbar that is usually fed from the transmission bus through a step-down transformer (Figure 6.24).

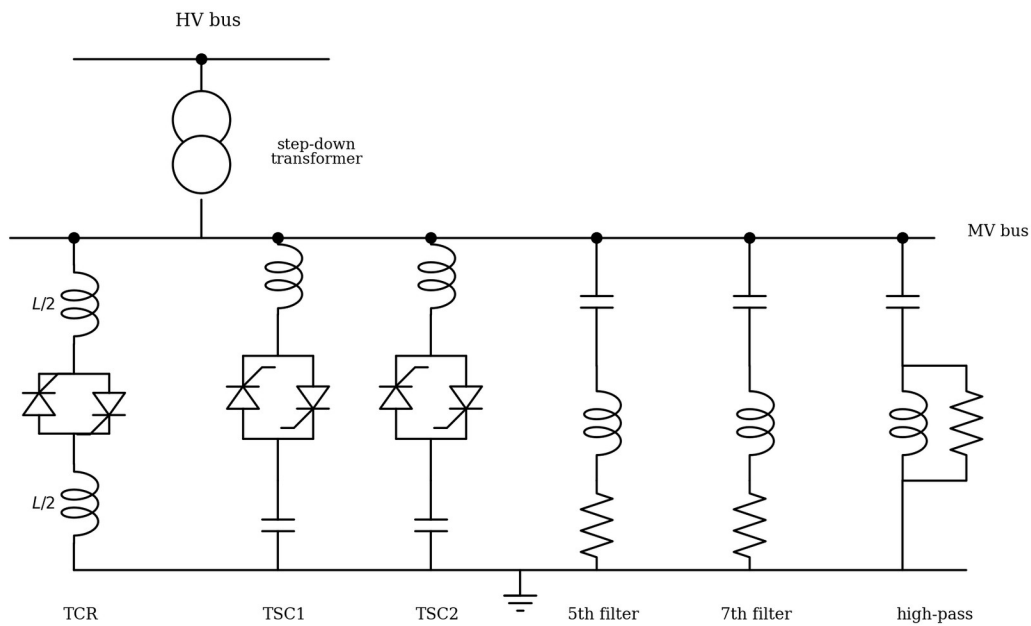


Figure 6.24. Single-line diagram of a transformer-coupled TSC-TCR SVC with 5th, 7th and high-pass filters.

Two broad families exist. *Transmission SVCs* regulate the voltage at a substation or along a long corridor. They raise transfer limits, improve transient stability, damp electromechanical oscillations (and occasionally subsynchronous torsional modes), and limit dynamic overvoltages after load rejection. *Industrial SVCs* sit close to arc furnaces, rolling mills, mine hoists and traction substations. There they reduce voltage flicker, balance single-phase loads and correct power factor, which requires very fast, phase-by-phase action.

6.5.2 V-I characteristic: ideal and practical

An ideal SVC would hold its bus voltage at V_{ref} for any current, capacitive or inductive, without losses, harmonics or delay: a horizontal line on the V-I plane. A practical SVC approaches this inside its control range and departs from it at the limits. Its characteristic is built by combining the characteristics of its branches, as Figure 6.25 shows for an FC-TCR. The fixed capacitor contributes a straight line of slope $-1/B_C$ on the capacitive side, and the TCR a fan of lines on the inductive side. The controller selects, at each voltage, the TCR susceptance that places the operating point on the regulating line.

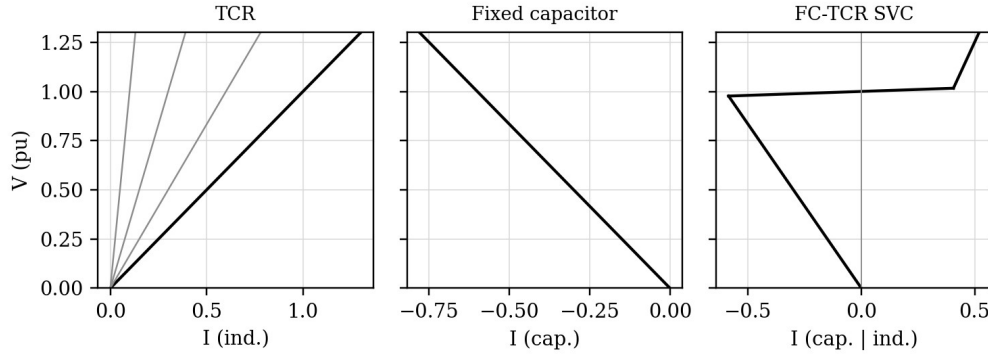


Figure 6.25. Building the SVC characteristic: TCR fan, fixed-capacitor line, and the resulting FC-TCR characteristic.

Inside the control range the SVC follows

$$V = V_{ref} + X_{SL} I_{SVC} \quad (6.34)$$

where V is the regulated bus voltage, V_{ref} the voltage at zero SVC current, X_{SL} the slope (droop) reactance, typically 1 to 5% on the SVC rating, and I_{SVC} the SVC reactive current, positive when inductive. Outside the range the SVC behaves as a fixed susceptance:

$$I_{SVC} = -B_{C,max} V \text{ (capacitive limit)}, I_{SVC} = B_{L,max} V \text{ (inductive limit)} \quad (6.35)$$

where $B_{C,max}$ and $B_{L,max}$ are the maximum net capacitive and inductive susceptances. The limits are lines through the origin, not vertical lines. At the capacitive limit the SVC current, and with it the support it can give, collapses in proportion to the voltage.

6.5.3 Operating point and the role of the slope

The steady operating point is the intersection of Equation (6.34) with the system load line, Equation (6.1). Solving the pair,

$$I_{SVC} = \frac{E_{th} - V_{ref}}{X_{th} + X_{SL}}, V = \frac{X_{SL} E_{th} + X_{th} V_{ref}}{X_{th} + X_{SL}} \quad (6.36)$$

Figure 6.26 shows how the operating point moves as the Thevenin voltage varies. When the system voltage would otherwise be high, the SVC absorbs reactive power, and when it would be low, the SVC injects it.

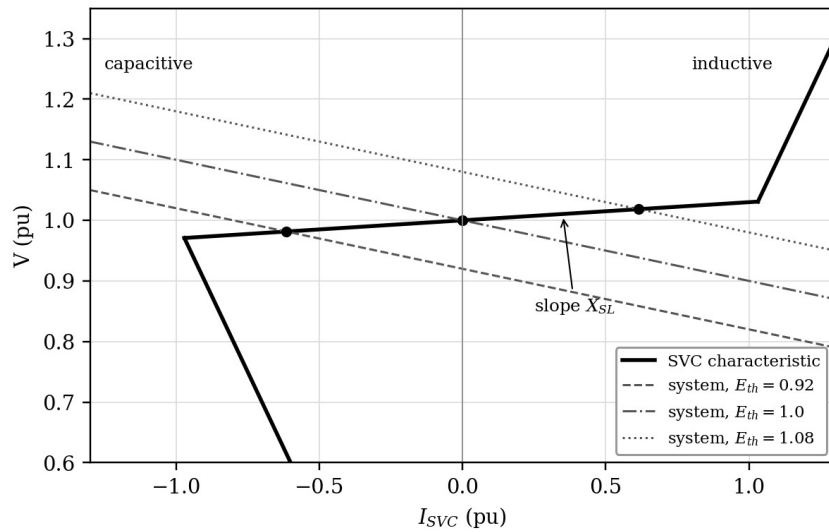


Figure 6.26. SVC operating points as the intersection of the SVC characteristic with system load lines for three Thevenin voltages.

Why not set $X_{SL}=0$ for perfect regulation? Equation (6.36) answers part of the question: the current needed to absorb a given excursion ΔE_{th} is $\Delta E_{th}/(X_{th}+X_{SL})$, so a non-zero slope reduces the required rating (Figure 6.27). For a stiff bus ($X_{th}=0.05$ pu) a 3% slope cuts the rating by 37.5%. There are three further reasons. The slope lets several SVCs, or an SVC and nearby generators, share the regulating duty in a stable way instead of fighting each other. It keeps the SVC away from its limits for ordinary voltage variations, preserving dynamic reserve for disturbances. And it makes the voltage loop gain less sensitive to network strength (Section 6.5.7). The voltage regulation obtained is $\Delta V/\Delta E_{th}=X_{SL}/(X_{th}+X_{SL})$.

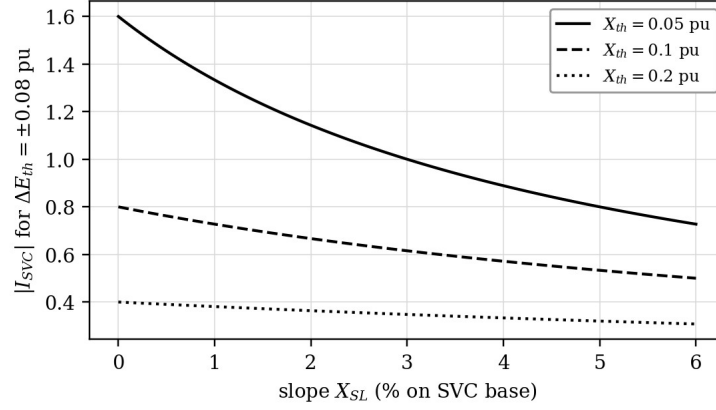


Figure 6.27. SVC current required to absorb a ± 0.08 pu excursion of the Thevenin voltage, as a function of slope, for three network strengths.

Example 6.6: SVC operating point, slope and saturation.

The Thevenin voltage at a bus varies between 0.92 and 1.08 pu with $X_{th}=0.06$ pu. An SVC with $V_{ref}=1.0$ pu is installed. Find (a) the SVC current range for $X_{SL}=0$; (b) the range for $X_{SL}=0.03$ pu; (c) the bus voltage band in case (b); (d) the reactive power at the extremes; (e) the capacitive susceptance needed at $E_{th}=0.92$; (f) the actual operating point if $B_{C,max}=0.8$ pu.

Solution. (a) Equation (6.36) with $X_{SL}=0$: $I=\pm 0.08/0.06=\pm 1.333$ pu, and $V=1.0$ pu exactly.

(b) $I=\pm 0.08/0.09=\pm 0.889$ pu, a 33% reduction in rating.

(c) $V=1\pm 0.03\times 0.889$, i.e. 0.9733 to 1.0267 pu.

(d) $Q=VI$: -0.865 pu (capacitive) at $E_{th}=0.92$ and $+0.913$ pu (inductive) at $E_{th}=1.08$.

(e) $B=0.889/0.9733=0.913$ pu.

(f) The SVC saturates and behaves as $B=0.8$ pu. With $I=-BV$ in Equation (6.1), $V=E_{th}/(1-X_{th}B)=0.92/0.952=0.9664$ pu, $I=-0.773$ pu and $Q=BV^2=0.747$ pu. The voltage is below the droop line, and the support is 14% less than the regulating characteristic would have demanded.

Note. The droop equation determines the commanded operating line, while the capacitive and inductive susceptance limits determine whether that command is physically achievable. A calculated operating point outside the limits must be clipped to the appropriate boundary.

6.5.4 SVC configurations

Fixed capacitor-thyristor controlled reactor (FC-TCR)

The FC-TCR (Figure 6.28) extends the purely inductive TCR into the capacitive region by adding a permanently connected capacitor, which in practice is split into tuned filter branches. Its net susceptance is

$$B_{SVC}(\alpha)=B_C-B_{TCR}(\alpha) \quad (6.37)$$

where B_C is the total fundamental capacitive susceptance of all fixed branches and filters, and B_{TCR} is given by Equation (6.13). Maximum capacitive output occurs with the TCR blocked ($\alpha = 180^\circ$), and maximum inductive output at full conduction. To deliver a range from $+Q_{cap}$ to $-Q_{ind}$ at rated voltage, the ratings must be

$$Q_C = Q_{cap}, Q_{TCR} = Q_{cap} + Q_{ind} \quad (6.38)$$

where Q_C is the fixed-capacitor rating and Q_{TCR} the TCR rating at full conduction. The TCR is therefore always larger than the SVC's inductive range, because it must first cancel the capacitor.

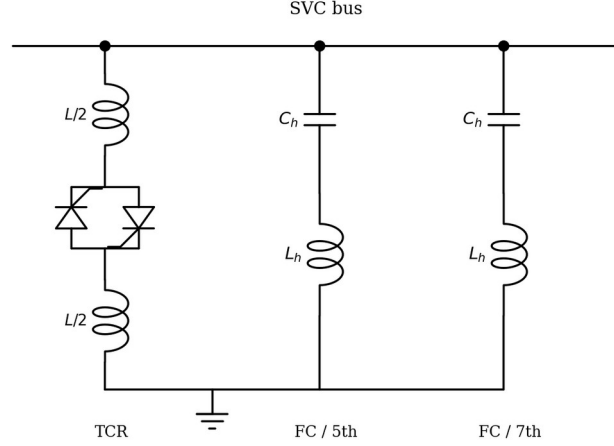


Figure 6.28. FC-TCR SVC with the fixed capacitance provided by 5th and 7th harmonic filters.

Figure 6.29 shows the V-I operating area and the net susceptance versus firing angle. The control is smooth and continuous, transients are absent because nothing is switched, and harmonics are absorbed by the filters that form the capacitor. The weakness is loss. To produce zero net output, the TCR must circulate a current equal to the capacitor current, so the SVC dissipates appreciable power even when it exchanges nothing with the network. The loss curve of Figure 6.31 falls from its highest value at full inductive output to its lowest at full capacitive output. This is why the FC-TCR is favoured for industrial loads, where fast and continuous control dominates, and is less attractive for transmission duties where the SVC idles near zero output for most of its life.

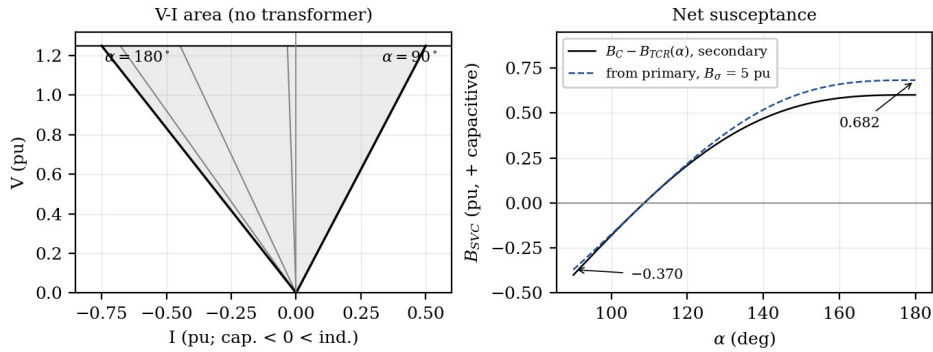


Figure 6.29. FC-TCR: V-I operating area without transformer (left) and net susceptance versus firing angle with and without a coupling transformer (right).

The FC-TCR controller performs four functions. A phase-locked loop provides synchronous timing referred to the voltage zero crossings of each branch. The required fundamental reactor current is computed by subtracting the known capacitor current amplitude from the reactive current reference. That current is converted to a firing angle through the inverse of Equation (6.13). Finally, the firing pulses are generated and distributed to the valves.

Thyristor switched capacitor-thyristor controlled reactor (TSC-TCR)

The TSC-TCR (Figure 6.24) replaces the fixed capacitor by N switched capacitor stages, each of rating Q_{TSC} , and retains one TCR for continuous fine control. For a capacitive demand Q , the controller switches in

$$N_{on} = \lceil \frac{Q}{Q_{TSC}} \rceil, Q_{TCR}(\alpha) = N_{on} Q_{TSC} - Q \quad (6.39)$$

where N_{on} is the number of TSCs in service, $\lceil \cdot \rceil$ denotes rounding up, and $Q_{TCR}(\alpha)$ the absorption the TCR must provide to cancel the surplus. Figure 6.30 compares the var output of the TSC steps and the TCR for both configurations. In the TSC-TCR the TCR current never exceeds one stage, so losses at zero output are small and rise only as capacitive output is called for (Figure 6.31). Redundancy is also better, since the loss of one TSC stage removes only part of the capacitive range.

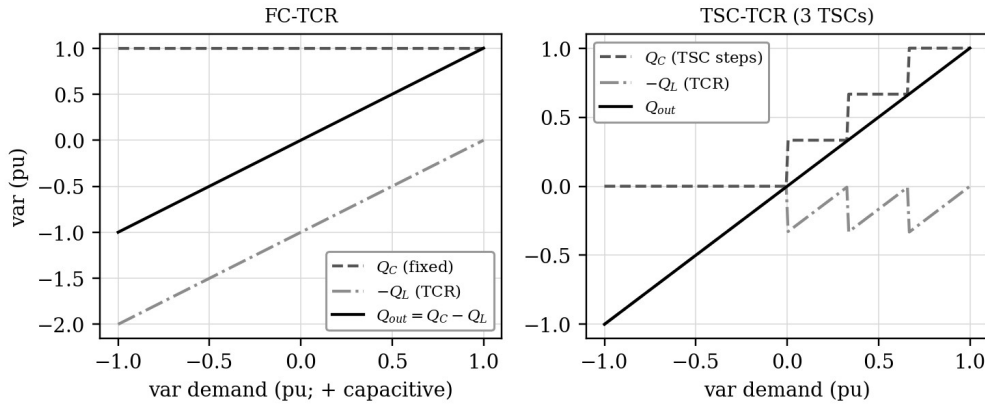


Figure 6.30. Var output versus var demand: FC-TCR (left) and TSC-TCR with three stages (right).

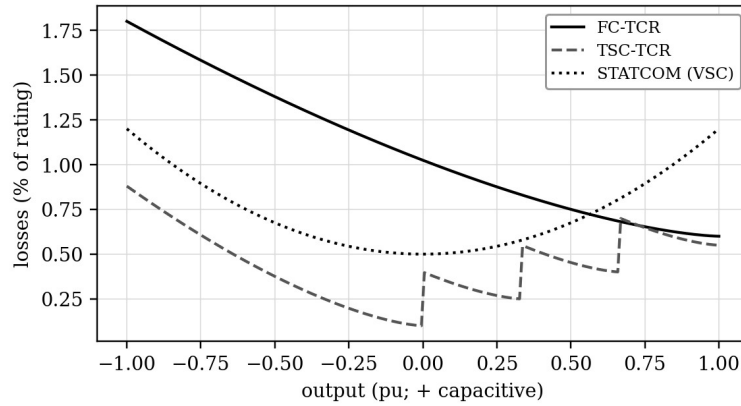


Figure 6.31. Illustrative loss characteristics of FC-TCR, TSC-TCR and VSC-based STATCOM of equal rating (simplified loss models; actual values depend on design).

For continuous coverage the TCR range must at least bridge one TSC step, and in practice exceed it slightly so that the switching-in and switching-out thresholds can be separated by a hysteresis margin:

$$B_{TCR,max} \geq B_{TSC}(1 + \epsilon) \quad (6.40)$$

where B_{TSC} is the susceptance of one TSC stage and ϵ a margin of a few percent. If this condition is violated the characteristic develops dead zones, ranges of susceptance that no combination of stages and firing angle can produce (Figure 6.32). Inside such a gap the SVC can only alternate between neighbouring stages, which defeats continuous control and causes repeated switching.

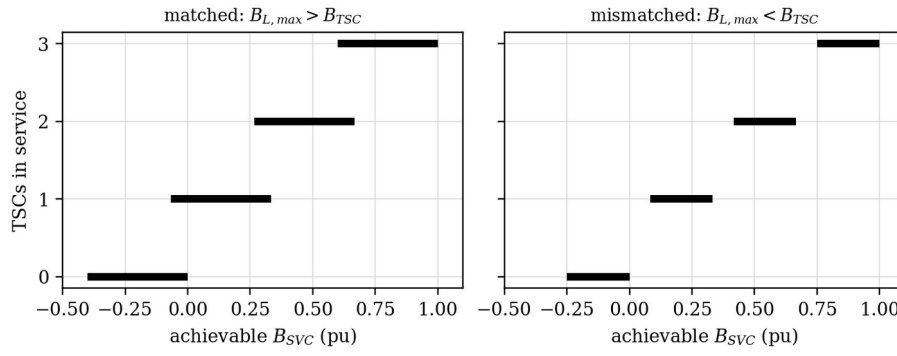


Figure 6.32. Achievable net susceptance of a three-stage TSC-TCR: matched TCR (left) and undersized TCR producing dead zones (right).

Other configurations

Table 6.4 summarizes the configurations in service. The mechanically switched capacitor with a TCR (MSC-TCR) is a low-cost variant for slow load cycles. The TSC-TSR SVC has no phase-controlled element and needs no filters, but it regulates only in steps. Saturated-reactor compensators are of historical interest only.

Table 6.4. Comparison of SVC configurations.

Configuration	Control	Harmonics	Losses at zero output	Typical use
FC-TCR	Continuous	TCR harmonics, absorbed by FC filters	High (TCR cancels FC)	Arc furnaces, rolling mills
TSC-TCR	Continuous	Small TCR (one stage) harmonics	Low	Transmission voltage control
TSC-TSR	Stepped	Negligible	Low	Slow voltage control, low cost
MSC-TCR	Continuous within TCR range, slow steps	TCR harmonics	Medium	Subtransmission, slow load cycles
Saturated reactor + FC	Inherent, fixed slope	Low (with slope correction)	High	Historical flicker compensators

6.5.5 SVC with a coupling transformer

Most transmission SVCs connect through a step-down transformer of leakage reactance X_σ , which places the thyristor valves at a convenient medium voltage. Seen from the high-voltage bus, the branch susceptance B on the secondary (positive when capacitive) appears in series with X_σ . The equivalent admittance of the series combination is $j/(1/B - X_\sigma)$, giving

$$B_{eq} = \frac{B}{1 - X_\sigma B} = \frac{B_\sigma B}{B_\sigma - B}, B_\sigma = \frac{1}{X_\sigma} \quad (6.41)$$

where B_{eq} is the susceptance seen from the primary and B_σ the transformer leakage susceptance magnitude. The transformer *enlarges* the capacitive end of the range ($B > 0$, denominator below one) and *shrinks* the inductive end ($B < 0$, denominator above one). The secondary voltage is raised on capacitive output and lowered on inductive output:

$$V_2 = \frac{V}{1 - X_\sigma B} \quad (6.42)$$

where V_2 is the secondary bus voltage referred to the primary. Two practical consequences follow. Secondary overvoltage at full capacitive output sets the voltage rating of the capacitors and valves. And because B_{eq} is a nonlinear function of B , the linearizer must include the transformer, or the regulator gain will vary with operating point. For $X_\sigma B \ll 1$, Equation (6.41) linearizes to $B_{eq} \approx B(1 + X_\sigma B)$, which is adequate for preliminary sizing (Example 6.7).

Example 6.7: SVC behind a coupling transformer.

An FC-TCR has $B_C = 0.8$ pu and $B_{L,max} = 0.9$ pu on the secondary, connected through a transformer with $X_\sigma = 0.2$ pu ($B_\sigma = 5$ pu). Find (a) the net secondary susceptance range; (b) the exact primary range; (c) the linearized values and their errors; (d) the secondary voltages at the limits with 1.0 pu on the primary; (e) the practical consequence.

Solution. (a) $+0.8$ pu (TCR blocked) to $0.8 - 0.9 = -0.1$ pu (full conduction).

(b) Equation (6.41): $0.8/(1 - 0.16) = 0.9524$ pu and $-0.1/(1 + 0.02) = -0.0980$ pu.

(c) $B(1 + X_\sigma B)$: 0.928 pu (-2.6%) and -0.0980 pu (-0.04%). The linear form is poor at the capacitive end.

(d) Equation (6.42): $V_2 = 1/(1 - 0.16) = 1.190$ pu at full capacitive output and $1/1.02 = 0.980$ pu at full inductive output.

(e) The transformer raises the capacitive range by 19%, but the capacitors and TSC valves must be rated for 1.19 pu secondary voltage. Writing the series combination as $B_\sigma B/(B_\sigma + B)$ would give a *reduced* capacitive range of 0.69 pu; that form has the wrong sign for a capacitive branch.

Note. Referring an SVC through its coupling transformer is not a simple turns-ratio exercise. Transformer leakage reactance modifies the effective susceptance and therefore the attainable capacitive and inductive ranges seen from the transmission bus.

6.5.6 Harmonic filters and resonance

A single-tuned filter branch is a series R-L-C circuit tuned to harmonic order h . At fundamental frequency it is capacitive, and its net reactance must deliver the rated filter output Q_F :

$$X_C - X_L = \frac{V^2}{Q_F}, X_L = \frac{X_C}{h^2} \Rightarrow X_C = \frac{h^2}{h^2 - 1} \cdot \frac{V^2}{Q_F} \quad (6.43)$$

where V is the voltage across the branch (phase voltage for star connection with per-phase Q_F , or equivalently line voltage with three-phase Q_F), X_C and X_L are the fundamental-frequency reactances of the filter capacitor and inductor, and h the tuning order. The resistance follows from the chosen quality factor Q_f as $R = h X_L / Q_f$. Filters are tuned a few percent below the nominal harmonic, for example $h = 4.9$ for the 5th, so that a loss of capacitance (ageing or failed elements, which raises the tuned frequency) or a fall in system frequency (which lowers the harmonic frequency) does not leave them tuned above the harmonic. The parallel resonance with the network lies just below the tuned frequency and would then move onto the harmonic. Figure 6.33 shows the impedance of typical 5th and 7th branches.

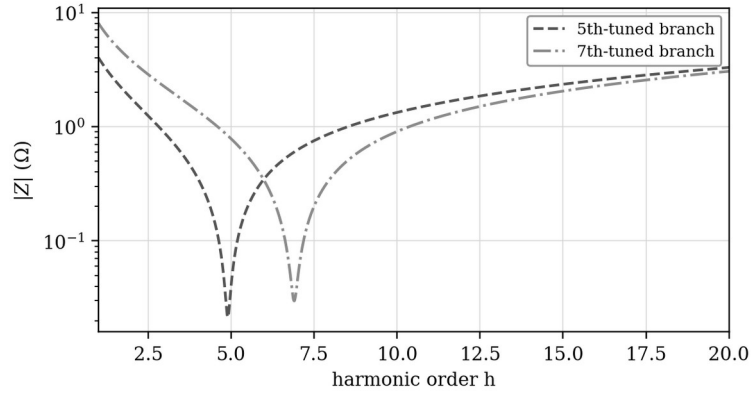


Figure 6.33. Impedance magnitude of 5th- and 7th-tuned filter branches ($Q_f = 40$) versus harmonic order.

Every capacitive branch also forms a parallel resonance with the network inductance. Below its tuned frequency a filter is capacitive, and with the system reactance X_{th} it resonates at approximately $h_p \approx \sqrt{X_C / (X_{th} + X_L)}$, where X_C and X_L are the branch reactances at fundamental frequency. Filter design must ensure that h_p does not coincide with a harmonic the TCR or the network can excite, under all switching states of the SVC and of the surrounding network.

6.5.7 SVC control

Control structure

Figure 6.34 shows the general control structure. Measured bus voltage and SVC current are rectified and filtered; the current, multiplied by the slope K_{SL} , is subtracted from the voltage reference, which implements Equation (6.34). A PI (or pure integral) voltage regulator produces a susceptance reference B_{ref} , to which supplementary signals such as power oscillation damping can be added. After limiting, B_{ref} is split by a distribution unit: TSC and TSR switching logic chooses the stages, and the TCR linearizer converts the residual susceptance into a firing angle. A firing-pulse generator synchronized by a PLL sends pulses to the valves.

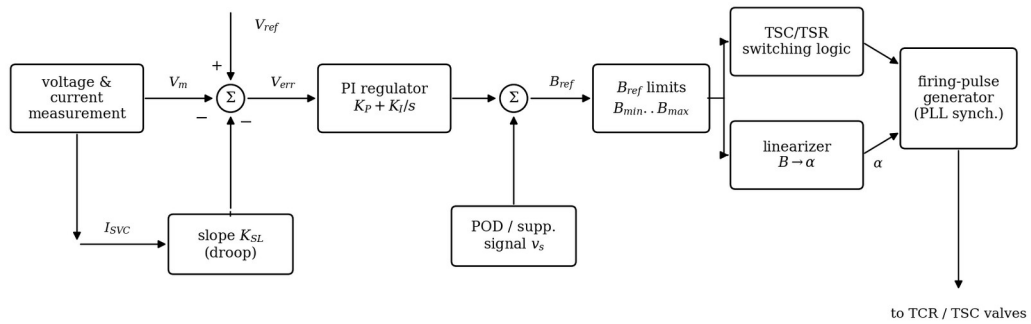


Figure 6.34. General SVC voltage control structure.

Regulator design against network strength

The closed-loop behaviour is dominated by the gain from SVC susceptance to bus voltage. Linearizing Equation (6.1) around $V \approx 1$ with $I_{SVC} = -B_{SVC}V$, a change ΔB in SVC susceptance (on the SVC rating) produces

$$\Delta V \approx K_N \Delta B, K_N = X_{th} = \frac{1}{E S C R} \quad (6.44)$$

where K_N is the network gain and $ESCR$ the effective short-circuit ratio at the SVC bus, expressed on the SVC rating. The thyristor actuator is modelled as a transport delay of mean value $T_b \approx T/4$ (5 ms at 50 Hz), approximated by $1/(1+sT_b)$. The slope feedback contributes a further gain K_{SL} around the loop. With the regulator

$$G_R(s) = K_C \frac{1+sT_C}{sT_C}, T_C = T_b \quad (6.45)$$

the zero cancels the actuator lag and the open-loop transfer function reduces to $K_C(K_N + K_{SL})/(sT_b)$. The closed loop is then first order with time constant

$$T_\omega = \frac{T_b}{K_C(K_N + K_{SL})} \quad (6.46)$$

The regulator gain is fixed at the weakest expected network, $K_{N,max} = 1/ESCR_{min}$, by choosing $K_C = 1/[2(K_{SL} + K_{N,max})]$, which places the loop crossover at about $1/(2T_b)$ and leaves an adequate phase margin despite the true delay. The time constant at any other strength is then

$$T_\omega = 2T_b \frac{K_{SL} + K_{N,max}}{K_{SL} + K_N}, t_{95} \approx 3T_\omega, \Delta V = \frac{K_N}{K_N + K_{SL}} \Delta V_{ref} \quad (6.47)$$

where t_{95} is the time to reach 95% of the final value and ΔV the steady change in bus voltage for a step ΔV_{ref} . In a stronger network K_N is smaller and the response becomes slower, as Figure 6.35 shows. This is the classic SVC tuning dilemma: a gain safe for the weakest network is sluggish in a strong one, while a gain tuned for the strong network can go unstable when lines trip. Modern controllers resolve it with a *gain supervisor* that detects oscillation in B_{ref} and reduces K_C automatically, or with adaptive gain scheduling based on an online ESCR estimate.

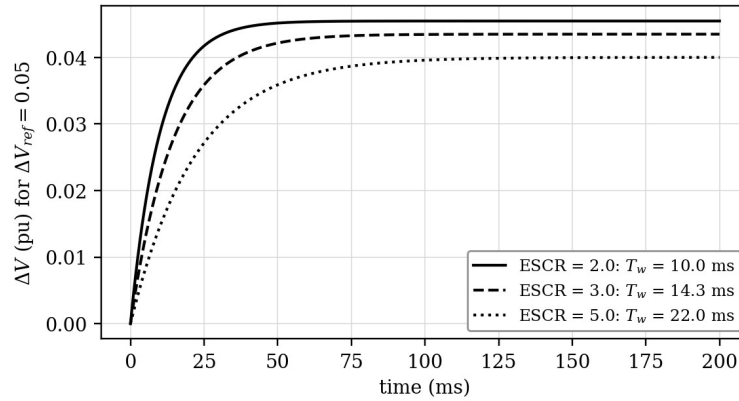


Figure 6.35. Closed-loop voltage response to a 0.05 pu reference step for a regulator designed at $ESCR = 2$, operating at $ESCR = 2, 3$ and 5 .

Standard models and supplementary controls

For power-system studies the IEEE and WECC define generic SVC models. The common structure contains a measurement lag, a voltage regulator (proportional with lag, or PI), a slope feedback, a thyristor-susceptance block with delay T_d and lag T_b , and susceptance limits. Some models also include a slow susceptance regulator that drifts the operating point back towards a preset susceptance after a disturbance, keeping the fast range in reserve for the next event.

Supplementary signals improve damping. Candidate inputs include line current, line active or reactive power, bus voltage angle, bus frequency, and the speed or accelerating power of a nearby generator. Line power and locally synthesized frequency are preferred because they are available at the SVC site and are sensitive to inter-area modes. The supplementary controller uses a washout filter, lead-lag compensation for the phase of

the target mode, and limits; its design is developed in Chapter 11. A well-known side effect deserves mention here. An SVC regulating voltage at a generator bus near a series-compensated line can interact with torsional modes, and the same supplementary channel, designed with the torsional frequency in mind, can add subsynchronous damping.

Example 6.8: Voltage regulator design against ESCR.

An SVC regulator is designed for $ESCR_{min}=2$ with $K_{SL}=0.05$ pu at 50 Hz. Find (a) $K_{N,max}$; (b) K_C ; (c) $G_R(s)$; (d) T_ω , t_{95} and ΔV for a 0.05 pu reference step at $ESCR = 5$; (e) the same at $ESCR = 2$; (f) the engineering conclusion.

Solution. (a) $K_{N,max}=1/2=0.5$ pu.

(b) $K_C=1/[2(0.05+0.5)]=0.909$ pu.

(c) $T_b=T/4=5$ ms; $G_R(s)=0.909(1+0.005s)/(0.005s)$.

(d) $K_N=0.2$; $T_\omega=2(5)(0.55/0.25)=22$ ms; $t_{95}=66$ ms; $\Delta V=0.2 \times 0.05/0.25=0.040$ pu.

(e) $T_\omega=10$ ms; $t_{95}=30$ ms; $\Delta V=0.0455$ pu.

(f) The response slows by a factor of 2.2 in the strong network. Gain scheduling or a gain supervisor is needed if fast response is required at all strengths.

Note. Regulator tuning cannot be separated from network strength. The same controller gains can produce quite different closed-loop behaviour as the Thevenin reactance changes, which is why weak-grid and contingency conditions must be included in an SVC control study.

6.5.8 Midpoint SVC on a long transmission line

For long lines the lumped-reactance result of Section 6.1.2 must be replaced by the distributed-parameter solution. A lossless line of surge impedance Z_c , phase constant β and length l has electrical length $\theta=\beta l$. With equal end voltages V and power angle δ , the uncompensated power and the open-circuit midpoint voltage are

$$P_u = \frac{V^2}{Z_c \sin \theta} \sin \delta = \frac{P_0}{\sin \theta} \sin \delta, V_{m0} = V \frac{\cos(\delta/2)}{\cos(\theta/2)} \quad (6.48)$$

where $P_0=V^2/Z_c$ is the surge impedance loading. With the midpoint held at V , each half behaves as a line of electrical length $\theta/2$ carrying angle $\delta/2$:

$$P_c = \frac{P_0}{\sin(\theta/2)} \sin \frac{\delta}{2}, \frac{P_{c,max}}{P_{u,max}} = \frac{\sin \theta}{\sin(\theta/2)} = 2 \cos \frac{\theta}{2} \quad (6.49)$$

For short lines $\cos(\theta/2) \rightarrow 1$ and the improvement approaches the factor of two found earlier; for a 500 km line at 50 Hz ($\theta=30^\circ$) it is 1.93. The reactive power the midpoint device must supply follows from the receiving-end reactive power of each half-line:

$$Q_{dev} = \frac{2P_0[\cos(\theta/2) - \cos(\delta/2)]}{\sin(\theta/2)} \quad (6.50)$$

which is negative (absorption) for $\delta < \theta$, when line charging exceeds the series reactive loss, and positive (injection) beyond. The SVC must therefore be able to absorb on light load and inject on heavy load, which is exactly the duty of the TSC-TCR. Seen from the midpoint, the network has the Thevenin reactance $X_{th}=(Z_c/2)\tan(\theta/2)$, the parallel combination of the two half-lines; this value is used in Equation (6.36) when the SVC has a finite slope (Example 6.30).

6.5.9 Load balancing and power-factor correction

An SVC with independently controlled phases can turn an unbalanced load into a balanced, unity-power-factor load on the supply. Let the load be represented by delta-connected admittances $Y_{ab}^L = G_{ab} + jB_{ab}^L$, and similarly for the other branches, with the convention that positive susceptance is capacitive. For a positive-sequence supply, the negative-sequence component of the line currents is proportional to $a^2 Y_{ab} + Y_{bc} + a Y_{ca}$, where $a = e^{j2\pi/3}$ and Y denotes the total (load plus compensator) admittance of each branch. Requiring this sum to vanish, and the total susceptance of each branch to be zero for unity power factor, gives the compensator susceptances (positive capacitive):

$$B_{ab}^c = -B_{ab}^L + \frac{G_{ca} - G_{bc}}{\sqrt{3}}, B_{bc}^c = -B_{bc}^L + \frac{G_{ab} - G_{ca}}{\sqrt{3}}, B_{ca}^c = -B_{ca}^L + \frac{G_{bc} - G_{ab}}{\sqrt{3}} \quad (6.51)$$

For a single-phase load $G + jB^L$ connected between a and b this reduces to the Steinmetz circuit: a capacitive branch $G/\sqrt{3}$ across b - c , an inductive branch $G/\sqrt{3}$ across c - a , and a branch $-B^L$ across a - b that cancels the load's own reactive power. The supply then sees a balanced resistive load drawing $P/(\sqrt{3}V_{LL})$ in each line. A TCR-FC per phase realizes these susceptances dynamically. This is the principle of traction and arc-furnace balancing compensators [18] (Example 6.31).

6.5.10 Industrial SVCs and field experience

Arc furnaces draw erratic, unbalanced reactive power that produces visible flicker at frequencies of a few hertz. Flicker reduction requires the compensator to track the furnace reactive power within roughly a half-cycle, phase by phase. FC-TCRs with open-loop (feedforward) control computed from the measured load current achieve flicker reduction factors of about two. Converter-based compensators, whose bandwidth is much higher, can do substantially better.

Table 6.5 lists representative installations. Ratings are as published by the suppliers; the Devers data are from the commissioning report [40].

Table 6.5. Selected SVC and TSC installations.

Installation	Year	Voltage	Range	Configuration
Radsted, Denmark	2007	132 kV	0 to – 65 Mvar	2 TCRs, 12-pulse
Greenbank and South Pine, Australia	2008	275 kV	+250 to – 100 Mvar (one TCR at South Pine)	1 TCR, 2 TSCs, 3 filters
Devers, California, USA	2006	525 kV, 60 Hz	SVC +330 to – 110 Mvar; with a 110 Mvar MSC the system spans +440 to – 110 Mvar	2 TCRs, 3 TSCs, 2 filters
Strathmore, Australia	2007	Transmission	+150 to – 75 Mvar	1 TCR, 1 TSC, 3 filters
Bom Jesus da Lapa, Brazil	2002	Transmission	+250 Mvar	1 TSC
Gerdau steel plant, Charlotte, USA	n.a.	13.2 kV	0 to 64 Mvar	TSC for arc furnace

6.6 The Static Synchronous Compensator (STATCOM)

6.6.1 Principle: a synchronous condenser without rotating mass

The devices studied so far vary an impedance. The STATCOM generates reactive power directly, without AC capacitors or reactors acting as reactive energy stores. A voltage-source converter (VSC), supplied from a DC capacitor, synthesizes a set of three-phase voltages V_{ca}, V_{cb}, V_{cc} in phase with the bus voltages and couples them to the bus through a small reactance, usually the leakage reactance of the coupling transformer (Figure 6.36). The analogy with a synchronous condenser is exact at fundamental frequency. Raising the excitation of a condenser makes its internal EMF exceed the bus voltage and it delivers capacitive current; lowering the excitation makes it absorb reactive power. The converter does the same by adjusting the magnitude of its output voltage, electronically and within a few milliseconds.

two-level three-phase VSC (switch = IGBT/IGCT with antiparallel diode)

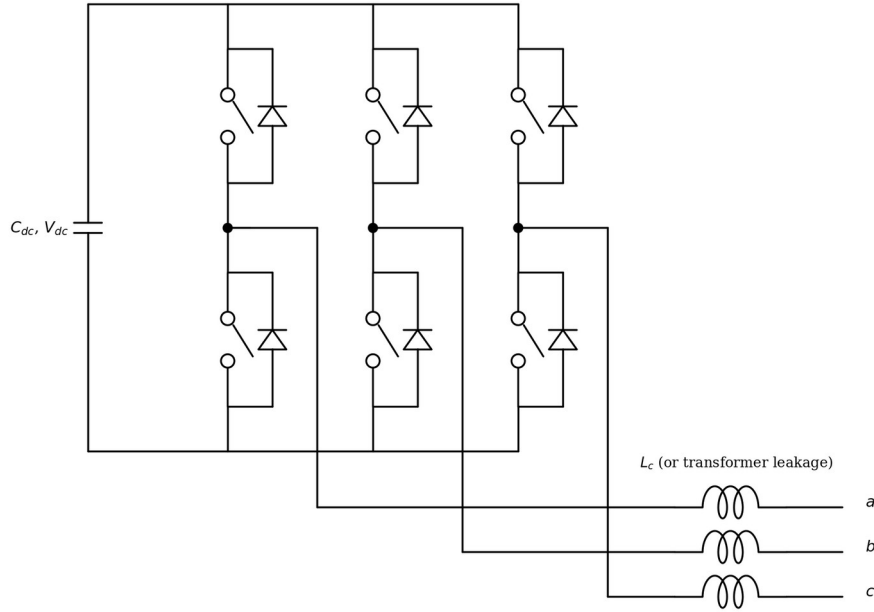


Figure 6.36. Two-level three-phase VSC with DC capacitor and coupling reactance, the basic STATCOM power circuit.

The terminology deserves precision. A *static synchronous generator* (SSG) is a converter that can exchange both active and reactive power because it has an energy source or store on its DC side. A STATCOM is an SSG operated without such a source: it exchanges reactive power only, apart from the small active power that covers its losses [22]. The voltage-source form is preferred over the current-source form for four reasons. Available turn-off devices (IGBTs, IGCTs) do not block reverse voltage well, which a current-source converter needs. Terminating a DC link in a charged capacitor is far less lossy than terminating it in a current-carrying reactor. A VSC needs only series inductance at its AC terminals, which the transformer provides, whereas a current-source converter needs AC filter capacitors. And the DC capacitor clamps the device voltages, giving inherent protection against line transients.

6.6.2 Current and power exchange

Refer the converter and bus phasors to the same side of the transformer and take the current positive from converter to bus. With $V = V \angle 0$, $V_c = V_c \angle \delta_c$ and a lossless coupling reactance X ,

$$I = \frac{V_c \angle \delta_c - V}{jX} \quad (6.52)$$

where I is the current phasor, V and V_c the RMS bus and converter phase voltages and δ_c the angle by which the converter voltage leads the bus voltage. The complex power delivered to the bus, $S = V I^*$, gives

$$P = \frac{V V_c}{X} \sin \delta_c, Q = \frac{V(V_c \cos \delta_c - V)}{X} \quad (6.53)$$

where P and Q are the active and reactive power delivered to the bus, per phase in SI units or three-phase in per unit. For $\delta_c = 0$ the exchange is purely reactive:

$$I_Q = \frac{V_c - V}{X}, Q = \frac{V(V_c - V)}{X} = \left(\frac{V_c}{V} - 1 \right) \frac{V^2}{X} \quad (6.54)$$

where I_Q is the reactive current, positive for capacitive injection. If $V_c > V$ the converter supplies capacitive vars and the current it delivers lags the bus voltage by 90° ; seen from the bus it draws a leading current, exactly like a capacitor. If $V_c < V$ it absorbs vars like a reactor (Figure 6.37). Because X is small, typically 0.1 to 0.15 pu, a change of only 10 to 15% in V_c swings the output across its full range. A small, deliberate phase shift δ_c makes the converter absorb active power from the bus ($\delta_c < 0$), which charges the DC capacitor and covers the converter losses.

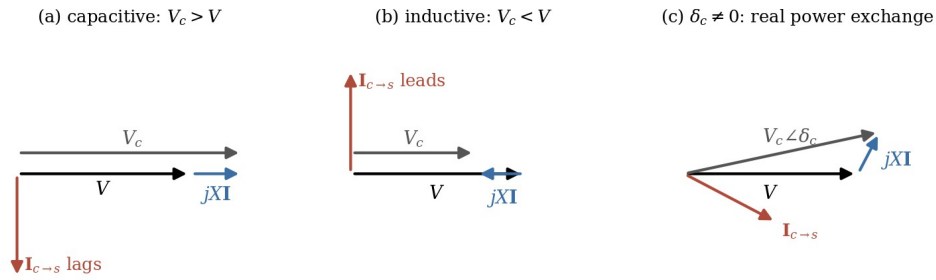


Figure 6.37. STATCOM phasor diagrams: (a) capacitive operation, (b) inductive operation, (c) phase-shifted operation with active power exchange.

Example 6.9: STATCOM reactive exchange.

A STATCOM is connected to a 132 kV bus through $X=0.15$ pu on 100 MVA. Find (a) I_Q and Q for $V_c=1.08$ pu, $V=1.0$ pu; (b) the current in amperes; (c) the mode of operation; (d) the unconstrained response if V falls to 0.92 pu with V_c unchanged; (e) the actual response with $I_{max}=1.0$ pu; (f) the converter voltage for 0.333 pu inductive; (g) the angle that supplies 1% losses.

Solution. (a) Equation (6.54): $I_Q = 0.08/0.15 = 0.533$ pu; $Q = 0.533$ pu = 53.3 Mvar.

(b) $I_B = 100/(\sqrt{3} \times 132) = 437.4$ A; $I = 233.3$ A.

(c) $V_c > V$: capacitive. Writing $Q = -53.3$ Mvar from $(1 - V_c/V)$ gives the power absorbed by the converter, not the power delivered to the bus.

(d) $I_Q = 1.067$ pu, $Q = 0.981$ pu.

(e) The limiter holds 1.0 pu: $V_c = 0.92 + 0.15 = 1.07$ pu, $Q = 0.92$ pu = 92 Mvar.

(f) $V_c = 1 - 0.333 \times 0.15 = 0.95$ pu.

(g) $\sin \delta_c = 0.01 \times 0.15/1.08$, so $\delta_c = -0.080^\circ$ (converter lagging, absorbing).

Note. The sign of reactive power follows the voltage difference across the coupling reactance. Before substituting numbers, state the current direction and the sign convention for injected reactive power. This prevents the most common STATCOM phasor-model error.

6.6.3 Instantaneous power balance and the role of the DC capacitor

The instantaneous power at the AC terminals is $p = v_{ca}i_a + v_{cb}i_b + v_{cc}i_c$. For balanced sinusoidal voltages and balanced currents in quadrature with them, this sum is zero at every instant. The reactive power is not stored anywhere; it circulates among the three phases through the converter switches. The DC capacitor therefore does not supply the reactive power. It sets the DC voltage from which the AC voltage is synthesized, and it absorbs only the ripple caused by switching harmonics, by unbalance, and by transient real-power exchange. This is why a STATCOM rated at hundreds of megavars needs a DC capacitor whose stored energy corresponds to only a few milliseconds of rated output (Section 6.6.7), and why the same converter MVA provides a symmetric dynamic range: a ± 100 Mvar STATCOM spans 200 Mvar of control with 100 MVA of converter.

6.6.4 V-I characteristic and operating area

Under voltage control the STATCOM follows the same droop law as the SVC, Equation (6.34), within its range. At the limits the difference appears (Figure 6.38). The converter current is bounded by device ratings, not by a susceptance, so the limits are *vertical* lines on the V-I plane: the STATCOM can deliver its full capacitive current down to very low bus voltages, limited eventually by its ability to stay synchronized and keep its DC voltage. An SVC at its limit is a fixed susceptance whose current falls in proportion to voltage.

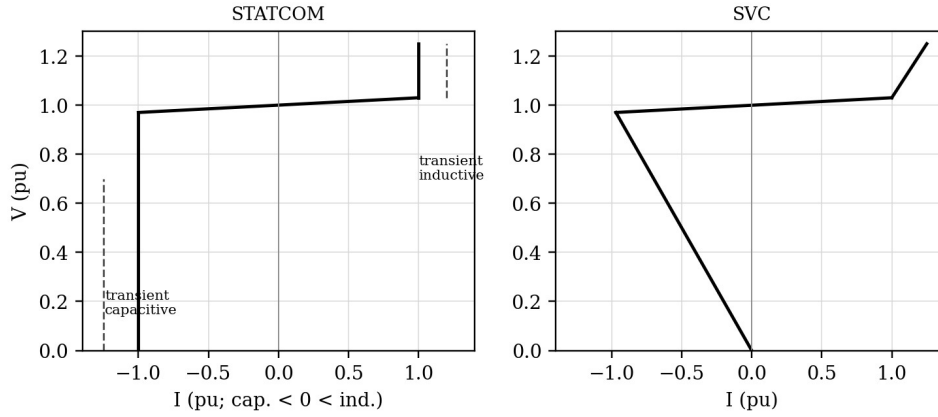


Figure 6.38. V-I characteristics: STATCOM with constant-current limits and transient ratings (left); SVC with susceptance limits (right).

Transient ratings differ on the two sides. The transient capacitive rating is set by the maximum current the switches can turn off. The transient inductive rating is set by the permissible junction temperature, since in inductive operation the antiparallel diodes carry much of the current and turn-off is not the binding limit. At a bus voltage V the current limit implies the reactive output limits

$$Q_{cap,max} = V I_{max}, Q_{SVC,max} = B_{C,max} V^2 \quad (6.55)$$

where I_{max} is the converter current limit. Figure 6.40 compares the two laws on equal nominal ratings. At $V = 0.7$ pu the STATCOM supplies 70% of rated vars and the SVC 49%; at 0.5 pu the ratio is two to one. The comparison is made at the respective capacitive limits. It does not mean that a STATCOM delivers constant Mvar during a voltage collapse: its current stays at the limit, so its output falls in proportion to V , while the SVC output falls with V^2 . The current limit also implies a circular P-Q capability, $P^2 + Q^2 \leq (V I_{max})^2$, but sustained operation away from the Q axis requires an energy source: a capacitor-only STATCOM lives on the narrow locus $P \approx -P_{loss}$. Figure 6.39 shows how the circle contracts with voltage. When a battery or supercapacitor is added, active and reactive services compete for the same semiconductor current, so a controller needs a priority rule during disturbances: voltage-support priority, active-power priority or a weighted allocation (Section 6.7.3).

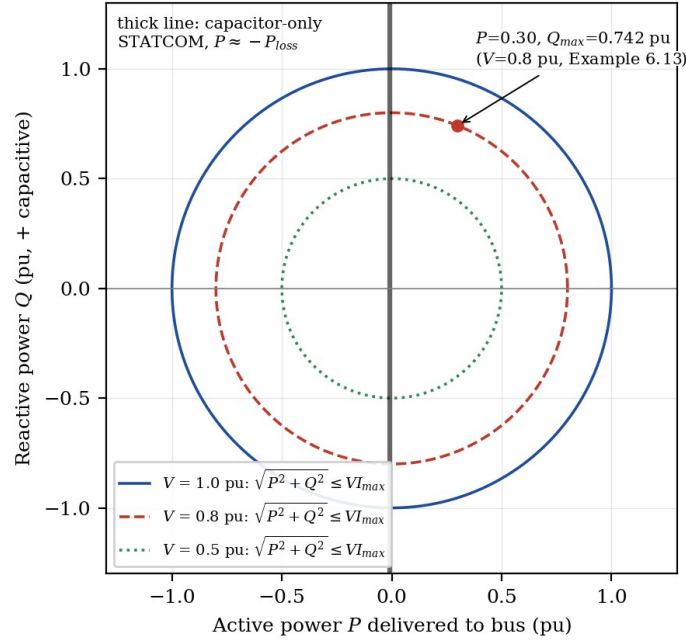


Figure 6.39. Current-limited P-Q capability of a VSC-based STATCOM at three bus voltages. The thick vertical line is the operating locus of a capacitor-only STATCOM.

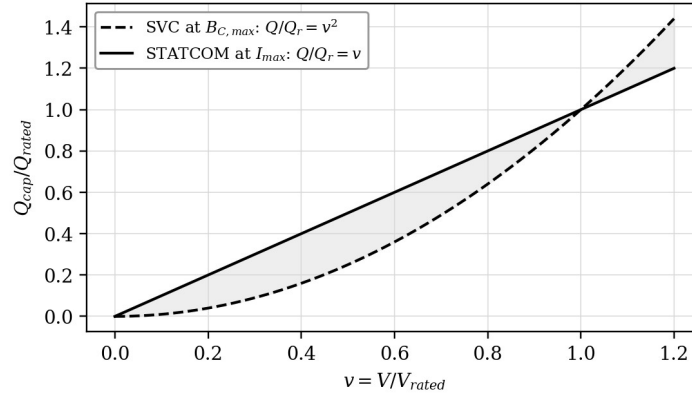


Figure 6.40. Maximum capacitive output versus bus voltage for SVC ($\propto V^2$) and STATCOM ($\propto V$) of equal rating.

6.6.5 Converter configurations

Two-level six-pulse converter

Operated with one switching per cycle (square-wave or six-step operation), each pole of the two-level converter connects its phase alternately to the positive and negative DC rails for 180° . The resulting phase-to-neutral voltage of the three-phase bridge is the six-step wave, whose Fourier series is

$$v_{cn}(t) = \frac{2V_{dc}}{\pi} \left(\sin \omega t + \frac{1}{5} \sin 5\omega t + \frac{1}{7} \sin 7\omega t + \frac{1}{11} \sin 11\omega t + \dots \right) \quad (6.56)$$

where V_{dc} is the DC voltage and the harmonics present are $n=6k \pm 1$ with amplitudes V_1/n . The fundamental is fixed by V_{dc} , so in six-step operation the AC voltage, and with it the reactive output, can only be changed by changing V_{dc} . That is the indirect control method of Section 6.6.6.

Multi-pulse converters

Combining $p/6$ six-pulse converters whose outputs are displaced by $360^\circ/p$ through phase-shifting transformers cancels all harmonics except $n=kp \pm 1$, with amplitudes still V_1/n . The voltage THD is therefore

$$THD_v = \sqrt{\sum_{k=1}^{\infty} \left[\frac{1}{(kp-1)^2} + \frac{1}{(kp+1)^2} \right]} \quad (6.57)$$

which evaluates to 31.1% for six pulses, 15.2% for 12, 7.6% for 24 and 3.8% for 48 (Figure 6.41). Each doubling of the pulse number approximately halves the THD. Truncating the series after the first two pairs underestimates THD by more than 10%, a common error. Harmonic *currents* are much smaller than harmonic voltages, because the coupling reactance X presents nX to the n th harmonic: $I_n = V_n/(nX) = V_1/(n^2 X)$. The first transmission STATCOMs, such as the TVA Sullivan unit, used 48-pulse converters built from eight three-phase bridges and an elaborate zig-zag transformer system.

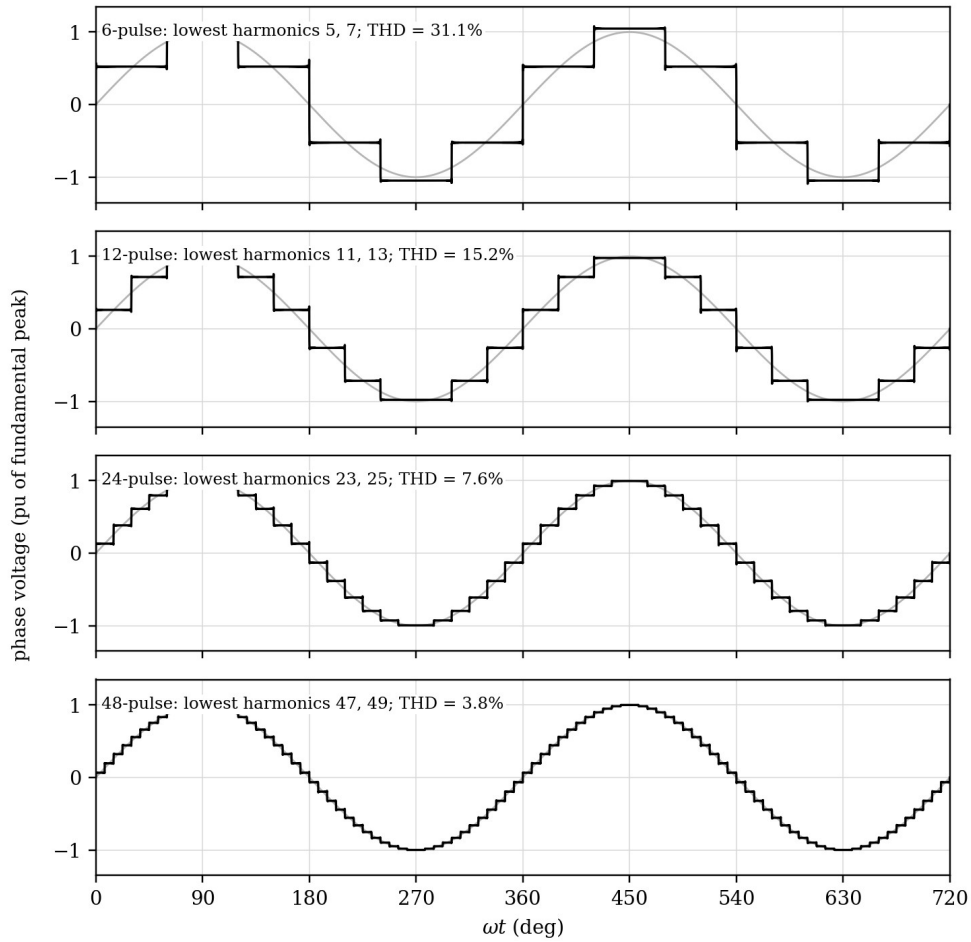


Figure 6.41. Phase voltage of 6-, 12-, 24- and 48-pulse converters with fundamental-frequency switching.

Three-level, PWM and multilevel converters

A three-level neutral-point-clamped (NPC) converter adds a zero level to each pole voltage. Holding the zero level for an interval 2β around each zero crossing gives the quasi-square wave of Figure 6.42, whose fundamental amplitude

$$V_1 = \frac{4}{\pi} \cdot \frac{V_{dc}}{2} \cos \beta \quad (6.58)$$

can be varied through β at constant DC voltage. This makes direct control possible without PWM. Pulse-width modulation achieves the same with a two-level bridge: the fundamental equals $m_a V_{dc}/2$ per pole in the linear range ($m_a \leq 1$), and harmonics are pushed to sidebands of the carrier frequency, at the cost of switching losses.

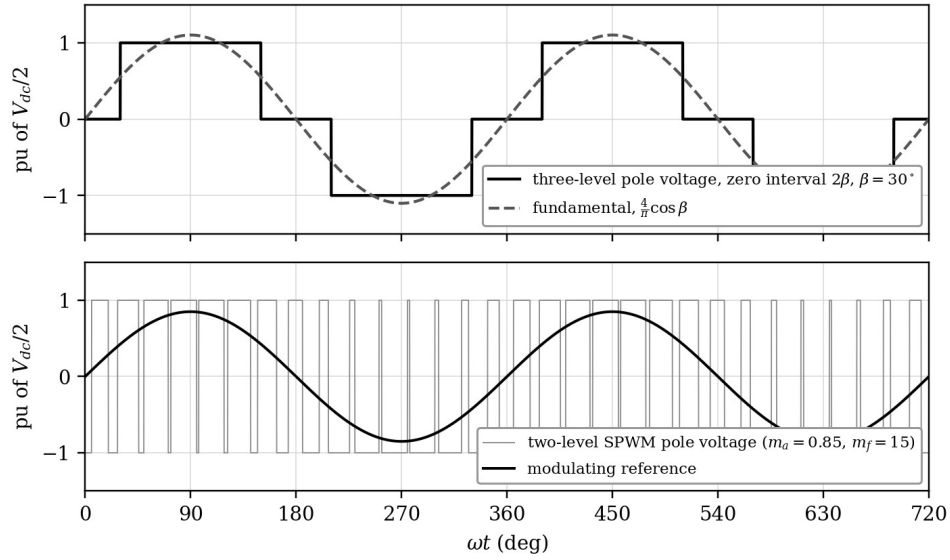


Figure 6.42. Three-level pole voltage with zero interval 2β (top) and two-level sinusoidal PWM (bottom).

Cascaded H-bridge (CHB) and modular multilevel converters (MMC), developed in Chapter 5, now dominate new transmission STATCOMs. Each phase is a chain of identical cells, each with its own floating capacitor, and the output is a fine staircase that needs little or no filtering at switching frequencies of only a few hundred hertz per cell. Because a STATCOM has no common DC link with an external circuit, Akagi's classification [14] distinguishes the single-star bridge-cell (SSBC) and single-delta bridge-cell (SDBC) arrangements. The delta-connected SDBC is especially attractive for STATCOMs serving unbalanced loads, because the delta provides a path for the zero-sequence circulating current that balances the cluster energies when negative-sequence current is supplied (Section 6.7.2). For power-system studies the familiar voltage source behind a coupling reactance remains the right external model, but the cell capacitor voltages and the cluster energies are additional internal states. They are invisible in the phasor diagram and become important under unbalanced voltage and negative-sequence current.

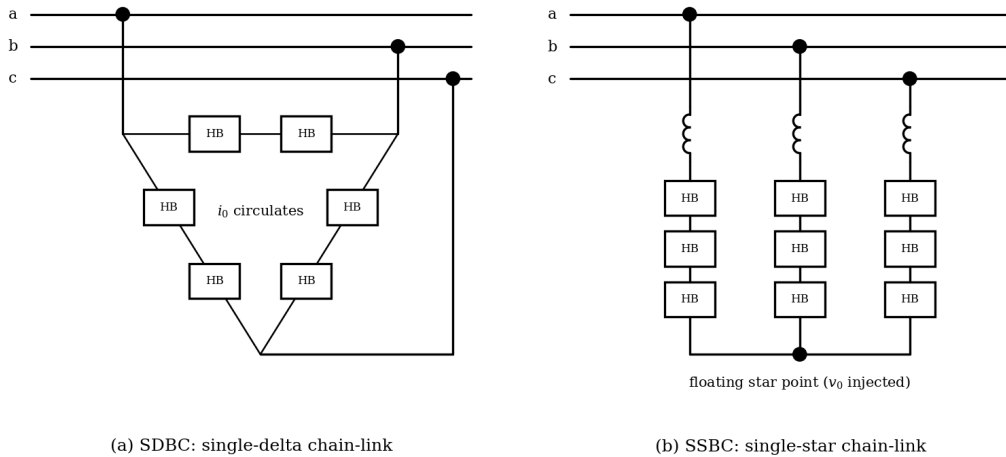


Figure 6.43. Chain-link STATCOM clusters without a common DC link: (a) single-delta bridge-cell (SDBC), (b) single-star bridge-cell (SSBC). HB denotes an H-bridge cell with its own capacitor; cluster inductors are omitted in (a).

Table 6.6. STATCOM converter topologies.

Topology	Switching	Lowest harmonics	Control of V_c	Remarks
Two-level, six-step	Fundamental	5, 7	Indirect (V_{dc})	Needs heavy filtering; basis of multi-pulse
Multi-pulse (12 to 48)	Fundamental	$k p \pm 1$	Indirect	Complex magnetics; early transmission STATCOMs
Three-level NPC	Fundamental or PWM	Depends on β or carrier	Direct (β or m_a)	Neutral-point balancing needed
Two-level PWM	Carrier (1 to 3 kHz)	Carrier sidebands	Direct (m_a)	Low-voltage and distribution STATCOMs
CHB / MMC	Low per cell, high effective	Very high	Direct	Modular, scalable, redundant cells; cell balancing control

Example 6.10: Multi-pulse converter harmonics.

For fundamental-frequency-switched converters with $V_n = V_1/n$ and coupling reactance 0.2 pu, find for 6, 12, 24 and 48 pulses (a) the lowest harmonics; (b) the exact voltage THD; (c) the THD if only two harmonic pairs are retained; (d) the lowest-order current harmonics as % of rated current; (e) current THD at rated current.

Solution. (a) 5, 7; 11, 13; 23, 25; 47, 49.

(b) Equation (6.57): 31.08%, 15.22%, 7.57%, 3.78%.

(c) 27.31%, 13.29%, 6.60%, 3.30%; truncation underestimates by about 12%.

(d) $I_n = 1/(n^2 X)$: 20.0% and 10.2%; 4.13% and 2.96%; 0.95% and 0.80%; 0.23% and 0.21%.

(e) 23.2%, 5.28%, 1.29%, 0.32%.

Note. Multi-pulse conversion reduces characteristic low-order harmonics by phase-shift cancellation, but it requires transformer complexity. Modern MMC-based transmission STATCOMs achieve waveform quality through a different mechanism, using many voltage levels and distributed submodule capacitors.

6.6.6 Control of the STATCOM

Internal and external control

The control system has two layers. The *external* (system) control decides what the STATCOM should do: regulate bus voltage along a droop, hold a reactive reference, damp oscillations or balance phases. It produces a reactive current reference $I_{Q,ref}$. The *internal* control makes the converter produce that current. It synchronizes with the AC system through a PLL, computes the required magnitude and angle of the converter voltage, and generates the gating pattern for the switches while keeping devices within their voltage and current limits. With its internal control in place, the converter behaves to the outside world as a controllable sinusoidal voltage behind the coupling reactance.

Indirect control (Type 2 converter)

When the converter operates with fixed modulation, as in six-step and multi-pulse designs, $V_c = k V_{dc}$ with k constant. The only handle is the angle δ_c . To increase capacitive output, the controller momentarily makes δ_c negative. By Equation (6.53) the converter then absorbs active power, the DC capacitor charges, V_{dc} and hence V_c rise, and Q increases. Once the target is reached, δ_c returns to the small negative value that just covers losses. The reverse sequence lowers V_{dc} . Figure 6.44 simulates this behaviour. The response is limited by the energy that must flow into or out of the capacitor (Example 6.33), and it is typically a few cycles.

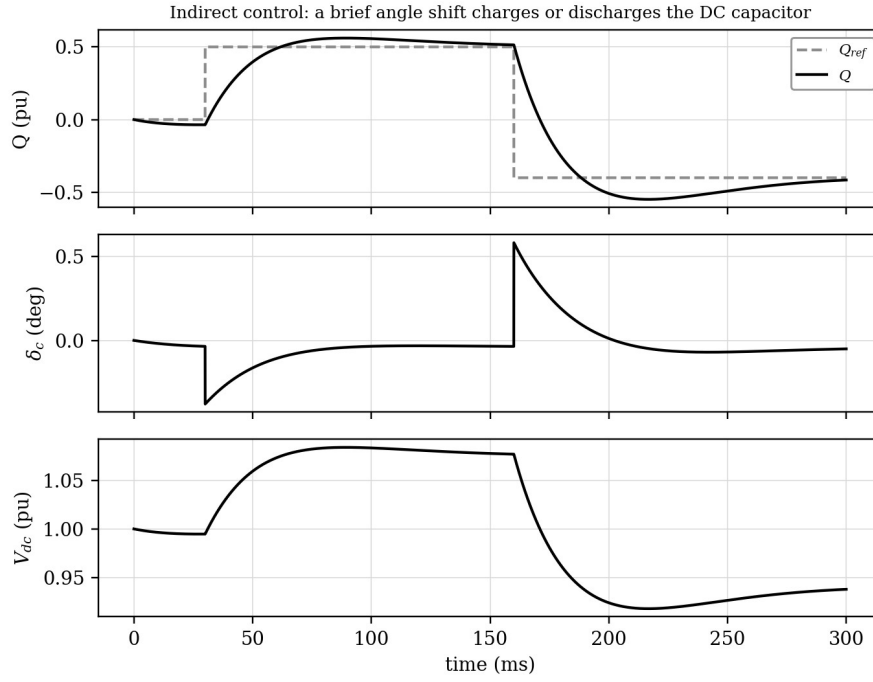


Figure 6.44. Indirect control: a reactive power step is achieved through a transient angle shift that charges or discharges the DC capacitor.

Direct control (Type 1 converter)

With PWM, three-level or multilevel converters, the magnitude of V_c can be varied directly at constant V_{dc} , through m_a or the zero interval β . A separate DC-voltage loop holds V_{dc} by adjusting δ_c , and therefore the small active current, while the reactive loop acts on the magnitude. Response is limited only by the current-control bandwidth, often within one cycle.

dq model and decoupled current control

In a synchronously rotating frame aligned with the positive-sequence bus voltage ($v_q=0$), with resistance R and inductance L of the coupling and the amplitude-invariant transformation, the converter current obeys

$$L \frac{di_d}{dt} = v_{cd} - v_d - Ri_d + \omega Li_q, L \frac{di_q}{dt} = v_{cq} - v_q - Ri_q - \omega Li_d \quad (6.59)$$

where i_d , i_q are the d and q components of the converter-to-bus current, v_{cd} , v_{cq} those of the converter voltage and v_d , v_q those of the bus voltage. The cross terms $\pm \omega Li$ couple the axes (Figure 6.45). The power delivered to the bus is

$$P = \frac{3}{2} (v_d i_d + v_q i_q), Q = \frac{3}{2} (v_q i_d - v_d i_q) \quad (6.60)$$

so with $v_q=0$, i_d controls active power and i_q reactive power, and capacitive injection corresponds to negative i_q .

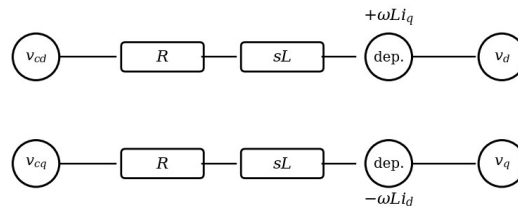


Figure 6.45. dq equivalent circuit of the STATCOM coupling, showing the speed-voltage coupling terms.

The decoupled controller (Figure 6.46) computes

$$v_{cd}^* = v_d + R i_d - \omega L i_q + u_d, v_{cq}^* = v_q + R i_q + \omega L i_d + u_q \quad (6.61)$$

where u_d and u_q are the outputs of PI regulators acting on the current errors $i_d^* - i_d$ and $i_q^* - i_q$. Substituting into Equation (6.59) leaves two independent first-order plants $L di/dt = u$, so each loop can be tuned separately. The reference i_q^* comes from the AC-voltage or droop regulator; i_d^* comes from a DC-voltage regulator, preferably acting on V_{dc}^2 , which is proportional to stored energy and makes the plant linear. A current limiter enforces $i_d^{*2} + i_q^{*2} \leq I_{max}^2$, normally giving priority to i_d^* so that DC voltage is never lost. Anti-windup, modulation-saturation handling and PLL stability in weak grids complete the design. The instantaneous power theory of Akagi and co-workers [15] provides the same decomposition in stationary coordinates and remains the usual basis for compensators that must also handle harmonics and unbalance.

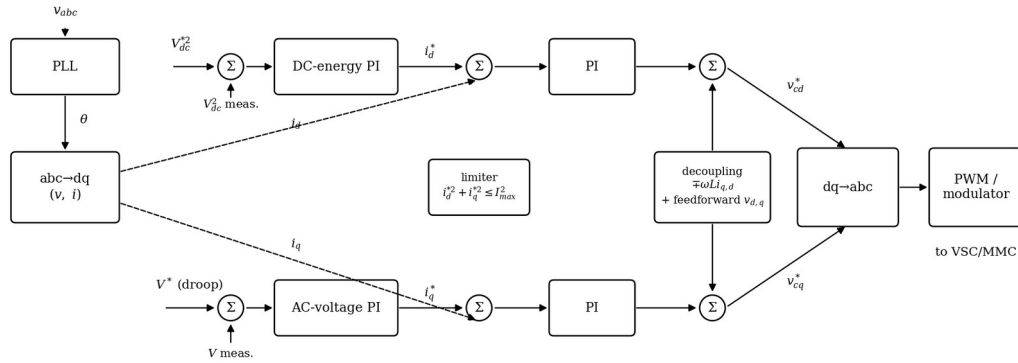


Figure 6.46. Decoupled dq control of a STATCOM with DC-energy and AC-voltage outer loops.

Example 6.11: dq quantities and converter voltage.

In per unit (amplitude-invariant dq, base $S_B = \frac{3}{2} V_{pk,B} I_{pk,B}$, so $P = v_d i_d$ and $Q = -v_d i_q$ with $v_q = 0$), a STATCOM has $v_d = 1.0$, $i_d = 0.02$, $i_q = -0.8$, $R = 0.005$ and $X = \omega L = 0.12$. The secondary is 6.6 kV line-to-line. Find (a) P and Q ; (b) the mode; (c) v_{cd} and v_{cq} in steady state; (d) $|v_c|$ and its angle; (e) the minimum V_{dc} for two-level SPWM; (f) with third-harmonic injection.

Solution. (a) $P = 0.02$ pu delivered to the bus and $Q = 0.8$ pu. (A STATCOM drawing its own losses from the bus would show a small negative i_d .)

(b) $i_q < 0$: capacitive.

(c) From Equation (6.59) in steady state: $v_{cd} = 1 + 0.005(0.02) - 0.12(-0.8) = 1.0961$; $v_{cq} = 0.005(-0.8) + 0.12(0.02) = -0.0016$.

(d) 1.0961 pu at -0.08° .

(e) $V_{pk} = 6.6 \sqrt{2}/\sqrt{3} = 5.389$ kV; converter peak 5.907 kV; $V_{dc} \geq 2 \times 5.907 = 11.81$ kV.

(f) $V_{dc} \geq \sqrt{3} \times 5.907 = 10.23$ kV.

Note. The dq equations are most useful when each term is given a physical meaning. The cross-coupling terms arise from the rotating reference frame and should not be mistaken for additional physical inductors. Feedforward decoupling improves current-loop behaviour but does not remove converter limits.

6.6.7 Sizing the DC capacitor

The DC capacitor is sized from *real*-power imbalance and ripple, not from the reactive rating. The energy balance of the converter is

$$\frac{d}{dt} \left(\frac{1}{2} C V_{dc}^2 \right) = p_{ac} - P_{loss} \quad (6.62)$$

where C is the DC capacitance, p_{ac} the active power entering the converter from the AC side and P_{loss} the converter losses. Two duties determine C .

Transient deficit. If a net discharge power ΔP persists for Δt (a loss surge, a delayed DC-voltage loop, or a deliberate brief active-power exchange) while V_{dc} is allowed to fall from $V_{dc,max}$ to $V_{dc,min}$,

$$C = \frac{2 \Delta P \Delta t}{V_{dc,max}^2 - V_{dc,min}^2} \quad (6.63)$$

Unbalance ripple. With a negative-sequence bus voltage V^- and positive-sequence current I^+ , the instantaneous power contains a component at twice the line frequency of amplitude $\hat{P}_2 \approx 3 V^- I^+$ (RMS phase quantities), that is, $\hat{P}_2 \approx (V^-/V^+) S$ for a converter operating at apparent power S . Integrating it in Equation (6.62) gives a peak-to-peak ripple

$$\Delta V_{dc,pp} \approx \frac{\hat{P}_2}{\omega C V_{dc0}} \quad (6.64)$$

where V_{dc0} is the mean DC voltage. Figure 6.47 illustrates the ripple. The ripple in turn produces a third-harmonic voltage on the AC side unless the control compensates, which is one reason negative-sequence current control is used in STATCOMs serving unbalanced loads. A capacitor sized by dividing a “reactive energy” $\sqrt{2} Q / 2\omega$ by $V_{dc} \Delta V_{dc}$, overstates the requirement by one to two orders of magnitude, because balanced reactive power does not flow through the capacitor at all (Example 6.12).

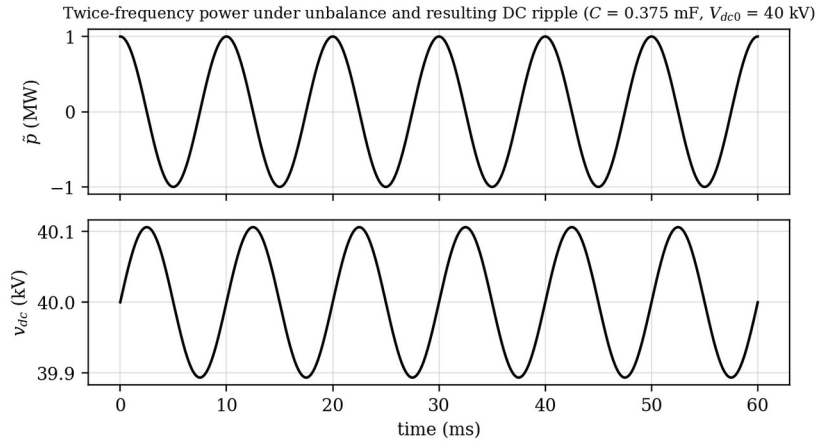


Figure 6.47. Twice-frequency power under unbalance and the resulting DC voltage ripple.

For an MMC STATCOM the aggregate result is necessary but not sufficient. In a chain-link cluster each cell capacitor carries a second-harmonic energy ripple even at purely reactive output; the arms of a double-star MMC also carry a fundamental-frequency component. Cell capacitance is therefore sized from cluster or arm energy ripple (Chapter 5 and Section 6.7.2).

Example 6.12: DC capacitor sizing.

A 20 Mvar STATCOM has a nominal DC voltage of 20 kV. Find (a) C for a deficit of 2 MW for 10 ms within 21 to 19 kV; (b) the stored energy and its equivalent in milliseconds of rated output; (c) the ripple caused by 5% negative-sequence voltage; (d) the capacitance needed for ripple alone within the same window; (e) why a “reactive energy” method gives a far larger value.

Solution. (a) Equation (6.63): $C = 2(2 \times 10^6)(0.01)/(21^2 - 19^2) \times 10^{-6} = 0.50 \text{ mF}$.

(b) $W = \frac{1}{2} C V_{dc}^2 = 100 \text{ kJ}$, equivalent to 5 ms of rated output.

(c) $\hat{P}_2 \approx 0.05 \times 20 = 1 \text{ MW}$. Equation (6.64): $\Delta V_{pp} = 10^6 / (314.16 \times 0.5 \times 10^{-3} \times 20000) = 318 \text{ V}$, or 1.6%.

(d) $C \geq 2 \hat{P}_2 / [\omega (V_{max}^2 - V_{min}^2)] = 0.080 \text{ mF}$; the transient deficit governs.

(e) Balanced reactive power never enters the capacitor (Section 6.6.3). Treating $\sqrt{2}Q/(2\omega) = 45 \text{ kJ}$ as energy that must cycle through the capacitor gives tens to hundreds of times the capacitance found in (a), depending on the ripple band assumed.

Note. A STATCOM DC capacitor is not sized by multiplying reactive power by time. In balanced steady-state reactive operation the average real power into the DC link is approximately the converter loss. Energy sizing follows real-power imbalance, voltage ripple, modulation headroom and, for MMCs, internal arm and submodule energy requirements.

6.6.8 Hybrid STATCOMs and loss characteristics

The losses of a STATCOM are roughly symmetric about zero output and, at zero output, are higher than those of a TSC-TCR (Figure 6.31), because the converter switches continuously. Where the duty is asymmetric, a smaller converter can be combined with passive elements (Figure 6.48). A fixed capacitor in parallel shifts the range towards capacitive output: a $\pm 50 \text{ Mvar}$ STATCOM with a 30 Mvar capacitor spans $+80$ to -20 Mvar at rated voltage. A fixed reactor shifts it towards inductive output. A STATCOM combined with TSCs and a TCR provides a wide range with a small converter rating. The price is that the passive part re-introduces the V^2 characteristic, so the low-voltage support of the hybrid lies between that of a pure STATCOM and a pure SVC. Commercial “hybrid SVC” and “SVC with STATCOM” designs use exactly this trade-off [37]–[39].

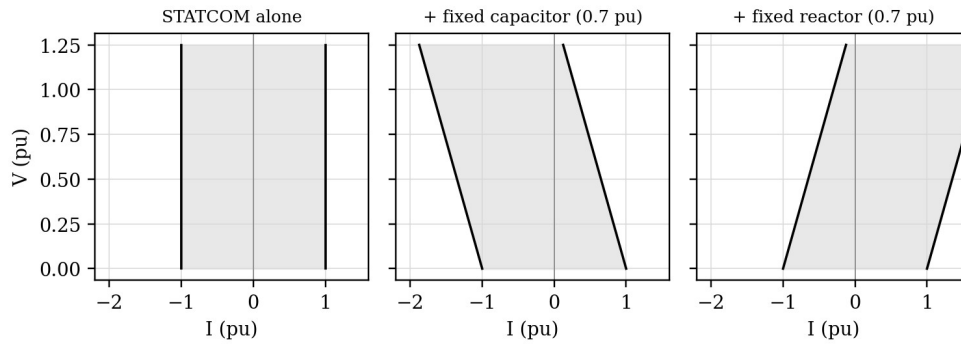


Figure 6.48. V-I operating areas of a STATCOM alone and hybridized with a fixed capacitor or fixed reactor.

6.6.9 Midpoint STATCOM with a current limit

Return to the two-machine system of Section 6.1.2 and let the midpoint compensator be a current source of fixed capacitive current I_Q , as a STATCOM is once it reaches its limit. The Thevenin equivalent at the midpoint is $V \cos(\delta/2)$ behind $X/4$ (the two half-lines in parallel), so the midpoint voltage becomes $V_m = V \cos(\delta/2) + I_Q X/4$. Substituting into the half-line power equation $P = (2V V_m / X) \sin(\delta/2)$,

$$P = \frac{V^2}{X} \sin \delta + \frac{V I_Q}{2} \sin \frac{\delta}{2} \quad (6.65)$$

where I_Q is the STATCOM's capacitive current. The STATCOM adds a $\sin(\delta/2)$ term to the uncompensated curve. By contrast, an SVC at its capacitive limit B_C is a fixed capacitor at the midpoint, and the same Thevenin argument gives

$$P = \frac{V^2}{X(1 - XB_C/4)} \sin \delta \quad (6.66)$$

which only scales the uncompensated curve and still peaks at $\delta = 90^\circ$. With equal ratings ($I_Q = B_C = 1$ pu) on a line of $X = 0.5$ pu, the STATCOM curve peaks at 2.36 pu near 95° and the SVC curve at 2.29 pu at 90° ; more importantly, the STATCOM curve remains higher at large angles, which enlarges the decelerating area available in a transient (Figure 6.3, Example 6.34).

6.6.10 Applications and control functions

Voltage regulation and support. The STATCOM's principal duty, executed through the droop law with a response limited by its current loop. Its capacitive output survives deep voltage sags far better than an SVC's.

Transient stability. During the first swing after a fault, temporarily raising V_{ref} , or simply forcing maximum capacitive current, maximizes the midpoint voltage and the decelerating area. The reference is then returned to normal.

Power oscillation damping. The reference voltage, or alternatively the reference current, is modulated in proportion to a signal derived from local frequency or line power, so that the STATCOM raises the transmitted power when the sending machine accelerates and lowers it when it decelerates. Current modulation avoids interference with the voltage regulator. With an energy store, active-power modulation gives much stronger damping, which is the motivation for E-STATCOMs (Section 6.7).

Var reserve control. A slow outer loop trims V_{ref} so that, in steady state, the STATCOM sits near zero output and the mechanically switched capacitors and reactors in the area carry the slowly varying base load. The full dynamic range then remains in reserve for disturbances.

Flicker and unbalance. Railway traction loads, which are large, single-phase and fluctuating, are a major application in Europe and Asia. STATCOMs with negative-sequence current control restore voltage balance and suppress flicker far more effectively than SVCs because of their bandwidth.

Case history: TVA Sullivan. The ± 100 MVA STATCOM commissioned in 1995 at the Sullivan substation of the Tennessee Valley Authority, designed and built by Westinghouse under EPRI sponsorship, used GTO-based 48-pulse converters with indirect control. It regulated the 161 kV bus, reducing the duty of transformer tap changers in the area, and its operating record established the credibility of the technology [6].

6.6.11 STATCOM versus SVC

Figure 6.49 compares typical dynamic responses, and Table 6.7 collects the comparison. Response times must always be specified with their metric: valve delay, current-loop rise time and full voltage recovery are different quantities.

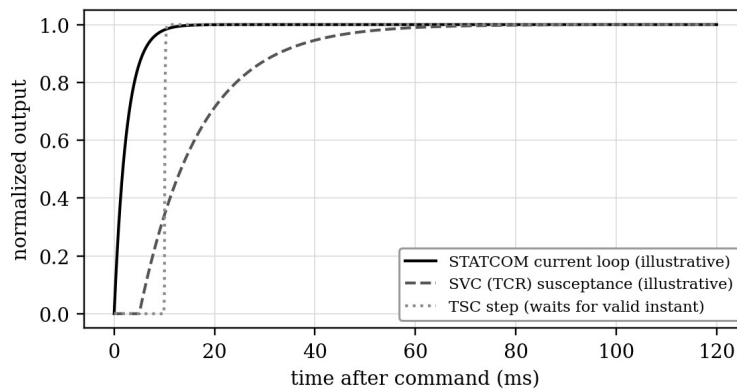


Figure 6.49. Illustrative response to a step in reactive reference: STATCOM current loop, TCR susceptance and TSC switching (stated assumptions; not a measured comparison).

Table 6.7. Comparison of SVC and STATCOM.

Attribute	SVC (TSC-TCR)	STATCOM (VSC/MMC)
Principle	Controlled susceptance	Controlled voltage source behind reactance
Output at limit	$Q \propto V^2$	$Q \propto V$ (constant current)
Low-voltage support	Collapses with voltage	Maintained to deep sags
Response	Valve delay up to a half-cycle plus regulator	Current loop within about one cycle
Harmonics	TCR harmonics, filters required	Low with multilevel/MMC; little filtering
Losses at zero output	Low (TSC-TCR), high (FC-TCR)	Moderate (switching losses)
Losses at full output	Moderate	Moderate to high
Footprint	Large (capacitors, reactors, filters)	Considerably smaller for the same dynamic range
Active power capability	None	Possible with DC energy storage
Overload	Limited by capacitor/reactor ratings	Limited by semiconductor ratings, short-time only
Cost (per Mvar)	Lower at high ratings	Higher, falling with MMC maturity

6.7 Recent Advances, Research Gaps and Emerging Directions

The TCR, TSC, SVC and STATCOM theory of this chapter is settled, and none of it is overturned by recent work. What has changed is the setting. Transmission systems now host large amounts of wind, solar and battery generation connected through converters, synchronous machines are being retired, and the short-circuit strength at many buses is falling. A modern shunt compensator is therefore asked to do more than hold a voltage: it may have to ride through faults in a weak grid, supply real power for a few seconds, behave as a voltage source rather than a current source, and avoid unstable interaction with neighbouring converters. This section shows, for a research student, which new states, constraints and equations enter the problem when the classical STATCOM model of Section 6.6 is extended in each of these directions. Each subsection ends with open questions; Table 6.9 at the end collects them.

6.7.1 Why the problem has changed: system strength

The strength of a bus is measured by its short-circuit ratio,

$$SCR = \frac{S_{sc}}{S_{rated}}, ESCR = \frac{S_{sc} - Q_C}{S_{rated}} \quad (6.67)$$

where S_{sc} is the three-phase short-circuit level at the bus, S_{rated} the rating of the converter-connected plant or compensator and Q_C the fixed shunt capacitance at the bus, which raises the apparent fault level without stiffening the voltage. Section 6.5.7 showed that the SVC voltage-loop gain is proportional to $1/ESCR$. For converter-interfaced plant the consequences go further. A grid-following converter synchronizes with a phase-locked loop (PLL) to a voltage that it also disturbs, and as SCR falls towards 1 to 2 the coupling between its own current and the measured angle can destabilize the PLL and the outer loops. The STATCOM is both a remedy and a participant: it supports voltage in a weak grid, but its own controller is subject to the same weak-grid interactions. This dual role drives most of the research below.

6.7.2 Modular multilevel STATCOMs: internal energy as a design variable

Section 6.6.5 introduced the chain-link (cascaded H-bridge) clusters used in modern STATCOMs, and Figure 6.43 showed the two arrangements without a common DC link. Each cluster is a series string of H-bridge cells with floating capacitors. A cluster is a single-phase element, so even at purely reactive output its instantaneous power pulsates. With cluster voltage $\sqrt{2}V \cos \omega t$ and current $\sqrt{2}I \sin \omega t$, the cluster power is $VI \sin 2\omega t$ and its energy swings by VI/ω peak to peak. If the cell voltages may ripple by a fraction ε peak to peak about their mean, the stored energy summed over the three clusters must satisfy

$$\frac{E_{\text{stored}}}{S} \geq \frac{1}{2\omega\epsilon} \quad (6.68)$$

where S is the converter rating. At 50 Hz and $\epsilon=10\%$ this lower bound is 15.9 kJ/MVA (Figure 6.50b). Installed values are typically about twice this once margins for negative-sequence duty, overload, cell bypass and control headroom are added. Capacitors dominate the volume and cost of a chain-link STATCOM, so reducing the energy requirement is an active research topic, for example by injecting harmonic or circulating components that reshape the energy ripple.

Cluster energy balancing under negative-sequence current. When a STATCOM supplies negative-sequence current I^- , for example to balance a traction or arc-furnace load (Section 6.5.9), the average power of each cluster is no longer zero, even though the total is. The three cluster capacitors would drift apart in voltage unless a zero-sequence component is added that redistributes power among clusters without reaching the grid. The two arrangements do this in different ways [14], [34], [35].

In the delta (SDBC) a zero-sequence current I_0 circulates inside the delta. With line-to-line cluster voltage phasor V_Δ and negative-sequence cluster current I_Δ^- , the three cluster powers balance when

$$I_0 = -\frac{V_\Delta I_\Delta^{*-}}{V_\Delta^*}, |I_0| = |I_\Delta^-| \quad (6.69)$$

an exact result with no singularity. The price is extra cluster current: the worst cluster carries up to $|I^+| + 2|I^-|$.

In the star (SSBC) the neutral floats, so a zero-sequence voltage V_0 is injected instead. The balance condition is

$$V_0 I^{+*} + V_0^* I^- = -V^+ I^{-*}, \frac{r}{1+r} \leq \frac{|V_0|}{V^+} \leq \frac{r}{1-r}, r = \frac{|I^-|}{|I^+|} \quad (6.70)$$

where the bound depends on the angle of I^- . As $r \rightarrow 1$ the required V_0 grows without limit, and when I^+ is small (a STATCOM near zero positive-sequence output asked to balance a load) the star cannot do the job at all. Figure 6.50a compares the two laws. This is the quantitative reason the delta arrangement is preferred for load-balancing STATCOMs, and why star arrangements suit balanced transmission duty.

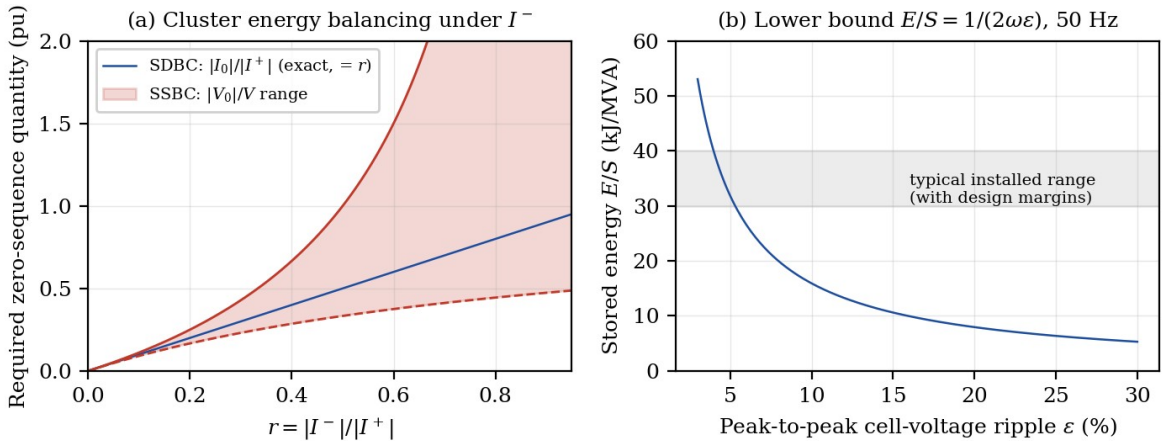


Figure 6.50. (a) Zero-sequence current (SDBC) and zero-sequence voltage (SSBC) needed to balance cluster energies when negative-sequence current is supplied; the SSBC band spans all angles of I^- . (b) Lower bound on stored energy per MVA of a chain-link STATCOM versus permitted cell-voltage ripple.

Open questions. Joint optimization of cell capacitance, losses and negative-sequence capability; energy-ripple shaping without extra semiconductor stress; fault-tolerant operation with bypassed cells while keeping cluster balance; lifetime models that couple capacitor and semiconductor ageing to the actual duty cycle.

6.7.3 Energy-storage STATCOMs: separating power and energy

Adding a battery, supercapacitor or other store to the DC side turns the STATCOM into a static synchronous generator in the sense of Section 6.6.1. Supercapacitor-equipped, grid-forming STATCOMs are now offered by several manufacturers [28], [37], [38], and Narula et al. showed the stability benefit of one at an offshore-wind connection [27]. CIGRE Technical Brochure 935 surveys the integration options, models and services [20]. Two constraints must be written separately, because they are set by different hardware:

$$P^2 + Q^2 \leq (V I_{max})^2, \left| \int_0^t P_{ES} d\tau \right| \leq E_{usable} \quad (6.71)$$

The first is the converter current limit already met in Section 6.6.4; at depressed voltage it shrinks the whole P-Q circle (Figure 6.39). The second is the energy store, where P_{ES} is the DC-side power (AC-side power plus losses) and E_{usable} the energy available inside the permitted state-of-charge or voltage window. For a supercapacitor allowed to fall from V_{max} to V_{min} , $E_{usable} = 1/2C(V_{max}^2 - V_{min}^2)$, so a voltage window of 2:1 uses 75% of the stored energy. A device can have a large converter and a small store (seconds of real power) or the reverse, and Figure 6.51a shows how quickly the duration falls as power rises.

Inertia-like and frequency support. If the converter injects power in proportion to the rate of change of frequency, it emulates a machine of inertia constant H on its rating S :

$$P_{in} = \frac{2HS}{f_0} \frac{df}{dt}, \Delta E = \frac{2HS}{f_0} \Delta f \quad (6.72)$$

where f_0 is the nominal frequency, Δf the frequency excursion over which support is given and ΔE the energy delivered over it. The power is set by the rate of change of frequency (RoCoF) and the energy by the depth of the excursion (Figure 6.51b). Note that ΔE is far smaller than the kinetic energy HS of the machine being emulated, because only the excursion is supported.

Damping. Section 6.6.10 noted that reactive-power modulation damps oscillations only through the voltage sensitivity of loads and line power. Modulating real power acts on the swing equation directly, so a small store gives much stronger damping per MVA. Beza and Bongiorno showed an adaptive controller that extracts the oscillatory component and commands both P and Q [16]; a fair comparison must charge the damping benefit against the converter current and energy it consumes.

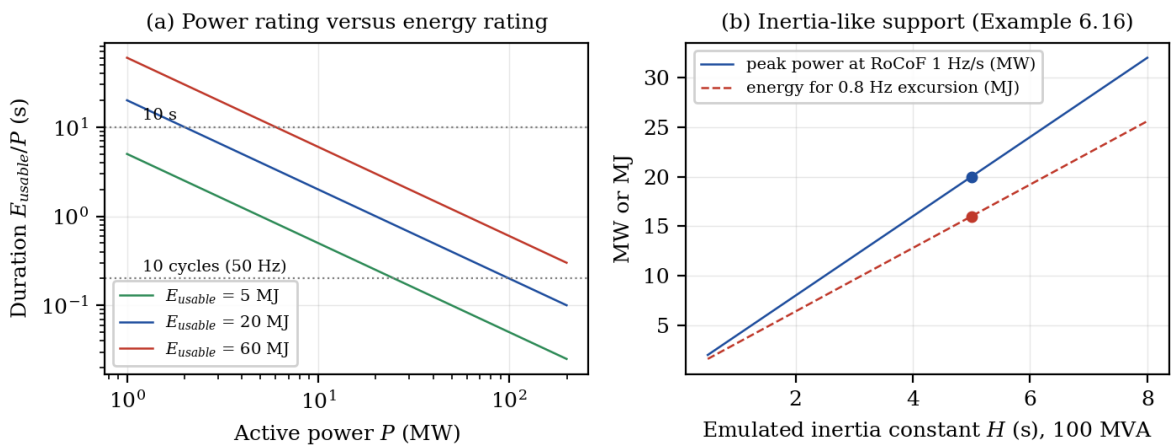


Figure 6.51. (a) Real-power duration available from a given usable energy. (b) Peak power and energy for inertia-like support from a 100 MVA energy-storage STATCOM.

Open questions. Unified P-Q priority rules during faults that respect both constraints of (6.71); state-of-charge management that keeps reserve for the next event; degradation-aware dispatch; and metrics such as damping ratio gained per MJ of storage, which allow a fair comparison with synchronous condensers.

6.7.4 Grid-forming STATCOMs and current limiting

Figure 6.52 contrasts the two control philosophies. A grid-following controller measures the grid angle with a PLL and controls current; it needs a voltage to follow. A grid-forming controller keeps its own internal voltage angle and magnitude as controller states and lets the current be whatever the network demands. Two common laws are droop control [32],

$$\omega = \omega_0 + m_p(P^* - P), E = E_0 + m_q(Q^* - Q) \quad (6.73)$$

and the virtual synchronous machine (VSM) [31], which adds inertia and damping:

$$2H \frac{d\omega}{dt} = P^* - P - D(\omega - \omega_0), \frac{d\theta}{dt} = \omega \quad (6.74)$$

in per unit, where m_p and m_q are droop gains, D a damping coefficient and θ the internal angle. Power-synchronization control [36] and virtual-oscillator control are other members of the family. A grid-forming converter synchronizes through the power it exchanges, $P = E V_g \sin \delta / X$, exactly as a synchronous machine does. It therefore stays stable in very weak grids where a PLL struggles, and it responds to a voltage or phase jump within the same cycle, because its terminal voltage is held by the control rather than built up through a current loop.

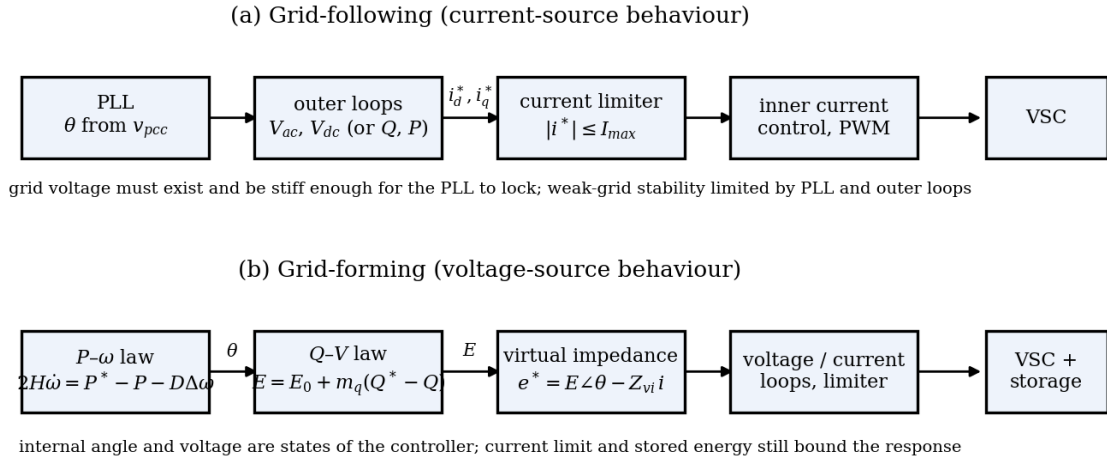


Figure 6.52. Grid-following and grid-forming STATCOM control structures. In both cases the current limiter and, for real-power services, the stored energy bound the response.

A capacitor-only STATCOM can be grid-forming for voltage and reactive power, but it cannot hold an internal frequency against the grid for long, because any sustained real-power exchange would discharge its DC link. Sustained frequency or inertia-like support therefore needs storage, which is why grid-forming and energy-storage STATCOMs are usually discussed together [20], [28].

Current limiting is the central difficulty. A voltage source behind a reactance X of 0.15 pu would deliver $(E - V_g)/X$, more than 6 pu for a bolted fault. Semiconductors allow perhaps 1.1 to 1.5 pu. Baeckeland *et al.* classify the remedies as direct limiters, which saturate the current reference and make the converter a current source while limited, and indirect limiters, which keep voltage-source behaviour by subtracting a virtual-impedance drop from the voltage reference [25], [26]:

$$e^* = E \angle \theta - Z_{vi} i, |Z_{vi} + jX| \geq \frac{E_0}{I_{max}} \quad (6.75)$$

The second relation sizes a *static* virtual impedance so that even a bolted terminal fault stays within I_{max} . Figure 6.53 shows the trade-off. Saturation uses the full current capability but abandons voltage-source behaviour. A static virtual impedance keeps it but under-uses the converter in shallow faults. A threshold virtual impedance, active only above a current threshold, sits between the two. The ratio X_{vi}/R_{vi} matters as well: a more inductive virtual impedance enlarges the transient-stability margin but reduces small-signal

damping. Wang, Xu and Li proposed a variable virtual impedance for a grid-forming energy-storage STATCOM that changes this ratio during transients [24].

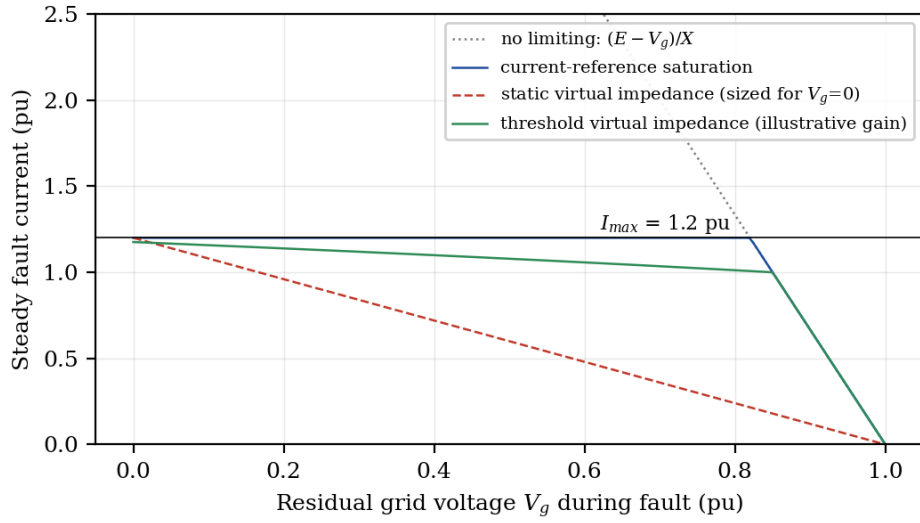


Figure 6.53. Steady fault current versus residual grid voltage for a grid-forming converter ($E = 1$ pu, $X = 0.15$ pu, $I_{max} = 1.2$ pu) under different current-limiting methods. The static virtual impedance is sized by Equation (6.75) with $X_{vi}/R_{vi} = 5$.

Open questions. Retaining voltage-source behaviour during unbalanced faults; avoiding windup of the internal angle when the limiter is active (a form of loss of synchronism); coordination of several grid-forming devices whose limiters act differently; and grid-code requirements, which are still being written [33].

6.7.5 Converter interaction and impedance-based stability

A STATCOM near a wind farm, an HVDC terminal or another FACTS device can interact with them at frequencies far above the electromechanical range. The standard tool is the impedance-based criterion [30]. Around an operating point, the STATCOM is represented by its small-signal output admittance $Y_c(s)$ and the network by its impedance $Z_g(s)$. The terminal voltage then obeys

$$V(s) = \frac{V_g(s)}{1 + Z_g(s)Y_c(s)} \quad (6.76)$$

so if both subsystems are stable on their own, the interconnection is stable when the minor-loop gain $Z_g(s)Y_c(s)$ satisfies the Nyquist criterion. A useful sufficient condition is passivity: a converter whose admittance satisfies $\text{Re}\{Y_c(j\omega)\} \geq 0$ at all frequencies cannot be destabilized by any passive network [29].

Digital control spoils passivity. For a current-controlled converter with filter inductance L , proportional gain K_p , voltage feedforward through a filter $H_{ff}(s)$ and a computation-plus-modulation delay T_d ,

$$Y_c(s) = \frac{1 - H_{ff}(s)e^{-sT_d}}{sL + K_p e^{-sT_d}} \quad (6.77)$$

With $T_d = 1.5T_s$ the real part of Y_c becomes negative between about $f_s/6$ and $f_s/2$, where $f_s = 1/T_s$ is the sampling frequency [29]. Figure 6.54 plots this for $f_s = 5$ kHz: the non-passive band runs from 0.84 to 2.5 kHz. An ordinary inductive grid crosses the converter impedance well below this band and leaves an adequate phase margin. A grid with a resonance inside the band, here 1.2 kHz from cable or filter capacitance, can drive the interconnection unstable even though both parts are individually stable. This is the mechanism behind several reported harmonic instabilities of converter-based plant.

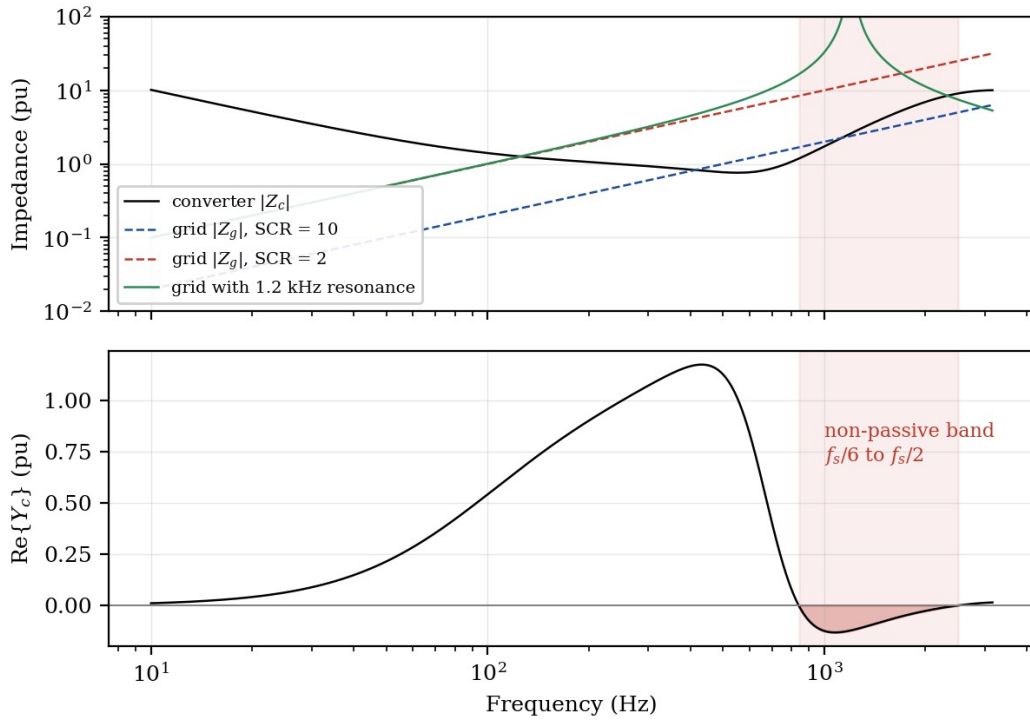


Figure 6.54. Upper: converter output impedance and three grid impedances. Lower: real part of the converter admittance, negative (non-passive) between $f_s/6$ and $f_s/2$ for a 1.5-sample delay ($f_s = 5$ kHz, $X = 0.15$ pu, 400 Hz current-loop bandwidth, 100 Hz feedforward filter).

Figure 6.55 places these phenomena on one frequency axis together with the model needed to study them. A phasor (RMS) model of the kind used in Sections 6.5.7 and 6.6.9 is valid for electromechanical and voltage-control studies up to a few hertz. It cannot establish the stability of a current loop or a harmonic interaction. The model bandwidth must be chosen from the phenomenon, not from the software that happens to be available.

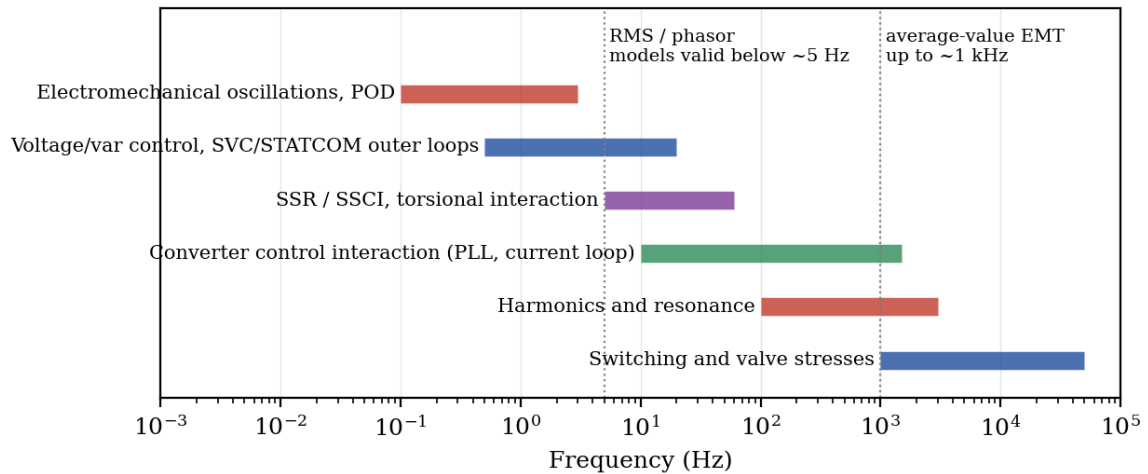


Figure 6.55. Frequency ranges of shunt-compensator phenomena and the model types that can represent them.

Open questions. Impedance models that remain valid when limiters and mode switching are active; frequency-coupled (dq) impedance models of MMC STATCOMs including cluster-energy control; online impedance estimation for adaptive tuning; and design rules that make passivity hold over the widest band at acceptable cost.

6.7.6 Hybrid, coordinated and data-assisted control

Commercial hybrid compensators combine a converter with switched capacitor or reactor branches so that the converter supplies fast, continuous control while the passive part carries the steady base (Section 6.6.8). The dispatch problem can be written as minimizing losses subject to keeping dynamic reserve:

$$\min_{Q_{conv}, N} P_{loss, conv}(Q_{conv}) + \sum_k P_{loss, k}(N_k) \text{ subject to } Q_{conv} + \sum_k N_k Q_k = Q_{ref}, |Q_{conv}| \leq Q_{conv, max} - Q_{reserve} \quad (6.78)$$

where N_k is the number of passive stages of size Q_k in service and $Q_{reserve}$ the converter range held back for disturbances. The same structure, extended over several devices and a time horizon, appears in coordinated voltage control involving SVCs, STATCOMs, synchronous condensers, HVDC converters and renewable-plant controllers.

Model-predictive, adaptive and machine-learning controllers are being studied for these tasks. Their value has to be demonstrated against a well-tuned conventional controller, with stability margins, constraint handling, tolerance of network changes, communication needs and real-time implementability stated. A data-driven controller is not better merely because it is newer, and a controller that cannot be certified for protection and grid-code compliance will not be installed.

6.7.7 Validation and evidence

Claims about a new STATCOM controller should climb a ladder of evidence (Figure 6.56). An analytical model gives insight; RMS simulation tests the system-level effect; sensitivity studies check behaviour across operating points and network strengths; EMT simulation, first with average-value and then switching converter models, tests fast dynamics; controller-hardware-in-the-loop (C-HIL) testing exposes sampling, delay, saturation and protection logic; and power-hardware-in-the-loop or field testing closes the loop. CIGRE Technical Brochure 663 gives guidance on STATCOM specification and testing [21]. Digital twins and online parameter estimation can support commissioning and condition monitoring, but they do not replace validated physical models and protection studies.

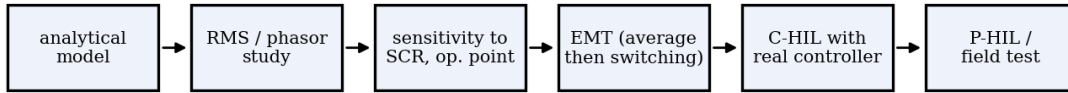


Figure 6.56. Evidence ladder for new shunt-compensator controllers.

6.7.8 Devices and packaging

Semiconductor technology affects switching frequency, conduction loss, valve redundancy, cooling and fault behaviour, but no device family is best at every rating. Silicon IGBTs, press-pack IGBTs and IGCTs dominate high-power STATCOMs, while silicon-carbide devices are most relevant at lower voltage and higher switching frequency. Comparisons should be made at the level of the complete converter and station, including losses, insulation, cooling, redundancy and maintainability, rather than device by device.

6.7.9 From classical concepts to research questions

Table 6.8 maps each classical concept of this chapter onto the state or constraint that recent work adds, and Table 6.9 lists gaps where a measurable contribution is possible.

Table 6.8. Extension of classical shunt-compensation concepts to current research questions.

Classical concept	Added state or constraint	Research question	Suitable validation
Voltage source behind X (Section 6.6.2)	Current saturation, weak-grid dynamics	How can voltage-forming behaviour be kept during faults?	EMT and C-HIL
DC capacitor (Section 6.6.7)	Cluster and cell energies, I_0	How can energy ripple and	Average model, then

Classical concept	Added state or constraint	Research question	Suitable validation
	or V_0	imbalance be reduced?	switching EMT
Reactive current control	P-Q sharing with storage, state of charge	How should priorities change during disturbances?	Dynamic simulation with energy constraints
Single compensator	Converter-to-converter impedance interaction	Which controller and grid combinations give poorly damped modes?	Impedance scan, eigenvalues, EMT
Fixed regulator gains (Section 6.5.7)	Operating point and network-strength variation	Can adaptation keep performance across network strengths without losing certifiability?	Sensitivity study, real-time tests

Table 6.9. Research gaps and possible contributions in shunt compensation.

Theme	Limitation of current practice	Potential contribution	Measurable impact
SVC gain tuning	Fixed gain chosen for the weakest ESCR; sluggish in strong grids	Online ESCR estimation with gain scheduling	Constant t_{95} across the ESCR range
Chain-link cell sizing	Capacitance set by worst-case ripple	Ripple-shaping injection; joint optimization with losses	kJ/MVA reduced at fixed ripple
Negative-sequence duty	Delta needs $ I_0 = I^- $; star limited by V_0	Topologies or controls with lower balancing cost	Cluster current or voltage margin saved
Energy-storage STATCOM	Separate P and Q controllers	Unified, current-limited P-Q damping control with adaptive mode tracking	Damping ratio gained per MJ of storage
Grid-forming STATCOM	Current limiting removes voltage-source behaviour	Virtual-impedance or hybrid limiting, also for unbalanced faults	Fault ride-through with preserved voltage-source behaviour
Hybrid SVC/STATCOM	Heuristic coordination	Optimal dispatch of passive and converter parts	Loss reduction at equal dynamic reserve
Wide-band interaction	Studies at fundamental frequency only	Passivity-based design from 1 Hz to a few kHz	Positive damping across the band

6.7.10 Worked examples on recent topics

Example 6.13: Energy-storage STATCOM: P-Q capability at depressed voltage. A 100 MVA energy-storage STATCOM with $I_{max} = 1.0$ pu operates at $V = 0.8$ pu. During a fault it is commanded to deliver $P = 0.30$ pu. Find the largest capacitive Q it can supply, and the result if the voltage falls to 0.5 pu.

Solution. The current limit gives $\sqrt{P^2 + Q^2} \leq V I_{max} = 0.8$ pu, so $Q_{max} = \sqrt{0.8^2 - 0.30^2} = 0.742$ pu, or 74.2 Mvar. At 0.5 pu, $Q_{max} = \sqrt{0.5^2 - 0.30^2} = 0.400$ pu, or 40 Mvar.

Note. The circle shrinks in proportion to voltage, so reserving real power for frequency support costs proportionally more reactive capability in a deep sag. A priority rule must be chosen, unless a short-time overcurrent rating is available.

Example 6.14: Negative-sequence duty of delta and star chain-link STATCOMs. A chain-link STATCOM supplies rated positive-sequence reactive current $|I^+| = 1.0$ pu and is asked to supply negative-sequence current $|I^-| = 0.3$ pu to balance a load. Find (a) the circulating current needed in a delta (SDBC) arrangement and the worst-case cluster current; (b) the range of zero-sequence voltage needed in a star (SSBC) arrangement; (c) the same for $|I^-| = 0.8$ pu.

Solution. (a) From (6.69), $|I_0| = 0.3$ pu. The worst cluster current, over all angles of I^- , is $1 + 2(0.3) = 1.6$ pu, so the cluster current rating rises by 60%. (b) With $r = 0.3$, (6.70) gives $0.231 \leq |V_0|/V^+ \leq 0.429$, so the cluster voltage rating must rise by up to about 43%. (c) With $r = 0.8$ the delta needs $|I_0| = 0.8$ pu, while the star needs up to $0.8/0.2 = 4.0$ times the phase voltage, which is impractical.

Note. This is why load-balancing STATCOMs use the delta arrangement. The star is attractive for balanced transmission duty, where it needs fewer cells.

Example 6.15: Sizing a static virtual impedance for current limiting. A grid-forming STATCOM has internal voltage $E_0 = 1.0$ pu, coupling reactance $X = 0.15$ pu and current limit $I_{max} = 1.2$ pu. Size a static virtual impedance with $X_{vi}/R_{vi} = 5$ so that a bolted terminal fault is limited to I_{max} , and find the fault current when the grid voltage falls to 0.5 pu.

Solution. (6.75) requires $|R_{vi} + j(5R_{vi} + 0.15)| = 1/1.2 = 0.833$ pu. Solving, $R_{vi} = 0.134$ pu and $X_{vi} = 0.672$ pu. Without limiting the bolted-fault current would be $1/0.15 = 6.67$ pu. At $V_g = 0.5$ pu, with E and V_g in phase, the current is $0.5/0.833 = 0.60$ pu.

Note. The static virtual impedance protects the converter but delivers only half of the available current in this shallow fault (Figure 6.53). Threshold and variable virtual impedances exist to recover that capability.

Example 6.16: Storage needed for inertia-like support. A 100 MVA energy-storage STATCOM on a 50 Hz system is to emulate $H = 5$ s. Find (a) the power at a RoCoF of 1 Hz/s; (b) the energy for support over a 0.8 Hz excursion; (c) the reactive capability left at 1.0 pu voltage with $I_{max} = 1.0$ pu while delivering (a); (d) the supercapacitor energy to install if it may operate between 100% and 50% of its rated voltage.

Solution. (a) (6.72): $P = 2(5)(100)(1)/50 = 20$ MW. (b) $\Delta E = 2(5)(100)(0.8)/50 = 16$ MJ. (c) $Q_{max} = \sqrt{1 - 0.2^2} = 0.980$ pu. (d) A 2:1 voltage window uses 75% of the stored energy, so at least $16/0.75 = 21.3$ MJ must be installed, before converter losses and ageing margins.

Note. The emulated machine would store $HS = 500$ MJ of kinetic energy, but only 16 MJ is delivered over the excursion. Sizing storage from HS overstates it about thirty-fold.

Key Equations Recap

Table 6.10. Key equations of Chapter 6.

Equation	Expression	Conditions
(6.4)	$P_c = (2V^2/X) \sin(\delta/2)$	Ideal midpoint regulation, lossless symmetric line
(6.6)	$Q_{sh} = (4V^2/X)[1 - \cos(\delta/2)]$	Ideal midpoint injection
(6.7)	$\beta = \alpha - 90^\circ$	Peak- versus zero-crossing-referenced firing
(6.9)	$i = (V_m/\omega L)(\cos \alpha - \cos \omega t)$	$\alpha \leq \omega t \leq 2\pi - \alpha$
(6.13)	$B_{TCR} = B_{max}[2(\pi - \alpha) + \sin 2\alpha]/\pi$	Fundamental, $90^\circ \leq \alpha \leq 180^\circ$
(6.17)	$I_{rms} = (V_m/\omega L) \sqrt{[(\pi - \alpha)(1 + 2\cos^2 \alpha) + 1.5 \sin 2\alpha]}/\pi$	Total RMS branch current
(6.20)	$I_n = \frac{4V_m}{\pi \omega L} \cdot \frac{\cos \alpha \sin n\alpha - n \sin \alpha \cos n\alpha}{n(n^2 - 1)}$	Odd $n \geq 3$, peak magnitude

Equation	Expression	Conditions
(6.22)	$i = (V_m / \omega L) (\cos \theta_c - \cos \theta)$	TSR closed at θ_c
(6.25)	$k = n^2 / (n^2 - 1), n = \sqrt{X_C / X_L}$	TSC magnification
(6.29)	$\cos \theta_c = 0, V_{C0} = \pm k V_m$	Transient-free TSC closing
(6.30)	$I_{osc} / I_{AC} = \sqrt{1 - \rho^2 (1 - 1/n^2)}$	Scheme 1, $-1 \leq \rho \leq 1$
(6.31)	$I_{osc} / I_{AC} = n [1 - (1 - 1/n^2) \rho]$	Scheme 2 (crest firing), $\rho < k$
(6.34)	$V = V_{ref} + X_{SL} I_{SVC}$	Within control range
(6.36)	$I_{SVC} = (E_{th} - V_{ref}) / (X_{th} + X_{SL})$	Operating point
(6.38)	$Q_{TCR} = Q_{cap} + Q_{ind}$	FC-TCR sizing
(6.41)	$B_{eq} = B / (1 - X_\sigma B)$	SVC behind transformer
(6.43)	$X_C = \frac{h^2}{h^2 - 1} \frac{V^2}{Q_F}, X_L = X_C / h^2$	Single-tuned filter
(6.47)	$T_\omega = 2 T_b (K_{SL} + K_{N,max}) / (K_{SL} + K_N)$	SVC regulator response
(6.49)	$P_{c,max} / P_{u,max} = 2 \cos(\theta/2)$	Midpoint compensation of long line
(6.51)	$B_{bc}^c = -B_{bc}^L + (G_{ab} - G_{ca}) / \sqrt{3}$, etc.	Load balancing
(6.53)	$P = V V_c \sin \delta_c / X, Q = V (V_c \cos \delta_c - V) / X$	STATCOM exchange
(6.57)	$T H D_v = \sqrt{\sum [(k p - 1)^{-2} + (k p + 1)^{-2}]}$	p -pulse converter
(6.60)	$P = \frac{3}{2} (v_d i_d + v_q i_q), Q = \frac{3}{2} (v_q i_d - v_d i_q)$	Amplitude-invariant dq
(6.63)	$C = 2 \Delta P \Delta t / (V_{dc,max}^2 - V_{dc,min}^2)$	DC capacitor, transient deficit
(6.65)	$P = (V^2 / X) \sin \delta + (V I_Q / 2) \sin(\delta/2)$	Current-limited midpoint STATCOM

Table 6.11. Key equations of Section 6.7.

Eq.	Expression	Conditions
(6.68)	$E_{stored} / S \geq 1 / (2 \omega \epsilon)$	Chain-link cell energy, reactive output
(6.69)	$ I_0 = I_\Delta $	Delta cluster balancing
(6.70)	$r / (1 + r) \leq V_0 / V^+ \leq r / (1 - r)$	Star cluster balancing
(6.71)	$P^2 + Q^2 \leq (V I_{max})^2,$ $ \int P_{ES} dt \leq E_{usable}$	Energy-storage STATCOM
(6.72)	$P_{in} = (2 H S / f_0) df / dt$	Inertia-like support
(6.75)	$ Z_{vi} + j X \geq E_0 / I_{max}$	Static virtual-impedance sizing
(6.76)	$V = V_g / (1 + Z_g Y_c)$	Impedance-based stability

Common Misconceptions

“Firing-angle formulas can be used whichever reference the problem gives.” They cannot. A TCR fired at “30°” conducts at 39% of full susceptance if the angle is peak-referenced ($\alpha = 120^\circ$), whereas a zero-crossing-referenced 30° lies outside the permitted range and would produce asymmetric current with a DC component. Always convert with Equation (6.7) first.

“TCR current THD has a maximum.” Individual harmonics have maxima; total branch THD increases monotonically towards blocking. The figure of about 36% often quoted is the THD at $\alpha = 120^\circ$, where the third harmonic peaks.

“Line current RMS of a delta TCR is $\sqrt{3}$ times the branch RMS.” This is true for the fundamental and for non-triplen harmonics only. Triplens circulate in the delta, so the line RMS current is $\sqrt{3}$ times the branch RMS excluding triplens.

“Firing a discharged TSC at the voltage zero is transient-free.” With a series reactor it minimizes, but does not eliminate, the oscillatory current, which then equals the steady-state current amplitude. Exactly transient-free closing needs crest firing with the capacitor at $\pm k V_m$.

“The series reactor of a TSC is chosen only for di/dt .” It also fixes the branch resonance, which must avoid characteristic TCR harmonics, and it magnifies capacitor voltage and output by k .

“A coupling transformer reduces the capacitive range of an SVC.” It does the opposite. In series with a net capacitive branch it enlarges the effective susceptance and raises the secondary voltage; only the inductive end shrinks.

“A midpoint compensator always doubles transfer capability.” Doubling requires an ideal, unlimited compensator on a short lossless symmetric line. A long line gives $2 \cos(\theta/2)$, and any real rating limits the gain further.

“The STATCOM’s DC capacitor stores its reactive energy.” Balanced reactive power circulates among the phases and never passes through the capacitor. The capacitor is sized from real-power imbalance and ripple.

“A STATCOM supplies rated Mvar at any voltage.” It supplies rated *current*; its var output still falls linearly with voltage, merely more slowly than an SVC’s.

Industry Spotlight

Industrial shunt compensation has developed along two complementary technology paths. Thyristor-based SVCs remain appropriate where large steady-state and dynamic reactive-power ranges, proven high-voltage equipment and economical bulk compensation are required. Converter-based STATCOMs are particularly attractive where rapid current control, compact installation, strong low-voltage reactive-current support, weak-grid performance or modular multilevel implementation is important. These technologies should therefore be compared by duty rather than by age. A practical specification must state the required V-I characteristic, continuous and short-time current ratings, harmonic performance, response time, transformer and filter duties, protection and bypass philosophy, losses, availability and validation tests. The preferred solution is the one that satisfies the required network performance over the specified operating envelope.

Chapter Summary

Shunt compensation controls voltage by injecting reactive current, and the effectiveness of that current is set by the Thevenin reactance of the bus. Ideal midpoint regulation doubles the steady-state transfer limit of a short symmetric line and raises it by $2 \cos(\theta/2)$ on a long line, at the cost of a reactive injection that grows rapidly with angle.

The TCR varies its fundamental susceptance continuously through the firing angle according to a nonlinear law, and it generates odd harmonics whose triplens are trapped in a delta and whose 5th and 7th are removed

by filters or twelve-pulse arrangements. The TSR and TSC switch whole stages; the TSC's series reactor fixes its resonance and magnification and governs a switching transient that the firing scheme keeps within the steady-state current. The SVC combines these branches behind a droop characteristic whose slope reduces rating and stabilizes the voltage loop, and whose regulator must be designed for the weakest expected network.

The STATCOM synthesizes a voltage instead of presenting an impedance. Its output is limited by current, so its support degrades linearly rather than quadratically in a voltage sag; its converter current can be controlled rapidly, although complete bus-voltage recovery depends on the network and outer control; and multilevel conversion can substantially reduce filtering requirements. Its DC capacitor handles only real-power imbalance and ripple. Hybrid arrangements trade between the two technologies, and energy storage and grid-forming control are extending the STATCOM from a reactive compensator towards a general-purpose grid-supporting source.

Solved Numerical Practice Problems

Worked examples have already been placed throughout the chapter immediately after the concepts they clarify. The examples collected here give broader design practice and cross-topic synthesis. Unless stated otherwise, firing angles are measured from the voltage zero crossing (Section 6.1.5), the supply frequency is 50 Hz ($\omega = 314.159$ rad/s), and per-unit quantities are on the stated three-phase base. Per-unit current and three-phase reactive power must use one consistent base: if V is the per-unit line-to-line voltage on a three-phase MVA base, a purely reactive shunt current has $|I| = |Q|/V$ in per unit. The factor $\sqrt{3}$ belongs only to conversions between physical line quantities, a distinction that matters in the midpoint-STATCOM examples.

Example 6.17: Single-phase TCR: currents, susceptance and valve stress.

A single-phase TCR with $L = 10$ mH is supplied at 240 V, 50 Hz and fired at $\beta = 30^\circ$ from the voltage peak. Find (a) the maximum var rating; (b) α and the total RMS current; (c) the RMS fundamental current, equivalent susceptance and reactive power; (d) the 3rd, 5th and 7th harmonic currents; (e) the THD; (f) the peak instantaneous current; (g) the valve voltage at the firing instant.

Solution. (a) $X_L = \omega L = 3.1416 \Omega$; $Q_{max} = V^2 / X_L = 240^2 / 3.1416 = 18.335$ kvar. Base $V_m / \omega L = 339.41 / 3.1416 = 108.04$ A.

(b) $\alpha = \beta + 90^\circ = 120^\circ$. Equation (6.17): $I_{rms} = 0.29411 \times 108.04 = 31.776$ A.

(c) $B/B_{max} = F(120^\circ) = 0.39100$, so $B_{TCR} = 0.391 / 3.1416 = 0.12446$ S, $I_1 = 240 \times 0.12446 = 29.870$ A and $Q = V^2 B_{TCR} = 7.169$ kvar.

(d) At 120° , $I_3 = 10.530$ A, $I_5 = 2.106$ A, $I_7 = 0.752$ A RMS.

(e) $THD = \sqrt{(31.776/29.870)^2 - 1} = 36.3\%$. This is the value at the third-harmonic peak, not a maximum THD.

(f) $i_{pk} = (V_m / \omega L)(1 + \cos \alpha) = 108.04 \times 0.5 = 54.02$ A.

(g) $v_T = V_m \sin \alpha = 339.41 \times 0.866 = 293.9$ V; the valve must nevertheless be rated for the full peak $V_m = 339.4$ V plus margin, since smaller α is used near full output.

Example 6.18: Six-pulse TCR at partial conduction: phase and line quantities.

A three-phase six-pulse TCR with 5.0 mH per delta branch operates at 415 V, 50 Hz with $\beta = 15^\circ$. Find (a) the maximum rating; (b) the branch RMS and fundamental currents; (c) the reactive power absorbed; (d) the branch harmonics up to the 13th; (e) the line-current harmonics; (f) the line RMS current and line THD.

Solution. (a) $X_L = 1.5708 \Omega$; $Q_{max} = 3 V^2 / X_L = 328.93$ kvar. Base $V_m / \omega L = 586.89 / 1.5708 = 373.63$ A.

- (b) $\alpha = 105^\circ$. $I_{rms} = 180.65$ A; $F(105^\circ) = 0.6742$, so $I_1 = 0.6742 \times 373.63/\sqrt{2} = 178.12$ A. Branch THD = 16.9%.
- (c) $Q = 3V^2 F/X_L = 221.75$ kvar.
- (d) $I_3 = 26.15$, $I_5 = 12.89$, $I_7 = 6.61$, $I_9 = 2.96$, $I_{11} = 0.765$, $I_{13} = 0.462$ A.
- (e) Triplens are absent. $I_{5\ell} = 22.33$ A (negative sequence), $I_{7\ell} = 11.44$ A (positive sequence), $I_{11\ell} = 1.32$ A, $I_{13\ell} = 0.80$ A; line fundamental 308.51 A.
- (f) The triplen content of the branch current is $\sqrt{\sum I_{3,9,15,\dots}^2} = 26.35$ A, so the line RMS current is $\sqrt{3} \sqrt{180.65^2 - 26.35^2} = 309.54$ A, not $\sqrt{3} \times 180.65 = 312.9$ A. Line THD = $\sqrt{(309.54/308.51)^2 - 1} = 8.2\%$, about half the branch THD.

Example 6.19: Inverting the susceptance law.

A 100 Mvar TCR is controlled through a linearizer. Find (a) the firing angles for 20, 40, 60 and 80% output; (b) the peak-referenced equivalents; (c) the incremental gain $d(B/B_{max})/d\alpha$ at each point; (d) the ratio of the largest to smallest gain; (e) three Newton iterations for 40%, starting at 120° ; (f) the error if a peak-referenced $\beta = 30^\circ$ is mistakenly inserted as α .

Solution. (a) Solving $F(\alpha) = q$: $\alpha = 133.40^\circ, 119.46^\circ, 108.64^\circ, 99.08^\circ$.

- (b) $\beta = 43.40^\circ, 29.46^\circ, 18.64^\circ, 9.08^\circ$.
- (c) From Equation (6.15), $-4 \sin^2 \alpha / \pi$ per radian: $-0.01173, -0.01685, -0.01995, -0.02167$ per degree.
- (d) $0.02167/0.01173 = 1.85$; over the full 95° to 165° range the variation exceeds 14, which is why a linearizer is essential.
- (e) $\alpha_{k+1} = \alpha_k - [F(\alpha_k) - 0.4]/F'(\alpha_k)$: $F(120^\circ) = 0.3910$, $F' = -0.9549/\text{rad}$, giving $\alpha_1 = 119.460^\circ$; the next iterates are 119.463° and 119.463° . Convergence is quadratic because F is smooth.
- (f) $\alpha = 30^\circ$ is outside 90° to 180° ; the formula would return $F = 1.94$, an impossible 194% of rating. Using the correct conversion, $\beta = 30^\circ$ gives 39.1%.

Example 6.20: Segmented TCR.

A 100 kvar, 415 V delta TCR is built as $m = 4$ sections of 25 kvar, only one of which is phase controlled. Find (a) the inductance per section; (b) the maximum third-harmonic branch current of one section; (c) the same for an unsegmented 100 kvar TCR; (d) the configuration and firing angle for 60 kvar; (e) the 3rd and 5th branch harmonics at 60 kvar for both designs; (f) the reduction factor.

Solution. (a) $X = 3(415)^2/25000 = 20.667 \Omega$, $L = 65.79$ mH.

- (b) $I_{3,max} = 0.13783 \times (586.89/20.667)/\sqrt{2} = 2.768$ A at $\alpha = 120^\circ$.
- (c) Four times larger: 11.07 A.
- (d) Two sections fully on (50 kvar) and the third supplying 10 kvar, i.e. 40% of its rating: $\alpha = 119.46^\circ$.
- (e) Segmented: $I_3 = 2.767$ A, $I_5 = 0.589$ A. Single TCR at 60% ($\alpha = 108.64^\circ$): $I_3 = 9.271$ A, $I_5 = 4.046$ A.
- (f) The worst-case reduction is exactly $m = 4$. At this particular operating point it is 3.35 for the 3rd and 6.9 for the 5th.

Example 6.21: TCR valve and reactor design.

A 60 Mvar, 13.8 kV, delta-connected TCR uses thyristors with a repetitive peak voltage of 8 kV, applied at 60% of rating, and reactors with $Q=80$. Find (a) branch reactance and inductance; (b) branch RMS and peak currents at full conduction; (c) RMS and average current per thyristor; (d) the peak valve voltage with 10% overvoltage; (e) the number of series thyristors with one redundant device; (f) the reactor losses at full conduction.

Solution. (a) $X = 3(13.8 \text{ kV})^2 / 60 \text{ Mvar} = 9.522 \Omega$; $L = 30.31 \text{ mH}$.

(b) $I_b = 13800 / 9.522 = 1449.3 \text{ A}$; $I_{pk} = 2049.6 \text{ A}$.

(c) $I_{T,rms} = I_{pk} / 2 = 1024.8 \text{ A}$; $I_{T,av} = I_{pk} / \pi = 652.4 \text{ A}$.

(d) $1.1 \times \sqrt{2} \times 13.8 \text{ kV} = 21.47 \text{ kV}$.

(e) $21.47 / (0.6 \times 8) = 4.47$, rounded to 5, plus one redundant: six thyristors per string in each direction.

(f) $R = X / Q = 0.1190 \Omega$; loss per branch $1449.3^2 \times 0.119 = 250 \text{ kW}$; 750 kW in total, 1.25% of rating.

Example 6.22: TCR quality factor and losses at partial conduction.

A single-phase TCR with $L = 18 \text{ mH}$ and $Q = 60$ is fed at 230 V and operated at $\alpha = 120^\circ$. Find (a) σ ; (b) the peak current; (c) the RMS current; (d) the fundamental current and vars; (e) the reactor loss; (f) the THD.

Solution. (a) $\sigma = 2(180^\circ - 120^\circ) = 120^\circ$; each thyristor conducts for 120° .

(b) $X = 5.6549 \Omega$; $i_{pk} = (325.27 / 5.6549)(1 + \cos 120^\circ) = 28.76 \text{ A}$ at $\omega t = 180^\circ$.

(c) $I_{rms} = 0.29411 \times 57.52 = 16.92 \text{ A}$.

(d) $I_1 = 0.391 \times 57.52 / \sqrt{2} = 15.90 \text{ A}$; $Q = 230^2 \times 0.391 / 5.6549 = 3.658 \text{ kvar}$.

(e) $R = X / Q = 0.0942 \Omega$; $P = 16.92^2 \times 0.0942 = 26.97 \text{ W}$, using the total RMS current.

(f) 36.3%.

Example 6.23: Binary-weighted TSR bank.

A three-phase delta TSR bank must provide 0 to 350 kvar in 50 kvar steps at 415 V. Find (a) the section ratings; (b) their inductances; (c) the branch currents; (d) the thyristor currents of the largest section; (e) the list of levels.

Solution. (a) Seven steps need three binary sections: 50, 100 and 200 kvar.

(b) $X = 3V^2 / Q$: 10.334, 5.167 and 2.583 Ω ; $L = 32.89$, 16.45 and 8.22 mH.

(c) 40.16, 80.32 and 160.64 A per branch.

(d) $I_{pk} = 227.18 \text{ A}$; $I_{T,rms} = 113.59 \text{ A}$; $I_{T,av} = 72.31 \text{ A}$; blocking 586.9 V peak.

(e) 0, 50, 100, 150, 200, 250, 300 and 350 kvar (eight states).

Example 6.24: Overcharged TSC: scheme 2.

A 12 kvar, 230 V TSC with $n=5$ is found charged to $V_{C0} = 380 \text{ V}$. Find (a) the scheme and angle; (b) I_{osc} / I_{AC} ; (c) the steady peak current; (d) the peak current bound; (e) the ratio if the opposite-polarity crest were used; (f) the valve stress after the next turn-off.

Solution. (a) $\rho = 1.168 > 1$: scheme 2, crest firing at $\theta_c = 90^\circ$ (positive crest, same polarity as V_{C0}).

(b) $5|0.96 \times 1.168 - 1| = 0.608$.

(c) $\sqrt{2} \times 12000 / 230 = 73.79 \text{ A}$.

(d) $73.79 \times 1.608 = 118.6 \text{ A}$.

(e) At the negative crest ρ effectively changes sign: $5|-1.121 - 1| = 10.6$, a destructive inrush.

(f) $(1+k)V_m = 664.1 \text{ V}$.

Example 6.25: Medium-voltage TSC: detuning and di/dt .

A star-connected 20 Mvar TSC on a 33 kV bus uses $n=4.5$. Find (a) the net, capacitive and inductive reactances per phase; (b) C and L ; (c) the steady peak current; (d) the characteristic impedance; (e) the di/dt for a mistimed closure at the worst instant; (f) the minimum L for a 20 A/ μ s valve.

Solution. (a) $X_{net}=33^2/20=54.45\ \Omega$; $X_C=57.28\ \Omega$; $X_L=2.829\ \Omega$.

(b) $C=55.57\ \mu\text{F}$; $L=9.00\ \text{mH}$.

(c) $\sqrt{2} \times 20\ \text{MVA} / (3 \times 19.05\ \text{kV}) = 494.8\ \text{A}$.

(d) $Z_n = \sqrt{L/C} = 12.73\ \Omega$.

(e) $V_m = 26.94\ \text{kV}$, worst $(1+k)V_m = 55.29\ \text{kV}$ across L : $di/dt = 6.14\ \text{A}/\mu\text{s}$.

(f) $L_{min} = 55.29\ \text{kV} / 20\ \text{A}/\mu\text{s} = 2.76\ \text{mH}$; the detuning reactor already exceeds it by a factor of 3.3.

Example 6.26: Binary TSC bank at low voltage.

A 415 V, delta-connected binary TSC bank has stages of 25, 50, 100 and 200 kvar with $n=4.5$. Find (a) the number of levels; (b) C and L for each stage; (c) branch and line currents of each stage; (d) the hysteresis band needed on a bus with $X_{th}=0.02\ \text{pu}$ on a 1 MVA base.

Solution. (a) $2^4 = 16$ states, 0 to 375 kvar in 25 kvar steps.

(b) $X_{net} = 3V^2/Q$; $C = 1/[\omega k X_{net}]$; $L = k X_{net}/(n^2 \omega)$: 25 kvar: 146.4 μF , 3.417 mH; 50 kvar: 292.8 μF , 1.709 mH; 100 kvar: 585.7 μF , 0.854 mH; 200 kvar: 1171.3 μF , 0.427 mH.

(c) Branch: 20.08, 40.16, 80.32, 160.64 A; line: 34.78, 69.56, 139.12, 278.24 A.

(d) One step changes voltage by $X_{th} \Delta Q = 0.02 \times 0.025 = 0.0005\ \text{pu}$; the band must exceed this. A practical band of $\pm 0.5\%$ (1% wide) is twenty times the step.

Example 6.27: FC-TCR design at 11 kV.

An FC-TCR on an 11 kV bus must provide +40 Mvar to -20 Mvar. The TCR is delta connected and the capacitance is star connected. Find (a) Q_C and Q_{TCR} ; (b) TCR inductance; (c) the single equivalent capacitance; (d) the components if Q_C is split into a 25 Mvar 5th filter ($h=4.9$) and a 15 Mvar 7th filter ($h=6.9$); (e) firing angles for 0, +20 and -10 Mvar.

Solution. (a) Equation (6.38): $Q_C = 40\ \text{Mvar}$, $Q_{TCR} = 60\ \text{Mvar}$.

(b) $X_L = 3(11\ \text{kV})^2/60\ \text{Mvar} = 6.05\ \Omega$, $L = 19.26\ \text{mH}$ per branch.

(c) $X_C = V_{LL}^2/Q_C = 3.025\ \Omega$ per phase, $C = 1052\ \mu\text{F}$.

(d) Equation (6.43): 5th: $X_C = 5.050\ \Omega$, $C = 630.3\ \mu\text{F}$, $X_L = 0.2103\ \Omega$, $L = 0.670\ \text{mH}$. 7th: $X_C = 8.240\ \Omega$, $C = 386.3\ \mu\text{F}$, $X_L = 0.1731\ \Omega$, $L = 0.551\ \text{mH}$.

(e) $Q_{TCR} = Q_C - Q$: 40, 20 and 50 Mvar, i.e. $F = 0.667, 0.333, 0.833$, giving $\alpha = 105.36^\circ, 123.59^\circ$ and 97.54° .

Example 6.28: TSC-TCR dispatch and dead zones.

A TSC-TCR SVC has three TSCs of 50 Mvar each and one TCR of 60 Mvar. Find (a) the range; (b) the dispatch for -45, 0, +80 and +130 Mvar; (c) the firing angles; (d) whether the overlap condition is met; (e) what happens if the TCR is reduced to 40 Mvar.

Solution. (a) -60 to +150 Mvar.

(b) Equation (6.39): -45: no TSC, TCR absorbs 45; 0: no TSC, TCR blocked; +80: 2 TSCs, TCR 20; +130: 3 TSCs, TCR 20.

- (c) $F=0.75$: $\alpha=101.40^\circ$; $F=0.333$: $\alpha=123.59^\circ$; blocked: 180° .
- (d) $60 > 50$ Mvar, so each sub-range overlaps the next by 10 Mvar, enough for a hysteresis band of a few Mvar.
- (e) With 40 Mvar the TCR covers only $N \times 50 - 40$ to $N \times 50$; the intervals 0 to 10, 50 to 60 and 100 to 110 Mvar become dead zones that the SVC cannot produce continuously (Figure 6.32).

Example 6.29: Complete design of a ± 150 Mvar SVC.

Design a ± 150 Mvar SVC for a 400 kV bus with a 400/11 kV transformer, using two delta TCRs, two star TSCs ($n=4.3$), a 40 Mvar 5th filter ($h=4.9$) and a 10 Mvar 7th filter ($h=6.9$), all on the 11 kV side (transformer reactance neglected for sizing). Find (a) TSC and TCR ratings; (b) TCR inductance; (c) TSC capacitance and inductance; (d) filter components; (e) TSC valve peak voltage after turn-off; (f) verification of both limits.

Solution. (a) Capacitive: filters 50 Mvar, so TSCs supply 100 Mvar, 50 Mvar each. Inductive: TCRs must absorb $150 + 50 = 200$ Mvar, 100 Mvar each.

- (b) $X_L = 3(11)^2/100 = 3.63 \Omega$, $L = 11.55$ mH per branch.
- (c) $X_{net} = 11^2/50 = 2.42 \Omega$; $k = 18.49/17.49 = 1.0572$; $X_C = 2.558 \Omega$, $C = 1244 \mu\text{F}$; $X_L = 0.1384 \Omega$, $L = 0.440$ mH.
- (d) 5th: $X_C = 3.156 \Omega$, $C = 1008 \mu\text{F}$, $L = 0.418$ mH. 7th: $X_C = 12.36 \Omega$, $C = 257.5 \mu\text{F}$, $L = 0.826$ mH.
- (e) $V_m = 8.98$ kV phase peak; $(1+k)V_m = 18.48$ kV per valve.
- (f) Full capacitive: $50 + 2 \times 50 = 150$ Mvar with TCRs blocked. Full inductive: $2 \times 100 - 50 = 150$ Mvar with TSCs off. Both met at rated voltage.

Example 6.30: Midpoint SVC on a 500 km, 400 kV line.

A lossless 400 kV, 50 Hz line, 500 km long, has $Z_c = 300 \Omega$ and $\beta = 0.06^\circ/\text{km}$. End voltages are 1.0 pu. Find (a) θ and SIL; (b) the uncompensated maximum power; (c) the midpoint voltage without compensation at $\delta = 60^\circ$ and 90° ; (d) the maximum power with the midpoint held at 1.0 pu and the ratio; (e) the midpoint reactive injection at $\delta = 60^\circ$, 90° and 120° ; (f) the midpoint Thevenin reactance.

Solution. (a) $\theta = 30^\circ$; $P_0 = 400^2/300 = 533.3$ MW.

- (b) $P_0/\sin 30^\circ = 1066.7$ MW.
- (c) Equation (6.48): $\cos 30^\circ/\cos 15^\circ = 0.8966$ pu and $\cos 45^\circ/\cos 15^\circ = 0.7321$ pu.
- (d) $P_0/\sin 15^\circ = 2060.6$ MW; ratio $2 \cos 15^\circ = 1.932$, slightly less than the lumped-line value of 2.
- (e) Equation (6.50): 411.7 Mvar at 60° (1030 MW), 1066.7 Mvar at 90° (1457 MW) and 1920 Mvar at 120° (1785 MW). The injection grows much faster than the power, which is why practical ratings stop well short of the ideal.
- (f) $(Z_c/2) \tan 15^\circ = 40.19 \Omega$, or 0.0251 pu on 100 MVA, 400 kV.

Example 6.31: Load balancing (Steinmetz compensation).

A 500 kVA, 0.8 lagging single-phase load is connected between lines a and b of an 11 kV, 50 Hz system. Find (a) G and B^L of the load; (b) the compensator susceptances; (c) element values; (d) branch var ratings; (e) the balanced line current; (f) a phasor check.

Solution. (a) $P = 400$ kW, $Q = 300$ kvar: $G = P/V^2 = 3.306$ mS, $B^L = -Q/V^2 = -2.479$ mS.

- (b) Equation (6.51): $B_{ab}^c = +2.479$ mS (capacitive), $B_{bc}^c = +G/\sqrt{3} = +1.909$ mS (capacitive), $B_{ca}^c = -1.909$ mS (inductive).

- (c) $C_{ab}=7.89\mu\text{F}$, $C_{bc}=6.08\mu\text{F}$, $L_{ca}=1.668\text{ H}$.
- (d) 300 kvar, 230.9 kvar and 230.9 kvar (reactive).
- (e) $I=P/(\sqrt{3}V)=20.99\text{ A}$ in each line.
- (f) Computing I_a, I_b, I_c with the compensator in place gives three currents of 20.99 A, each in phase with its phase voltage. The ab branch is needed only because the load is not resistive: it cancels the load's own 300 kvar, and without it the line currents would be neither balanced nor at unity power factor.

Example 6.32: STATCOM, SVC and hybrid at depressed voltage.

An SVC and a STATCOM are each rated 100 Mvar capacitive at 1.0 pu. A hybrid consists of a ± 50 Mvar STATCOM and a 30 Mvar fixed capacitor. Find (a) both outputs at 0.7 pu; (b) at 0.5 pu; (c) the voltage at which the SVC gives half rating, and the STATCOM output there; (d) the hybrid range at 1.0 pu; (e) at 0.6 pu.

Solution. (a) 49 and 70 Mvar; ratio 1.43.

- (b) 25 and 50 Mvar; ratio 2.0.
- (c) $v^2=0.5$; $v=0.707$; STATCOM 70.7 Mvar.
- (d) +80 to -20 Mvar.
- (e) Capacitive $50(0.6)+30(0.36)=40.8\text{ Mvar}$; inductive $50(0.6)-10.8=19.2\text{ Mvar}$. The capacitor contributes only 10.8 Mvar here, so hybridization costs low-voltage support.

Example 6.33: Indirect control: energy needed for a reactive step.

A STATCOM with fixed modulation ($V_c=kV_{dc}$) has $X=0.12\text{ pu}$ on 100 MVA and stores 4 MJ at nominal V_{dc} (where $V_c=V=1\text{ pu}$). Find (a) V_c for $Q=0.5\text{ pu}$; (b) the required rise in V_{dc} ; (c) the energy to be added; (d) the average active power to do this in 20 ms; (e) the angle required; (f) its sign.

Solution. (a) $V_c=1+0.5\times 0.12=1.06\text{ pu}$.

- (b) 6%.
- (c) $\Delta W=4(1.06^2-1)=0.494\text{ MJ}$.
- (d) $24.7\text{ MW}=0.247\text{ pu}$.
- (e) $\sin\delta_c=0.247\times 0.12/(1\times 1.03)$, using the mean converter voltage: $|\delta_c|=1.65^\circ$.
- (f) Negative: the converter voltage must lag the bus to absorb power. The response of an indirectly controlled STATCOM is therefore limited by the active power that a small angle can draw, a few cycles in practice.

Example 6.34: Midpoint STATCOM with a current limit.

A lossless line with $V_s=V_r=1.0\text{ pu}$ and total $X=0.5\text{ pu}$ has a midpoint STATCOM. Find (a) P , Q and I at $\delta=60^\circ$ with ideal regulation; (b) the same at 90° ; (c) δ_{lim} and P for $I_{max}=1.0\text{ pu}$; (d) P at 90° , 120° and 150° in current limit and the midpoint voltage; (e) the comparison with an SVC of $B_C=1\text{ pu}$; (f) the maximum power in each case.

Solution. (a) Equations (6.4) and (6.6): $P=4\sin 30^\circ=2.0\text{ pu}$ (uncompensated 1.732); $Q=I=8(1-\cos 30^\circ)=1.072\text{ pu}$.

- (b) $P=2.828\text{ pu}$; $Q=I=2.343\text{ pu}$.
- (c) $\cos(\delta_{lim}/2)=1-I_{max}X/4=0.875$; $\delta_{lim}=57.91^\circ$, $P=1.936\text{ pu}$.
- (d) Equation (6.65): 2.354, 2.165 and 1.483 pu, with $V_m=0.832$, 0.625 and 0.384 pu.
- (e) Equation (6.66): 2.286, 1.979 and 1.143 pu: the STATCOM is 3%, 9% and 30% higher.

- (f) STATCOM 2.361 pu at 94.9° ; SVC 2.286 pu at 90° ; uncompensated 2.0 pu. In per unit $I=Q/V$; writing $Q/(\sqrt{3}V)$ mixes line and phase quantities.

Example 6.35: Voltage restoration at a distribution bus.

A 415 V, 50 Hz feeder with $Z_s=0.8+j4\ \Omega$ per phase supplies a star load $Z_L=35+j27\ \Omega$ per phase (constant impedance). Find (a) the current and load voltage without compensation; (b) the drop across the feeder; (c) the shunt capacitive susceptance that restores 415 V at the load; (d) the compensator var rating; (e) its current; (f) the load power after restoration.

Solution. (a) $I=239.6/|35.8+j31|=5.060$ A; $V_L=223.65$ V (387.4 V line).

(b) $|IZ_s|=20.64$ V.

(c) Solving $|V_L(B)|=239.6$ V numerically: $B=18.11$ mS per phase, capacitive.

(d) $BV^2=1039$ var per phase, 3.12 kvar three-phase.

(e) 4.34 A.

(f) 1028 W per phase, 3.08 kW total. (If the load were modelled as constant power at its depressed-voltage value, a slightly different rating results; the load model must always be stated.)

Unsolved Practice Problems

- 6.1** A 35 kvar, 400 V, 50 Hz delta TCR is to be designed. Find the branch reactance and inductance, the branch and line currents at full conduction, and the firing angle for 40% of rated output.
- 6.2** For a TCR at $\alpha=130^\circ$, find the 3rd, 5th and 7th branch harmonics as percentages of the full-conduction fundamental, the branch-current THD and the delta line-current THD.
- 6.3** A single-phase TCR of 20 mH on 230 V is fired at $\alpha=140^\circ$. Find the total RMS current, the fundamental current, the reactive power and the peak instantaneous current.
- 6.4** For a twelve-pulse TCR at $\alpha=135^\circ$, which harmonics appear in the primary line current? Give the 11th, 13th, 23rd and 25th as percentages of the full-conduction fundamental.
- 6.5** Design a three-section binary TSR bank giving 0 to 175 kvar in 25 kvar steps at 415 V, delta connected. Give the reactance and inductance of each section.
- 6.6** A 60 kvar, 415 V, delta TSC with $n=4.8$ must deliver its rated output. Find X_{net} , X_C , C , L , k , the peak capacitor voltage and the peak valve voltage after turn-off.
- 6.7** A TSC with $n=4$ has a residual voltage of $+0.3V_m$. Find the two scheme-1 closing angles in the positive half cycle, I_{osc}/I_{AC} for scheme 1, and I_{osc}/I_{AC} for crest firing.
- 6.8** A discharged TSC with $n=4$ is closed by mistake at a voltage crest. Find I_{osc}/I_{AC} and an upper bound on the peak current in multiples of I_{AC} .
- 6.9** An SVC with $V_{ref}=1.0$ pu and $X_{SL}=0.04$ pu sits on a bus with $X_{th}=0.08$ pu. Find the SVC current, bus voltage and reactive power for $E_{th}=1.04$ pu and for $E_{th}=0.95$ pu.
- 6.10** An FC-TCR on a 13.8 kV bus must cover +60 to -30 Mvar. Find Q_{TCR} , the delta TCR inductance, the star capacitance and the firing angle for zero output.
- 6.11** A TSC-TCR has four 25 Mvar TSCs and a 30 Mvar TCR. Give the dispatch (TSCs in service and firing angle) for +10, +55, +90 and -20 Mvar.
- 6.12** An FC-TCR with $B_C=1.0$ pu and $B_{L,max}=1.4$ pu on its secondary is connected through a transformer with $X_\sigma=1/6$ pu. Find the exact capacitive and inductive limits seen from the primary.

- 6.13** A STATCOM has $X=0.1$ pu. Find V_c for 0.6 pu capacitive output at $V=1.0$ pu, the output at $V=0.85$ pu at $I_{max}=1.0$ pu, and V_c for 0.4 pu inductive output at 1.0 pu.
- 6.14** A midpoint STATCOM with $I_{max}=1.5$ pu serves a line with $X=0.6$ pu and 1.0 pu end voltages. Find δ_{lim} , the power at δ_{lim} , the power at 120° and the ideal maximum.
- 6.15** A STATCOM DC link must cover 1.5 MW for 15 ms while falling from 18 to 16 kV. Find C , and the ripple for $\hat{P}_2=0.6$ MW at a mean of 17 kV.
- 6.16** For a 24-pulse STATCOM with coupling reactance 0.15 pu, find the 23rd and 25th harmonic currents as a percentage of rated current.
- 6.17** A 10 kW resistive load is connected between lines b and c of a 415 V system. Design the balancing compensator and find the line current.
- 6.18** A utility must choose shunt compensation for a weak 220 kV bus feeding an arc furnace and a wind farm. Fault-induced sags to 0.5 pu occur, flicker limits are tight, and the site is space-constrained. Compare an FC-TCR, a TSC-TCR, a STATCOM and a hybrid STATCOM, and justify a recommendation.

Answer Key (Unsolved Problems)

- 6.1** 13.714 Ω ; 43.65 mH; 29.17 A; 50.52 A; 119.46° .
- 6.2** 12.26%, 0.75%, 2.04%; branch THD 51.6%; line THD 9.7%.
- 6.3** 5.86 A; 4.79 A; 1.103 kvar; 12.11 A.
- 6.4** Only $12k \pm 1$: 0.58%, 0.35%, 0.13%, 0.10%.
- 6.5** 25 kvar: 20.67 Ω , 65.8 mH; 50 kvar: 10.33 Ω , 32.9 mH; 100 kvar: 5.17 Ω , 16.4 mH.
- 6.6** 8.611 Ω ; 9.002 Ω ; 353.6 μF ; 1.244 mH; 1.0454; 613.5 V; 1200 V.
- 6.7** 17.46° and 162.54° (choose the instant at which the valve with the correct polarity is forward biased); 0.957; 2.875.
- 6.8** 4.0; $5 I_{AC}$.
- 6.9** +0.333 pu, 1.0133 pu, 0.338 pu absorbed; -0.417 pu, 0.9833 pu, 0.410 pu injected.
- 6.10** 90 Mvar; 20.21 mH; 1003 μF ; 105.36° .
- 6.11** +10: 1 TSC, $\alpha=113.83^\circ$; +55: 3 TSCs, 105.36° ; +90: 4 TSCs, 123.59° ; -20: none, 105.36° .
- 6.12** +1.200 pu and -0.375 pu (using $B_{eq}=B/(1-X_\sigma B)$).
- 6.13** 1.06 pu; 0.85 pu; 0.96 pu.
- 6.14** 78.39° ; 2.107 pu; 2.093 pu; 3.333 pu (ideal); in current limit the maximum is 2.217 pu at 98.4° .
- 6.15** 0.662 mF; 170 V peak-to-peak.
- 6.16** 1.26% and 1.07%.
- 6.17** $G=58.06$ mS; capacitive $B_{ca}=33.52$ mS (106.7 μF) across $c-a$; inductive $B_{ab}=33.52$ mS (94.95 mH) across $a-b$; each 5.77 kvar; line current 13.91 A.
- 6.18** Open-ended. A well-argued answer favours a STATCOM or hybrid STATCOM: constant-current support at 0.5 pu ($Q \propto V$ rather than V^2), the bandwidth needed for flicker, the small footprint and low filtering needs. An FC-TCR is acceptable for flicker but weak for sags and lossy at idle.

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